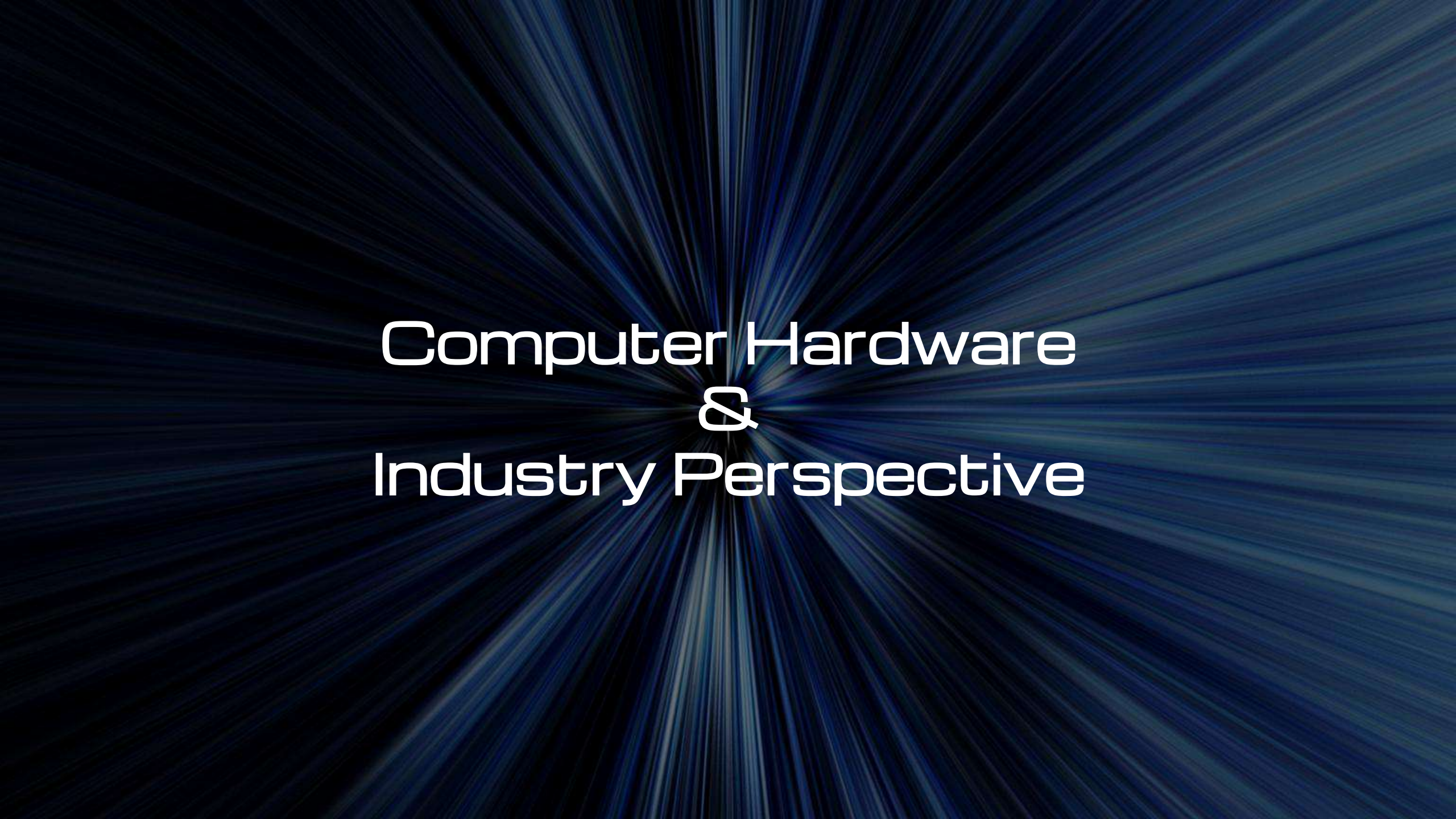




Computer Hardware & Industry Perspective

커리큘럼

주차	날짜	이론	실습	선각 보고서 주제
1주차	4/20 (일)	오리엔테이션, 하드웨어 구성 요소 개요	데스크탑 전체 분해, 주요 부품 구조 관찰	컴퓨터를 이해한다는 건 무엇일까?
2주차	4/26 (토)	메모리 & 저장장치 구조	RAM 분리/장착, SSD 분해 관찰	Computational Storage란 무엇인가?
3주차	5/3 (토)	발열과 냉각 구조	CPU 쿨러 제거 + 써멀 재도포, 팬 속도 측정	Arm SoC는 왜 발열에 더 강할까?
4주차	5/17 (토)	CPU 구조 & 최신 기술 + 분해 실습	CPU 히트스프레더 제거(뚜따)	CPU의 성능 향상이 둔화된 이유와 기술적 대안
5주차	5/24 (토)	GPU 구조 & 최신 기술 + 분해 실습	DLSS와 레이트레이싱 체험	AI는 GPU없이 존재할 수 있는가?
6주차	5/31 (토)	입출력 장치 & HCI	키보드/마우스 분해	HCI의 미래 : 뇌-컴퓨터 인터페이스
7주차	6/13 (금)	반도체 제작: 웨이퍼~패키징 전체 흐름		

※ 일정은 변동 가능하며, 팀 내 논의를 통해 유동적으로 조정될 수 있습니다.
※ 주차별로 배운 내용을 다음주까지 리포트(선각)를 작성해 발표합니다.

22:00

1교시 : 과제 발표

쉬는 시간 10분

2교시 : 이론

반도체가 없는 삶은 어떤 모습일까



ASML

**Standing on the
shoulders of giants**

이론

반도체 기업
반도체 제조 공정

반도체 기업 분류

A dark blue background with a light blue grid pattern that curves from the left towards the right, creating a sense of depth and perspective.

**Fabless
Foundry
IDM
Equipment Suppliers**

팹리스(Fabless)

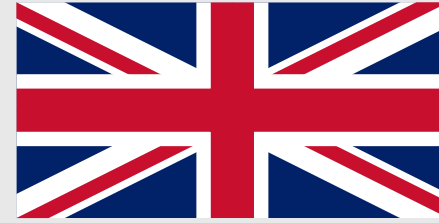
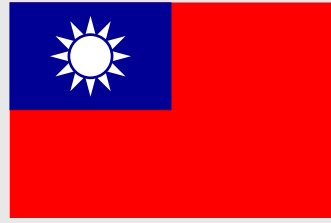
반도체 설계만 하고, 실제 제조(웨이퍼 가공)는 파운드리(Foundry) 업체에 외주 주는 기업

장점

- 초기 설비 투자 절감
- 설계에 집중 → 빠른 기술 전환 가능

역할 분담

- 팹리스 → 칩 설계·검증·패키징
- 파운드리 → 웨이퍼 가공(물리적 제조)



HISILICON



REALTEK



UNISOC®



NOVATEK



파운더리(Foundry)

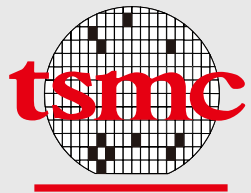
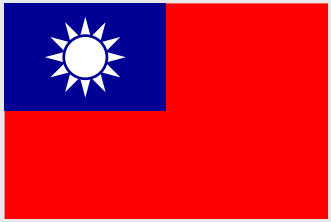
팹리스가 설계한 반도체 칩을 실제 웨이퍼 위에 제조(가공)해 주는 전문 업체

장점

- 막대한 설비투자 없이 첨단 공정 이용 가능
- 팹리스와 파운더리의 분업을 통해 설계·제조 효율 극대화

주요 역할

- 공정 개발(R&D)
- 웨이퍼 가공(리소그래피, 에칭, 증착 등)
- 테스트 및 패키징 전 초기 웨이퍼 단계까지



SAMSUNG
Foundry



intel foundry



UMC

종합반도체 회사(Integrated Device Manufacturer)

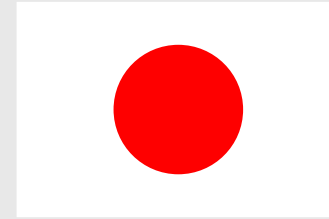
설계·제조·패키징까지 반도체의 모든 단계를 자체 수행하는 기업

장점

- 설계와 제조 공정 간 최적화 용이
- 기술 축적 및 IP 보호 유리

단점

- 없음



장비 공급업체(Equipment Suppliers)

웨이퍼 가공 공정에 필요한 주요 장비를 공급하는 기업

장점

- 첨단 리소그래피·식각·증착 기술 제공
- 다양한 공정 단계(노광, 에칭, 검사 등) 일괄 지원
- 글로벌 서비스 네트워크로 안정적 가동 보장

단점

- 높은 초기 장비 도입 비용 필요
- 복잡한 운영·유지보수 요구
- 공정별 특화 인력·교육 부담 증가

회사명	국가	주요 장비 분야
ASML	NL 네덜란드	DUV/EUV 리소그래피
Nikon	JP 일본	DUV 리소그래피 스캐너
Canon	JP 일본	스텝퍼(Stepper) 리소그래피
Ebara Corporation	JP 일본	CMP 장비, 진공 펌프
Hitachi High-Tech	JP 일본	식각, SEM/TEM 검사 장비
Ulvac	JP 일본	증착, 에칭, 진공 시스템
Screen Semiconductor Solutions	JP 일본	세정·코팅·현상, 열처리, CMP
Tokyo Electron (TEL)	JP 일본	PVD/CVD, 식각, 이온 주입, 클린/스트립
Advantest	JP 일본	ATE(자동시험장비), 반도체 테스트 시스템
Teradyne	US 미국	반도체·모듈 자동시험장비
Veeco Instruments	US 미국	MOCVD, PVD, 에피택시(Epitaxy)
Applied Materials	US 미국	증착(CVD/PVD), CMP, 열처리
Lam Research	US 미국	식각(Etch), 증착(Deposition)
Teradyne	US 미국	반도체·모듈 자동시험장비
Veeco Instruments	US 미국	MOCVD, PVD, 에피택시(Epitaxy)

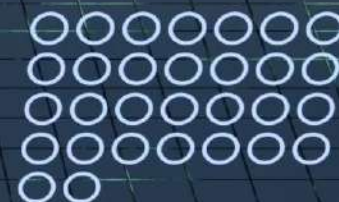
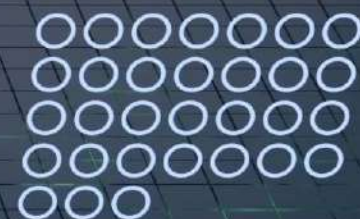
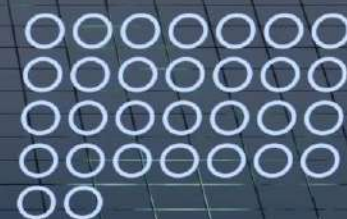
반도체 제조 공정

- Wafer processing (also called front end)
 - Wet cleans
 - Cleaning by solvents such as acetone, trichloroethylene or ultrapure water sometimes while spinning the wafer
 - Piranha solution
 - RCA clean
 - Wafer scrubbing
 - Spin cleaning
 - Jet spray cleaning
 - Cryogenic aerosol
 - Megasonics
 - Immersion batch cleaning
 - Surface passivation
 - Photolithography
 - Photoresist coating (often as a liquid, on the entire wafer)
 - Photoresist baking (solidification in an oven)
 - Edge bead removal
 - Exposure (in a photolithography stepper, scanner or mask aligner)
 - Post Exposure Baking (PEB) improves the durability of the photoresist
 - Development (removal of parts of the resist by application of a liquid developer, leaving only parts of the wafer exposed for ion implantation, layer deposition, etching, etc)
 - Hardmask creation
 - Ion implantation (in which dopants are embedded in the wafer creating regions of increased or decreased conductivity)
 - Etching (microfabrication)
 - Dry etching (plasma etching)
 - Reactive-ion etching (RIE)
 - Deep reactive-ion etching (DRIE)
 - Atomic layer etching (ALE)
 - Plasma ALE
 - Thermal ALE
 - Wet etching
 - Buffered oxide etch
 - Chemical vapor deposition (CVD)
 - Metal organic chemical vapor deposition (MOCVD), used in LEDs
 - Atomic layer deposition (ALD)
 - Physical vapor deposition (PVD)
 - Sputtering
 - Evaporation
 - Epitaxy
 - Molecular beam epitaxy (MBE)
 - Ion beam deposition
 - Plasma ashing (for complete photoresist removal/photoresist stripping, also known as dry strip, historically done with a chemical solvent called a resist stripper, to allow wafers to undergo another round of photolithography)
 - Thermal treatments
 - Rapid thermal processing (RTP), rapid thermal anneal
 - Millisecond thermal processing, millisecond anneal, millisecond processing, flash lamp anneal (FLA)
 - Laser anneal
 - Furnace anneals
 - Thermal oxidation
 - LOCOS
 - Laser lift-off (for LED production)
 - Electrochemical deposition (ECD). See Electroplating.
 - Chemical-mechanical polishing (CMP)
 - Wafer testing (where the electrical performance is verified using automatic test equipment, binning and/or laser trimming may also be carried out at this step)
- Die preparation
 - Through-silicon via manufacture (for three-dimensional integrated circuits)
 - Wafer mounting (wafer is mounted onto a metal frame using dicing tape)
 - Wafer backgrinding and polishing (reduces the thickness of the wafer for thin devices like a smartcard or PCMCIA card or wafer bonding and stacking, this can also occur during wafer dicing, in a process known as Dice Before Grind or DBG)
 - Wafer bonding and stacking (for three-dimensional integrated circuits and MEMS)
 - Redistribution layer manufacture (for WLCSP packages)
 - Wafer bumping (for flip chip BGA (ball grid array), and WLCSP packages)
 - Die cutting or wafer dicing
- IC packaging
 - Die attachment (The die is attached to a leadframe using conductive paste or die attach film.)
 - IC bonding: Wire bonding, thermosonic bonding, flip chip or tape automated bonding (TAB)
 - IC encapsulation or integrated heat spreader (IHS) installation
 - Molding (using special plastic molding compound that may contain glass powder as filler to control thermal expansion)
 - Baking
 - Electroplating (plates the copper leads of the lead frames with tin to make soldering easier)
 - Laser marking or silkscreen printing
 - Trim and form (separates the lead frames from each other, and bends the lead frame's pins so that they can be mounted on a printed circuit board)
- IC testing





940
Steps







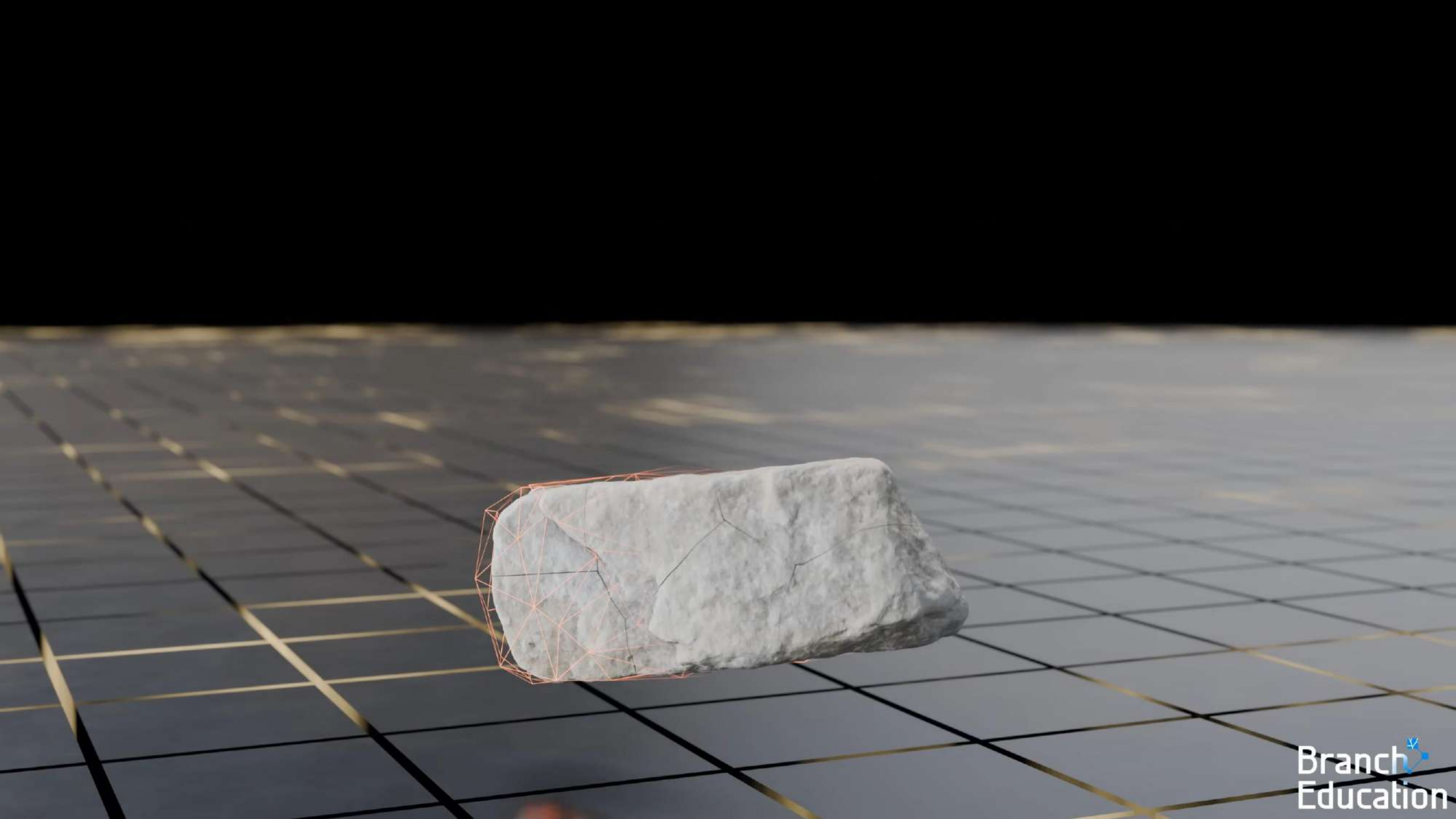
- 웨이퍼 준비
 - 실리콘 잉곳 절단 → 연마·세정
- 포토레지스트(PR) 코팅
 - 얇은 빵 반죽(감광액)을 웨이퍼 전체에 바르기
- EUV 노광(Exposure)
 - EUV 광원에서 나온 극자외선(빛)을 마스크 패턴대로 쏘기
 - (내부적으로는 tin 플라즈마→광 집광→마스크 투과 과정을 거침)
- 현상(Development)
 - 빛이 닿은(PR 반응한) 부분만 남기고, 나머지(PR이 필요 없는 영역) 제거
- 식각(Etching)
 - PR이 남은 부분을 보호막 삼아, 노출된 실리콘·금속을 깎아내기
- PR 제거(Strip)
 - 남은 포토레지스트를 화학이나 플라즈마로 완전히 벗겨내기
- 박막 증착(Deposition)

- 회로 연결에 필요한 금속·절연막을 얇게 덮기
- 평탄화(CMP)
 - 증착·식각으로 울퉁불퉁해진 표면을 ‘샌드페이퍼’처럼 깎아 매끈하게 만들기
- 어닐링(열처리)
 - 고온에서 가열하여 주입된 이온 확산·결정 구조 안정화
- 2~9 단계 반복
 - 수십에서 수백 개 ‘회로층’을 쌓아 올림
- 다이싱 & 패키징
 - 웨이퍼를 개별 칩(다이)으로 자르고, 본딩·몰딩·테스트 후 최종 패키지 완성

The image features a complex, abstract geometric design. It consists of numerous thin, wavy, parallel lines that create a sense of depth and movement. These lines are arranged in a way that suggests a three-dimensional structure, possibly a series of stacked, curved planes. The color palette is primarily dark, with the lines appearing as bright, glowing streaks against a black background. In the lower-left corner, the text "SAMSUNG TOMORROW" is visible in a light, sans-serif font.

SAMSUNG TOMORROW

잉곳 만들기 -> 자르기 -> 연마하기

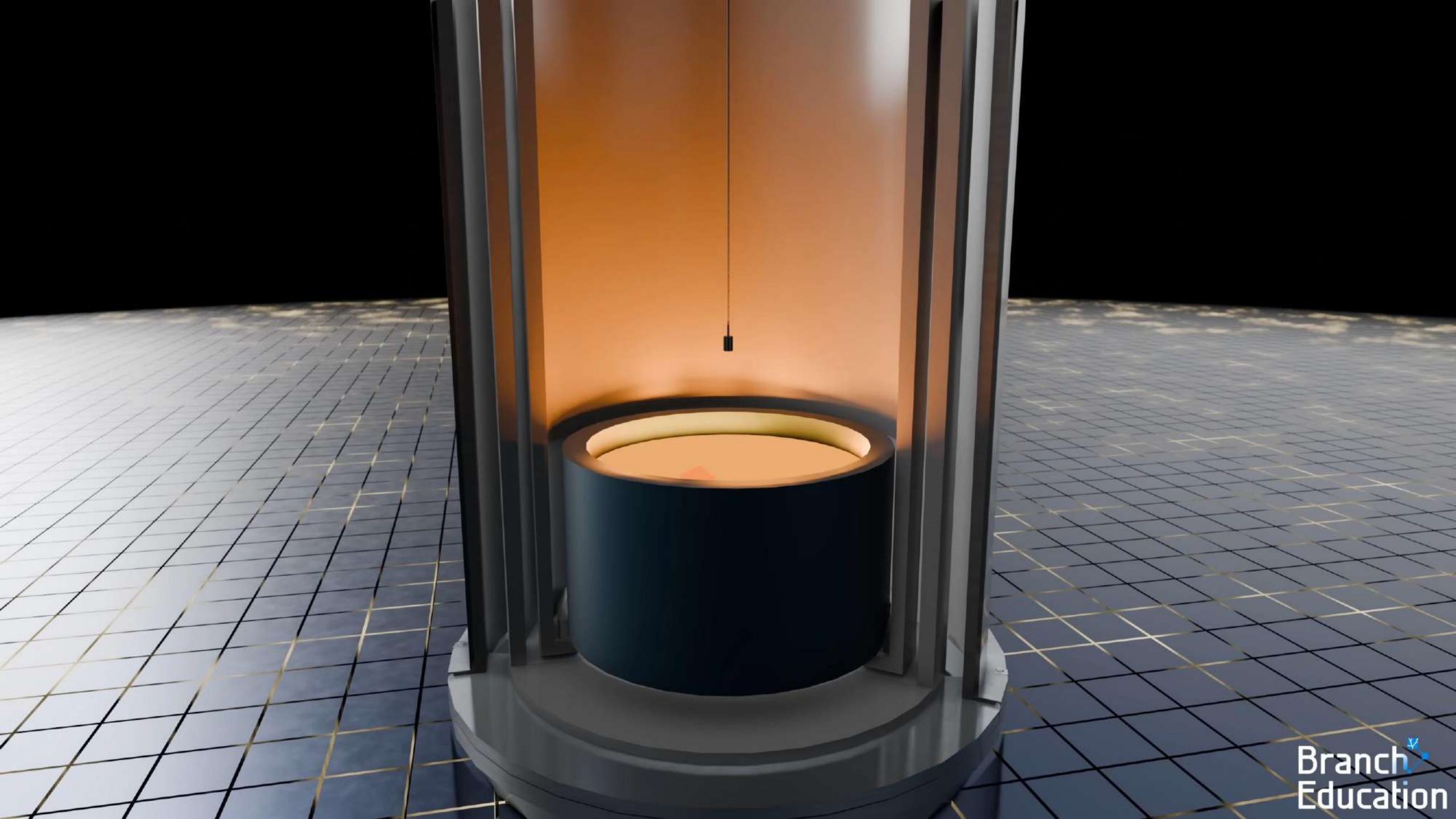






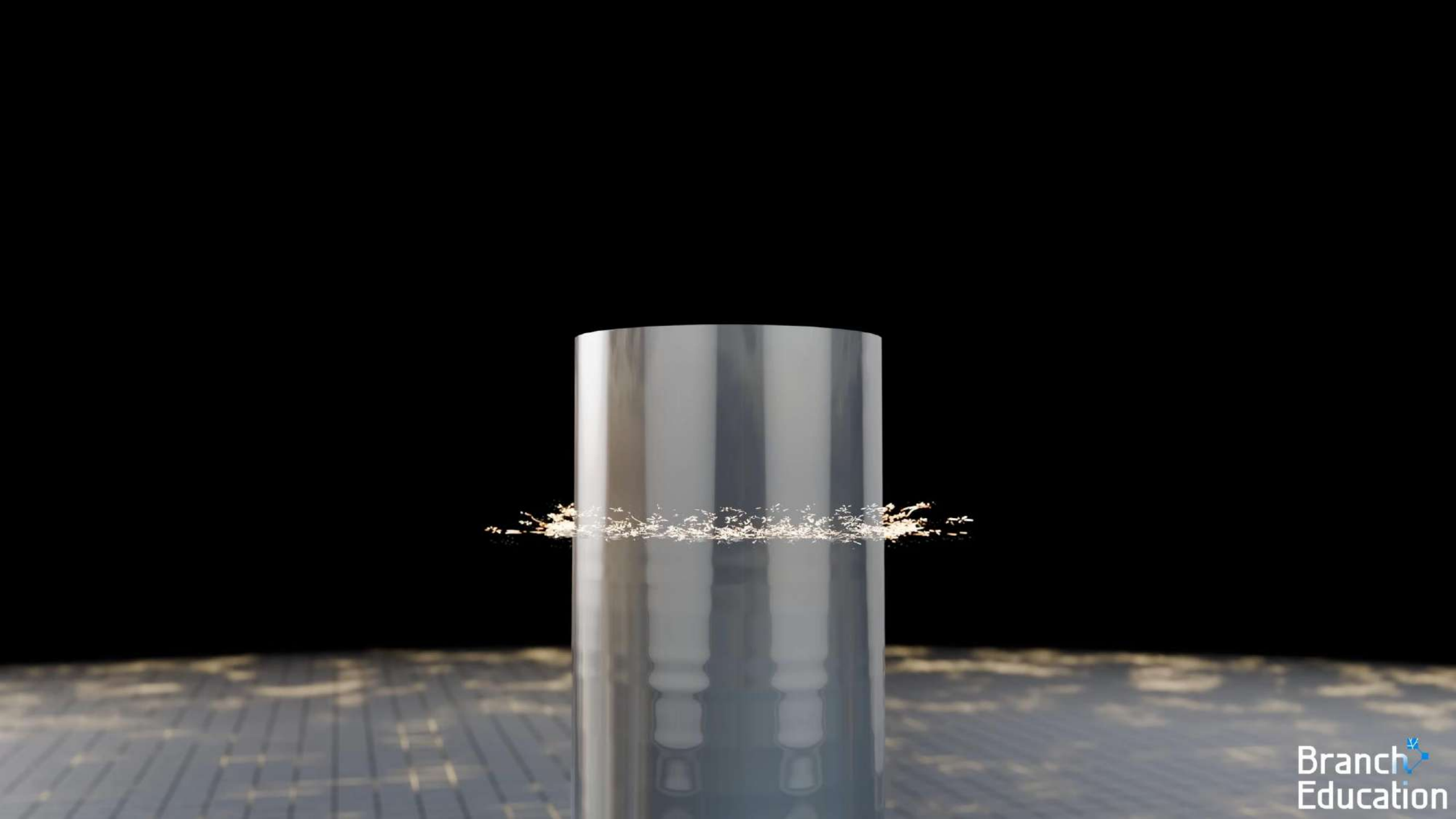




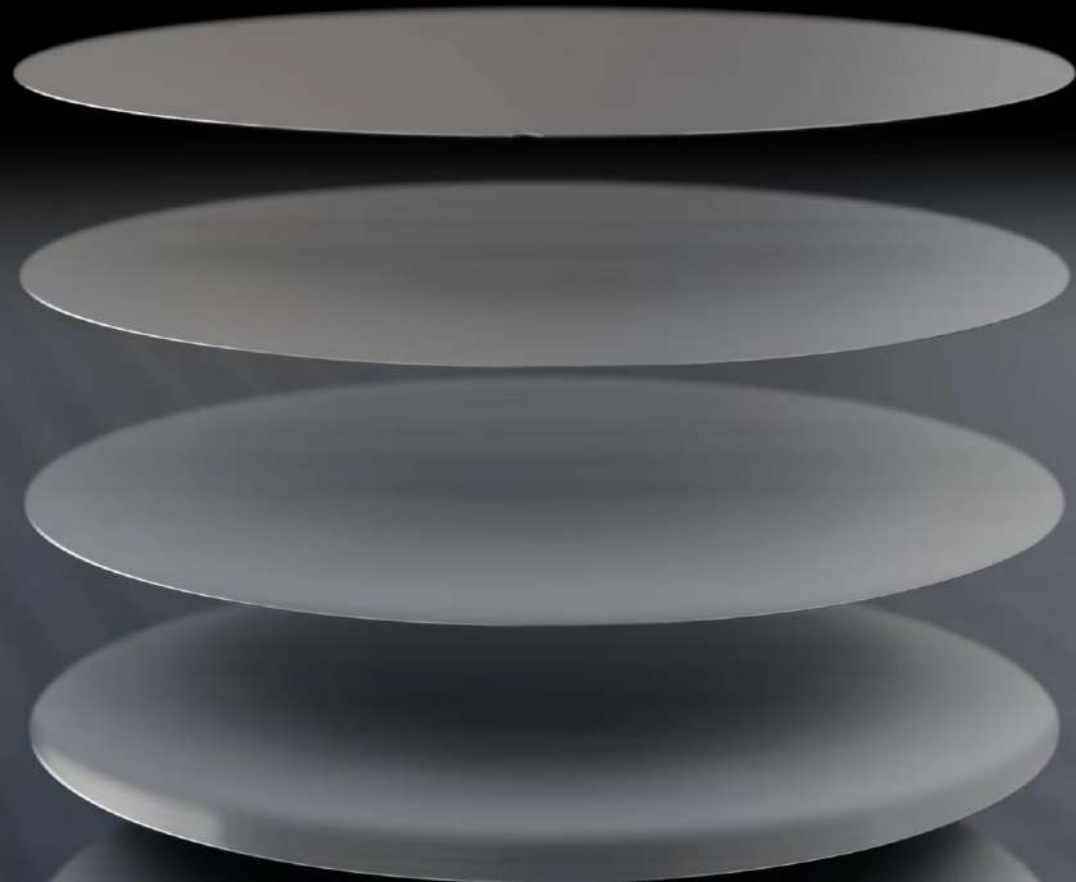












300mm

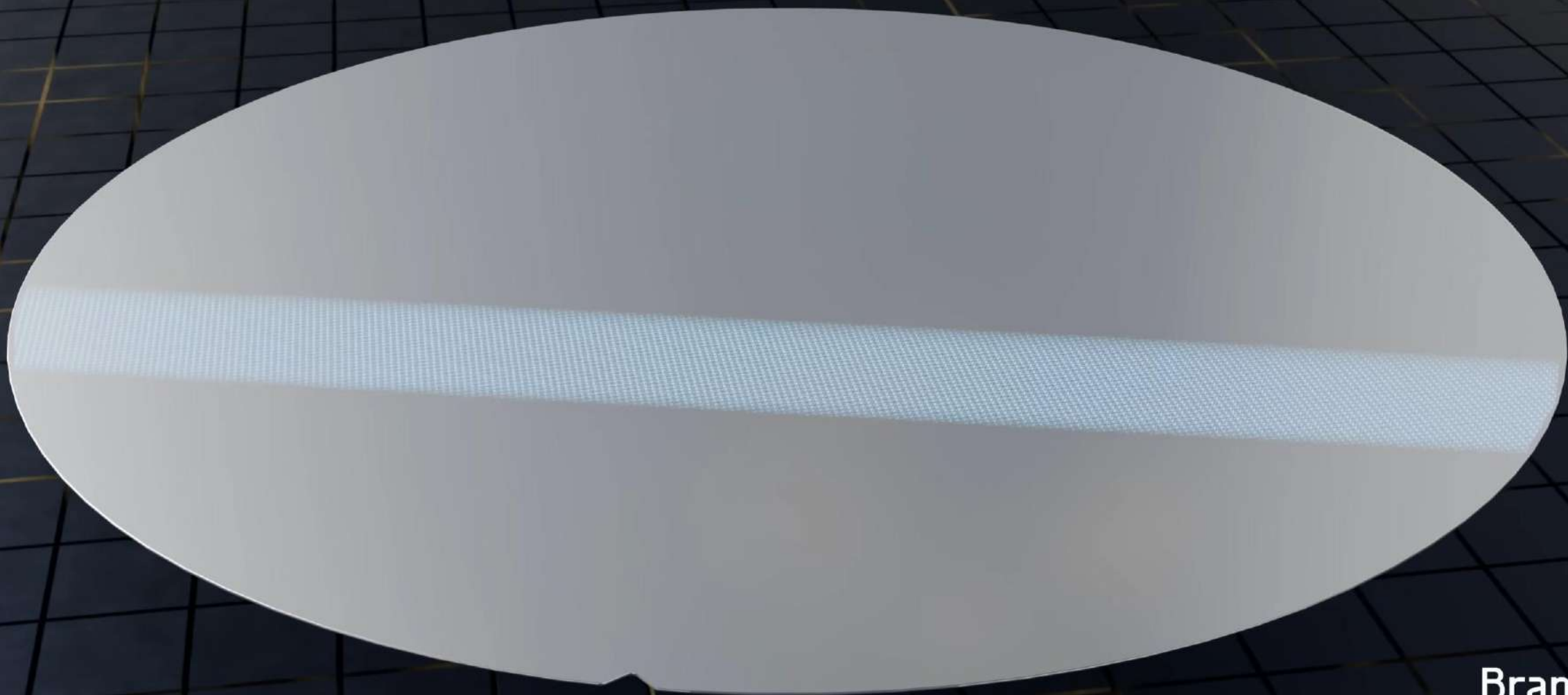


0.75mm

I

6QA719221BE | 15.5 | A 111 | G3



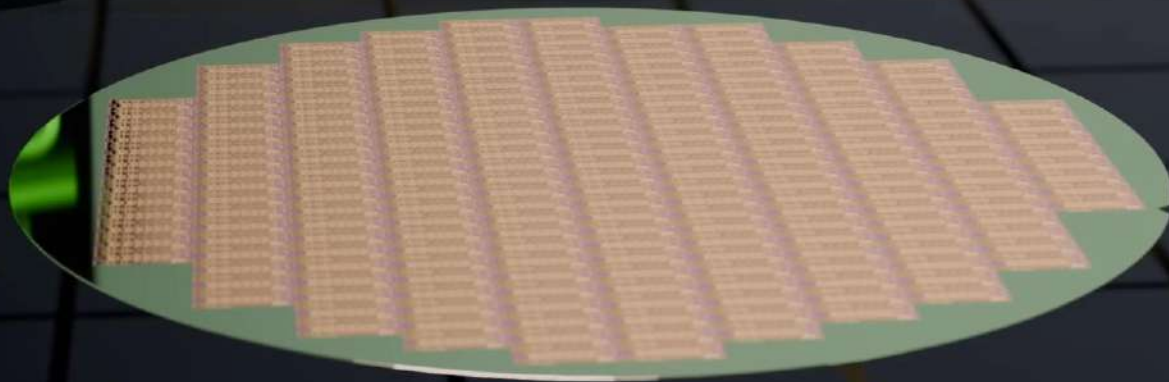


\$100



CPUs

$$\begin{array}{r} 230 \text{ chips} \\ \times \$552/\text{chip} \\ \hline \$126960 \end{array}$$



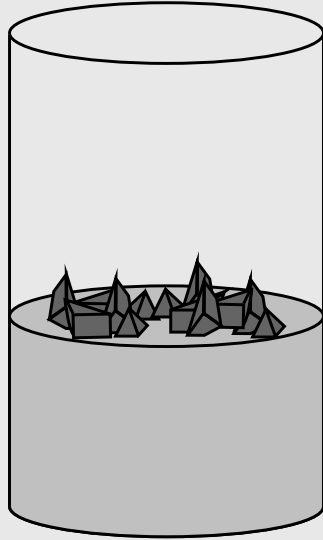
CPU's

$$\begin{array}{r} 230 \text{ chips} \\ \times \$552 / \text{chip} \\ \hline \$126960 \end{array}$$

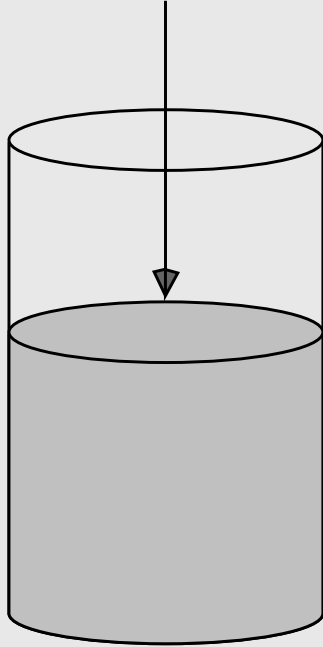
Gold

$$\begin{array}{r} 125 \text{ gram / wafer} \\ \times \$77 / \text{gram} \\ \hline \$9625 \end{array}$$

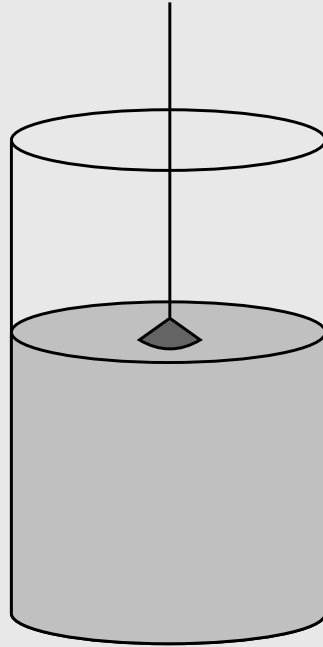




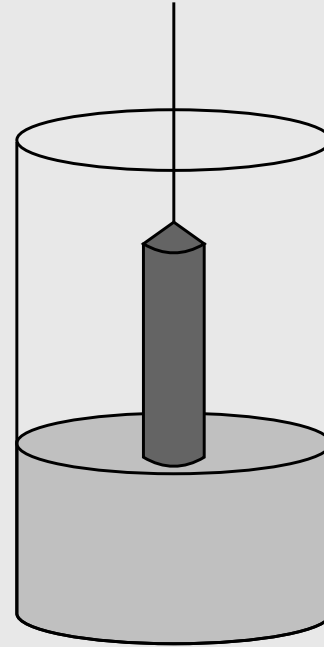
Melting of
polysilicon,
doping



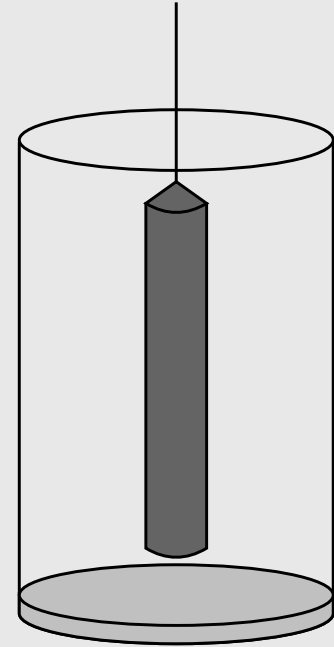
Introduction
of the seed
crystal



Beginning of
the crystal
growth



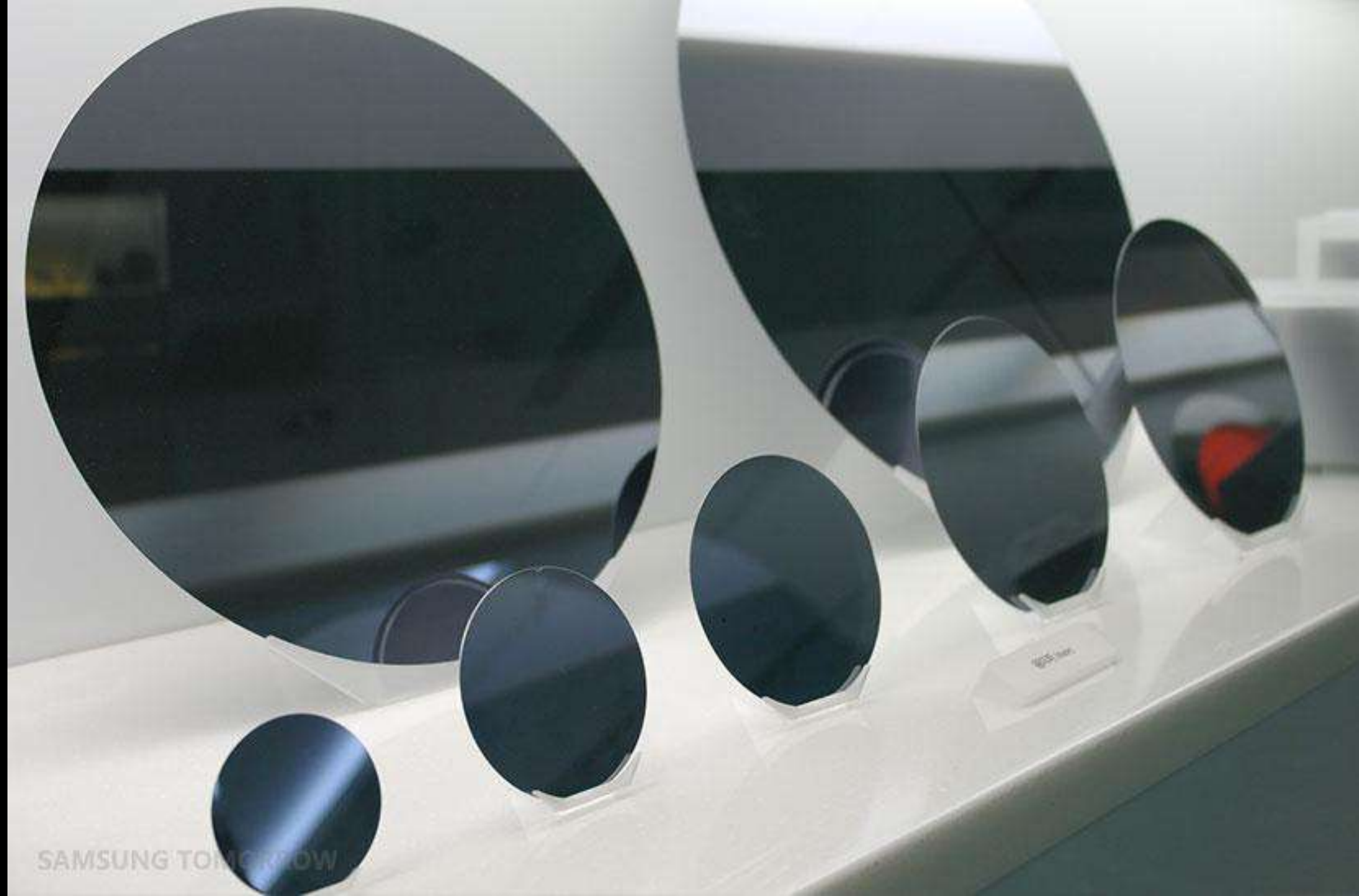
Crystal
pulling



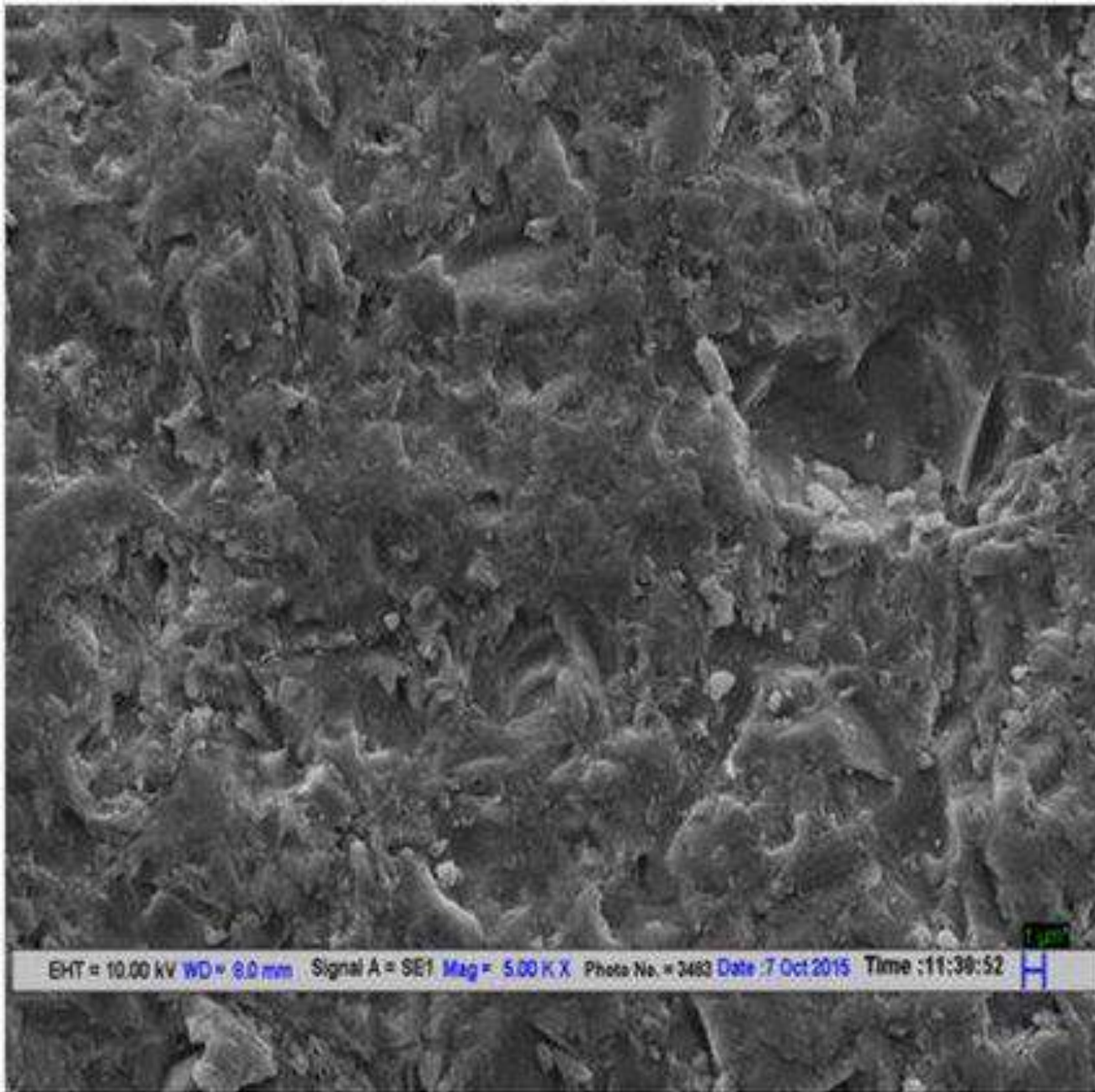
Formed crystal
with a residue
of melted silicon



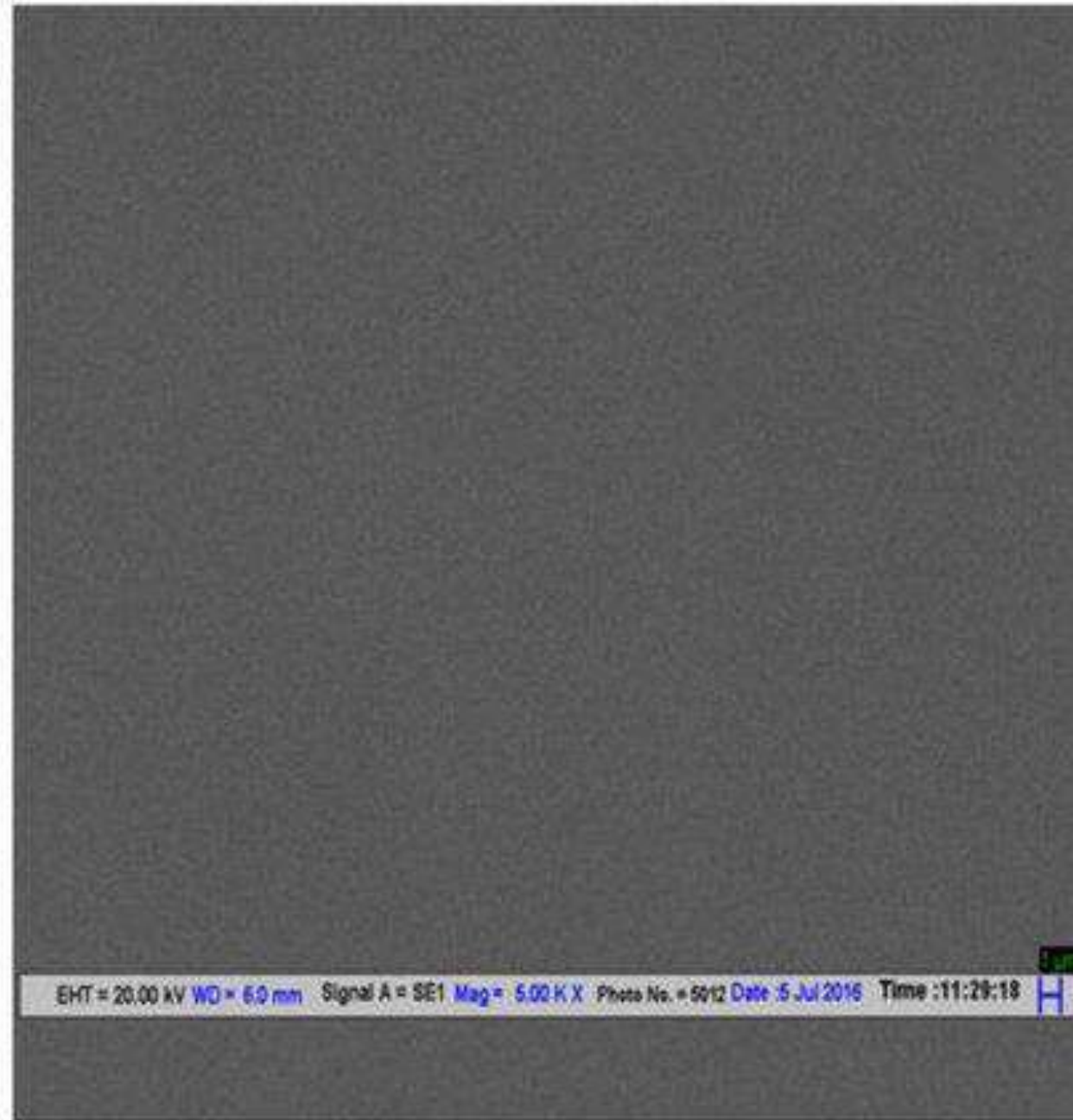
SAMSUNG TOMORROW



SAMSUNG TOMORROW



(a)

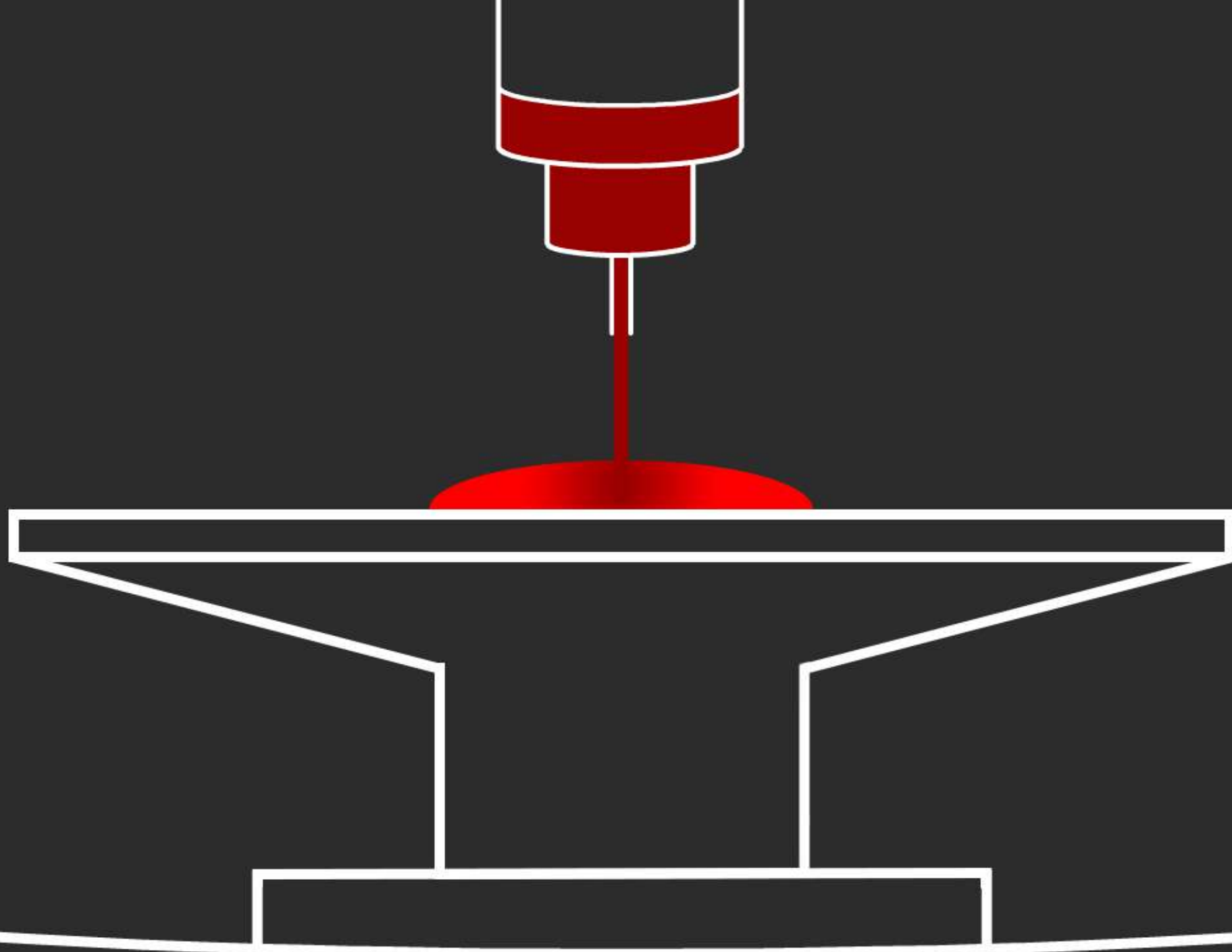


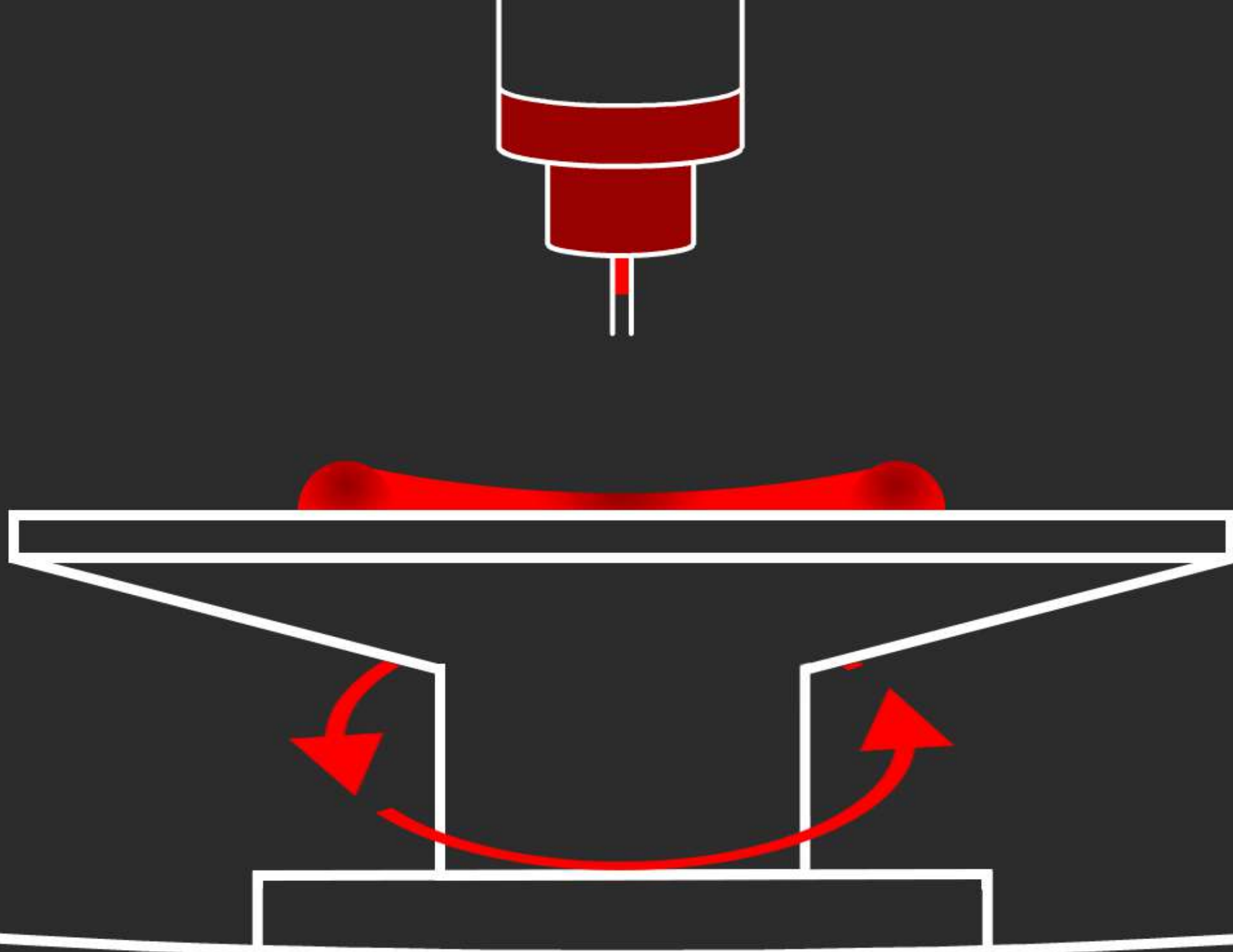
(b)

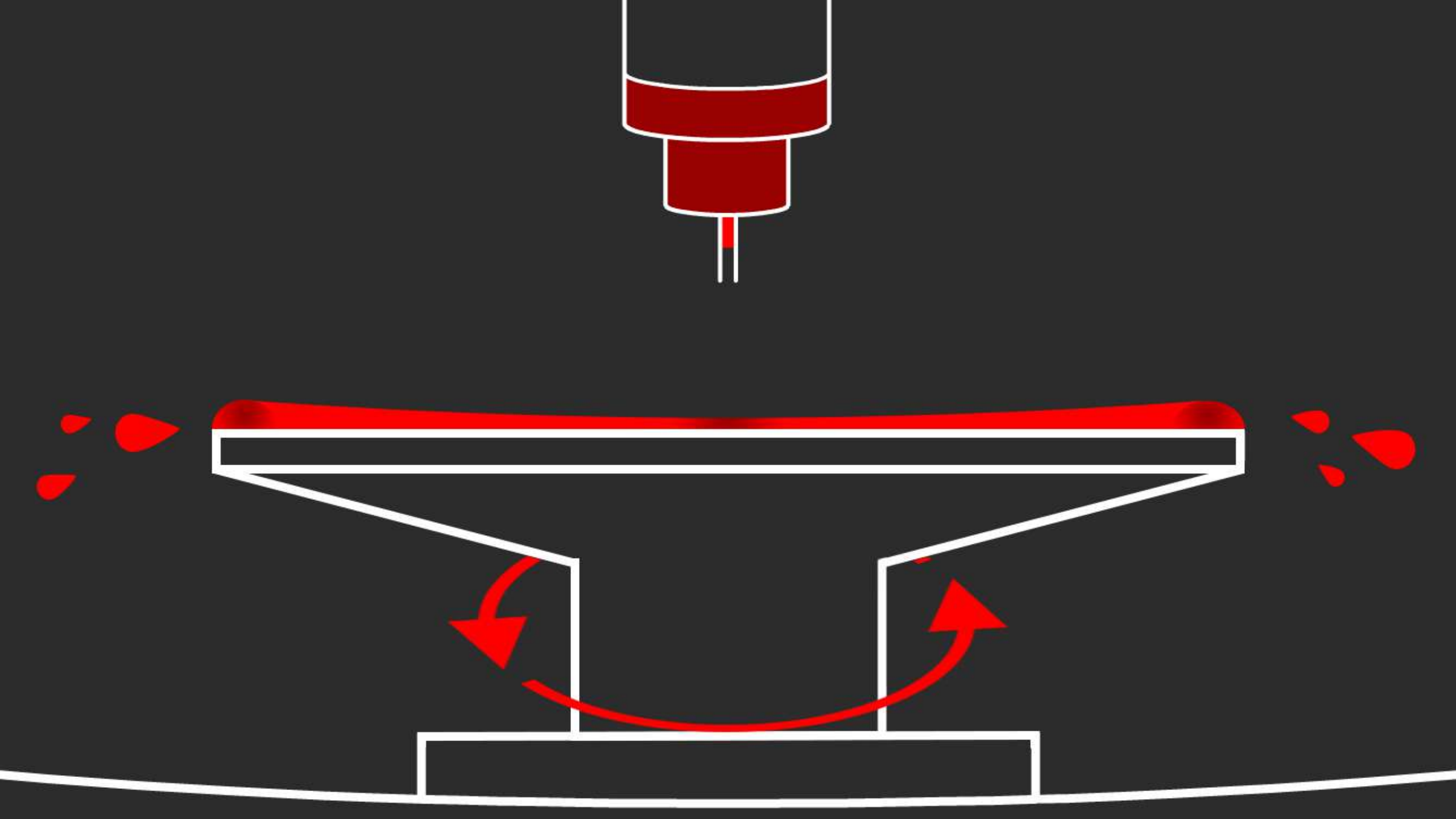
포토리지스트(PR) 코팅

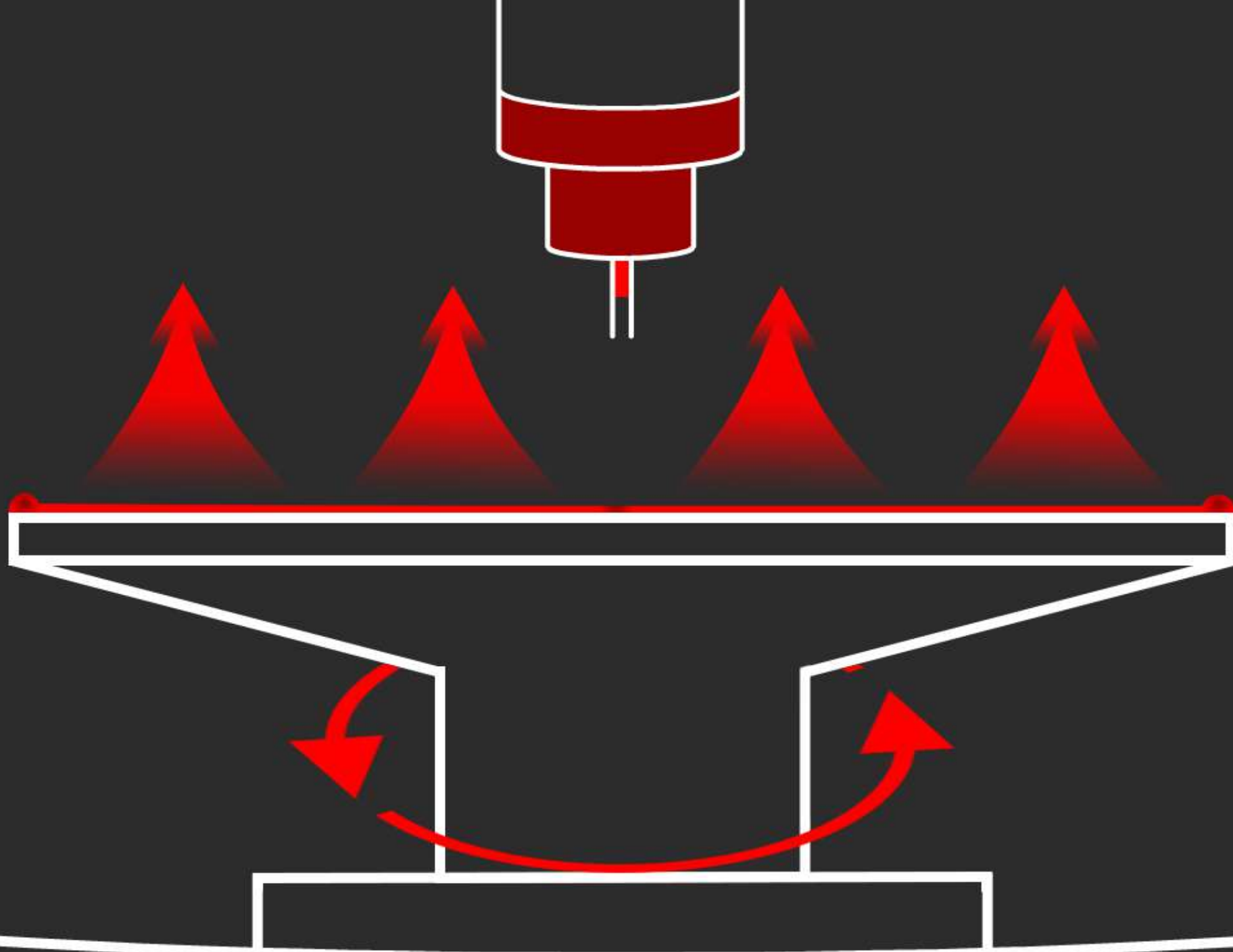
반도체 웨이퍼 위에 얇게 도포하는 '감광(光化)' 물질로,
빛(자외선·EUV 등)을 쬐면 화학적 성질이 변한다.

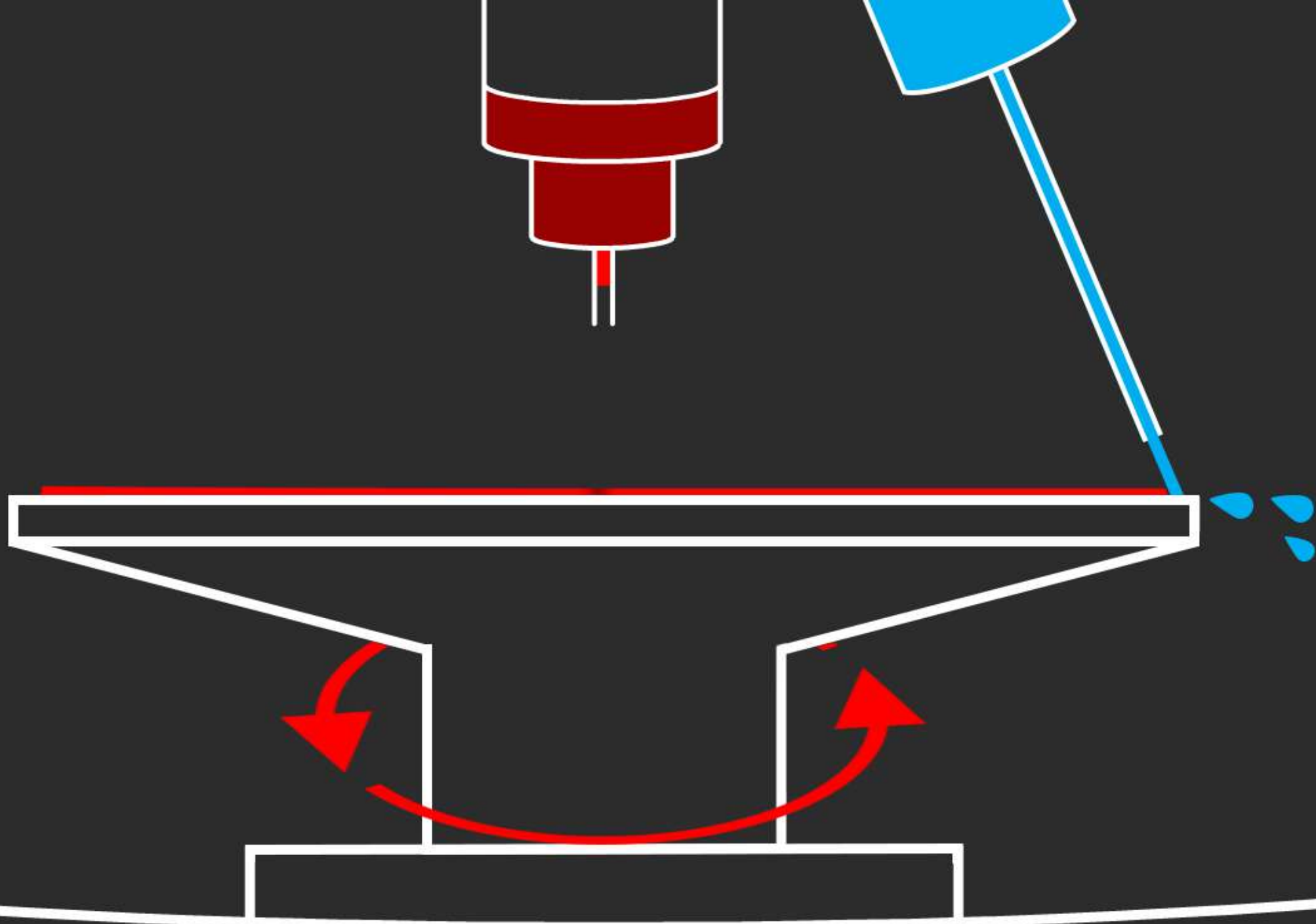
빛을 쬐인 부분과 그렇지 않은 부분을 구분해 회로 패턴을 만들 때 사용한다.











Photoresist Spin Coater



Photoresist Spin Coater

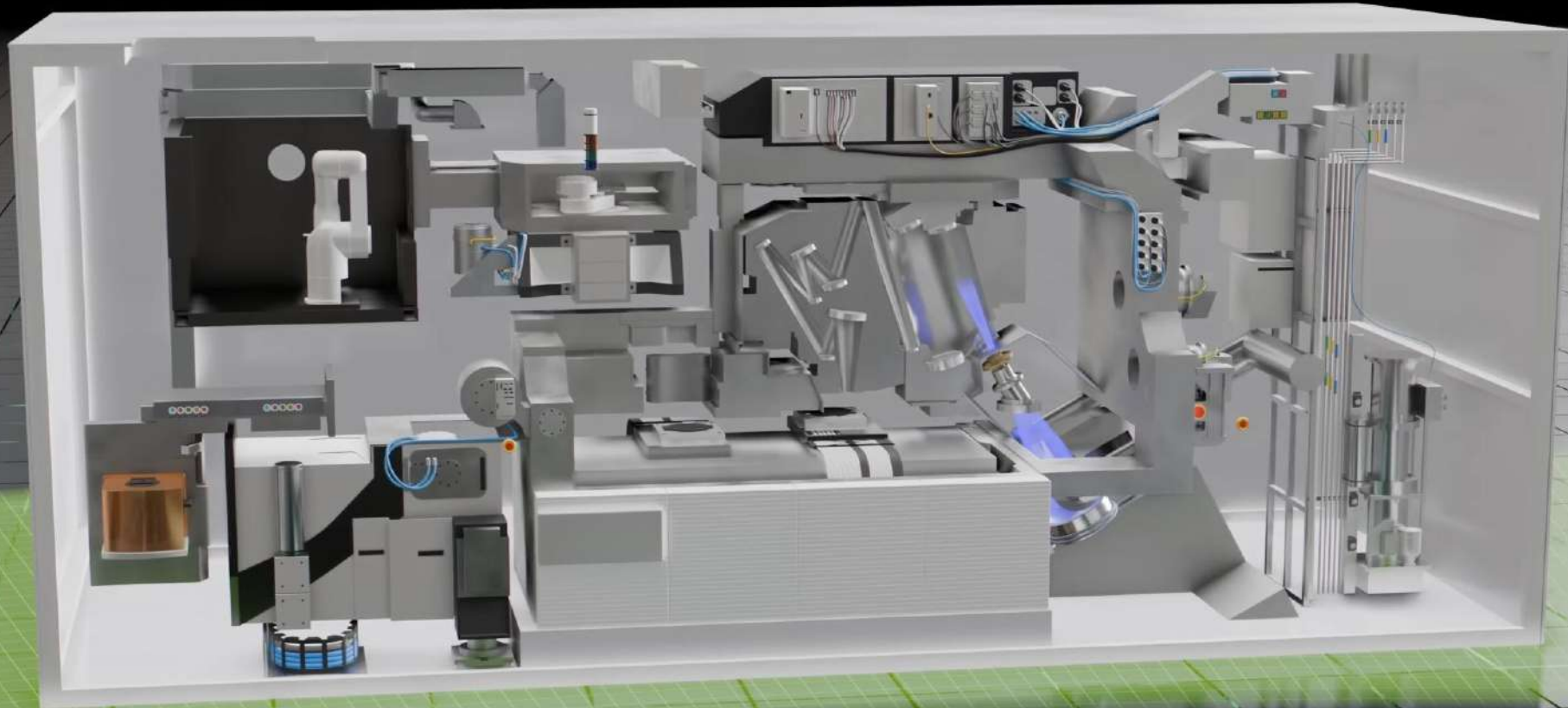


포토리소그래피(Photolithography)

빛을 이용해 마스크(mask)에 그려진 회로 패턴을 실리콘 웨이퍼 위에 정밀하게 전사하는 공정
E U V Exposure

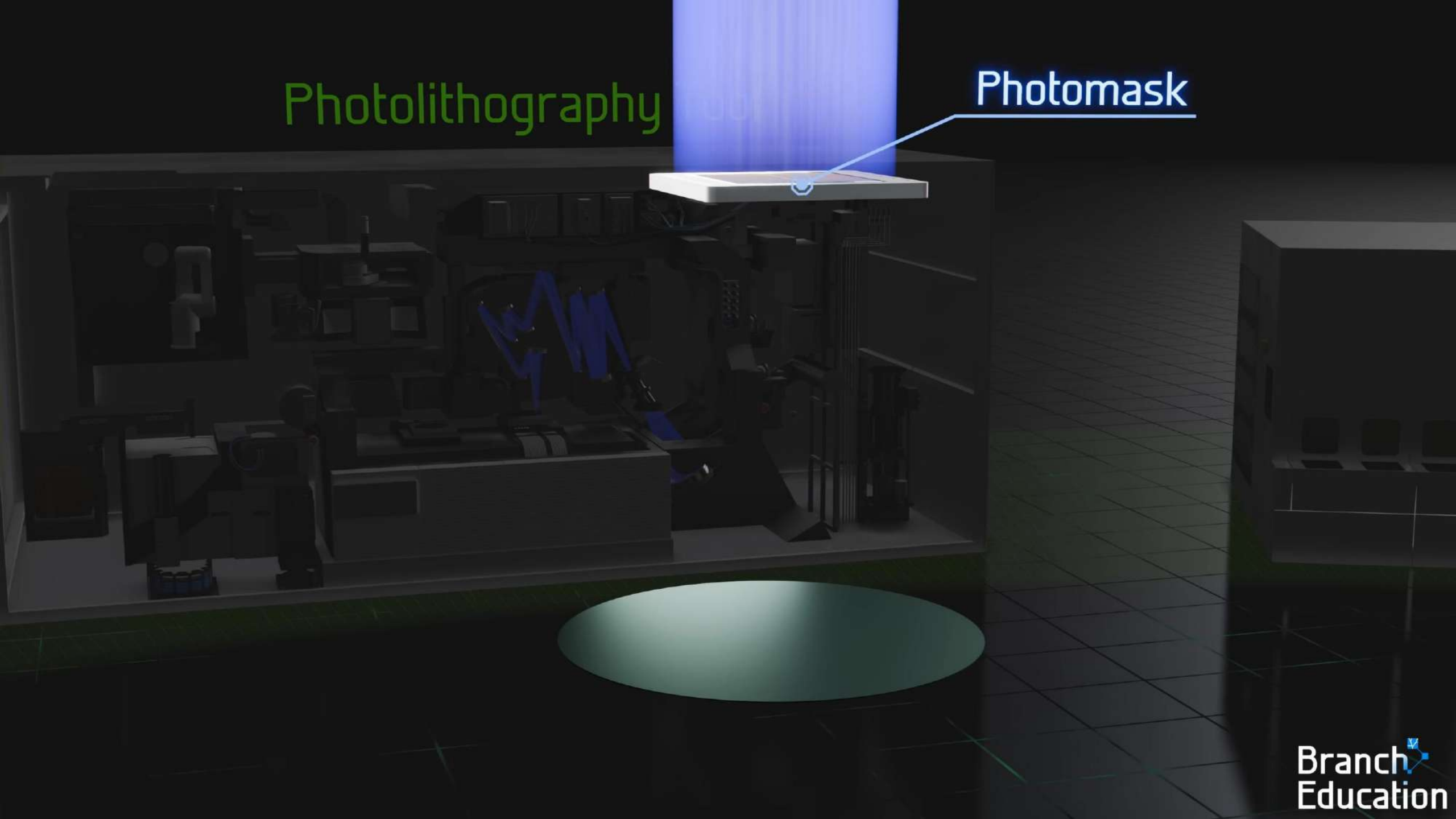
빛 만들기 -> 쏘기 -> 깎아낼 부분 약화시키기

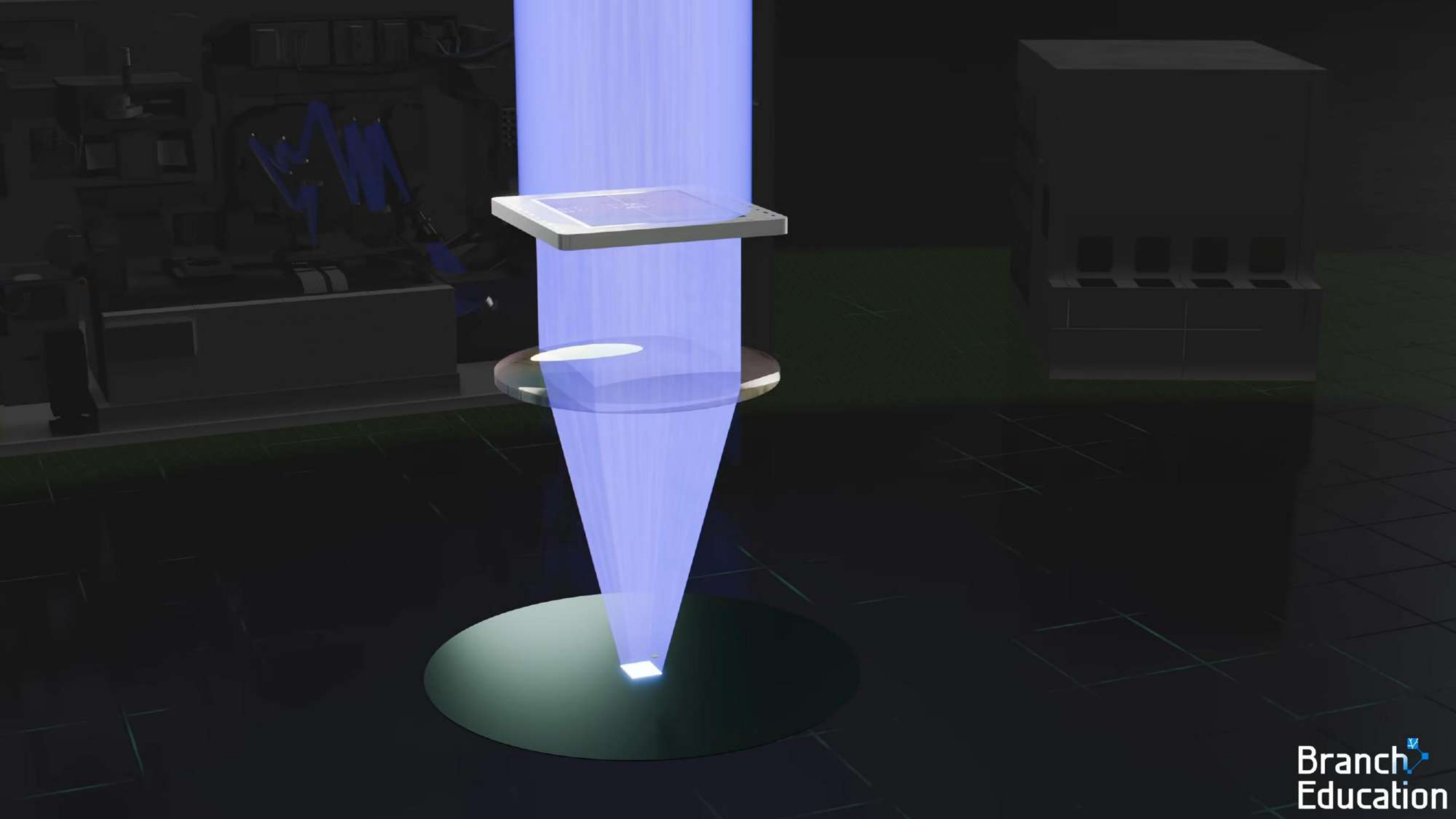
Photolithography Tool

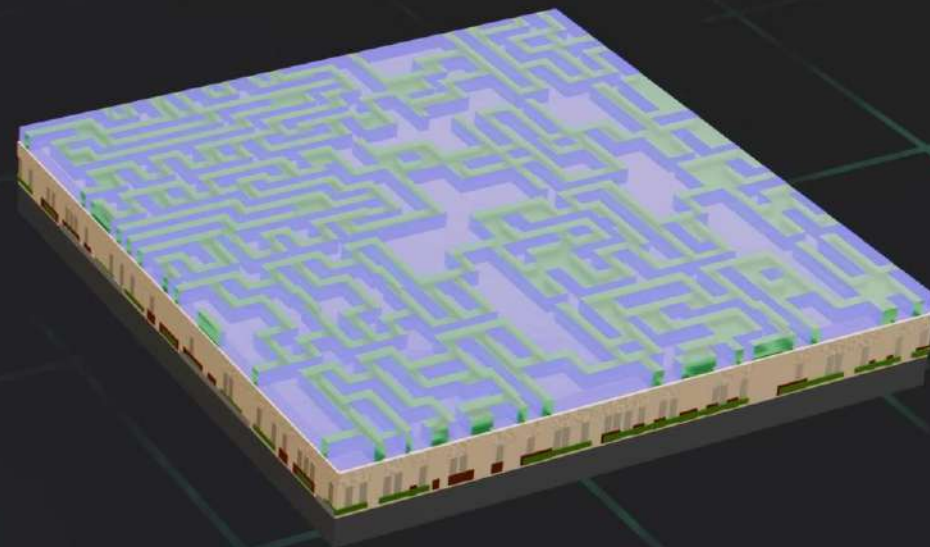
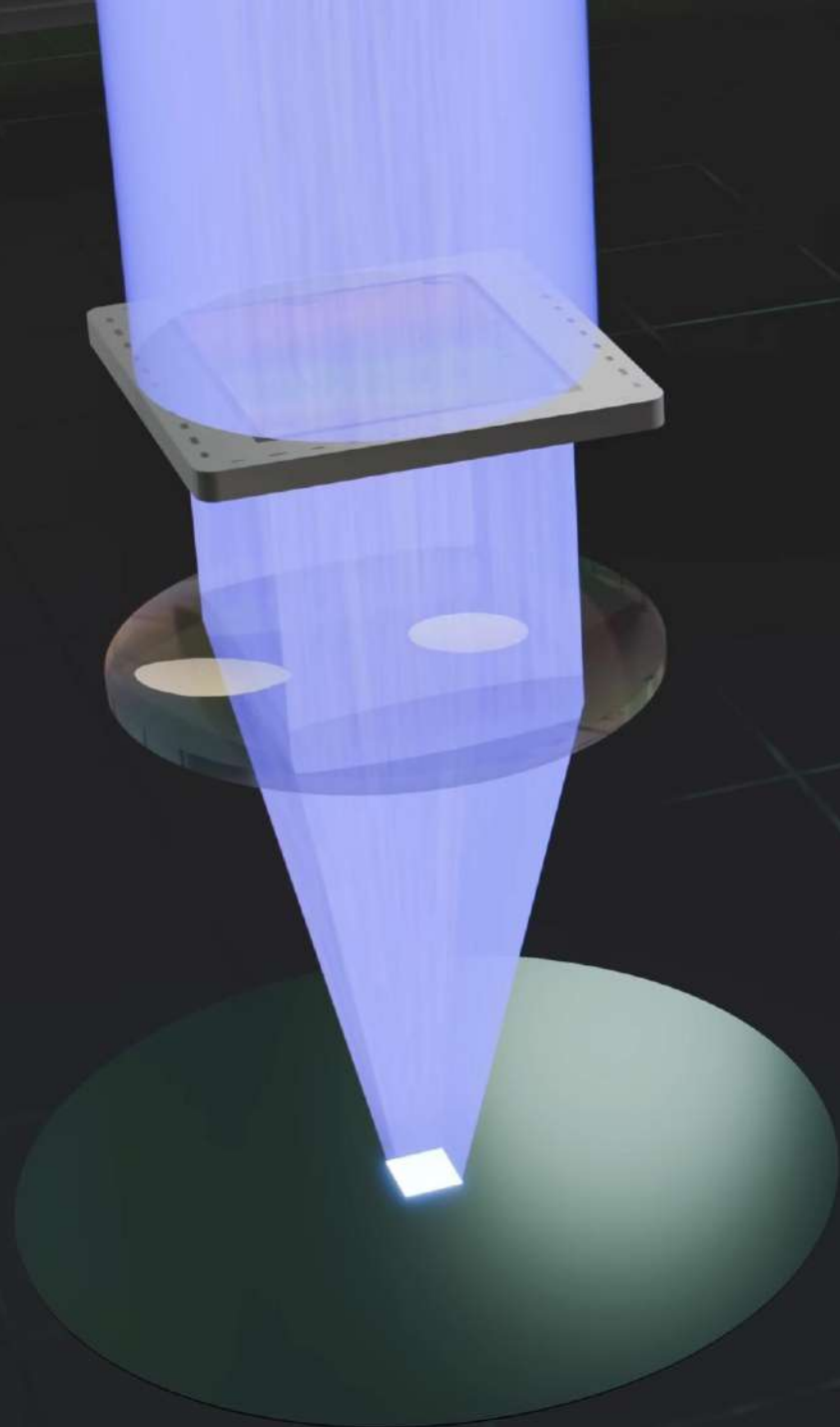


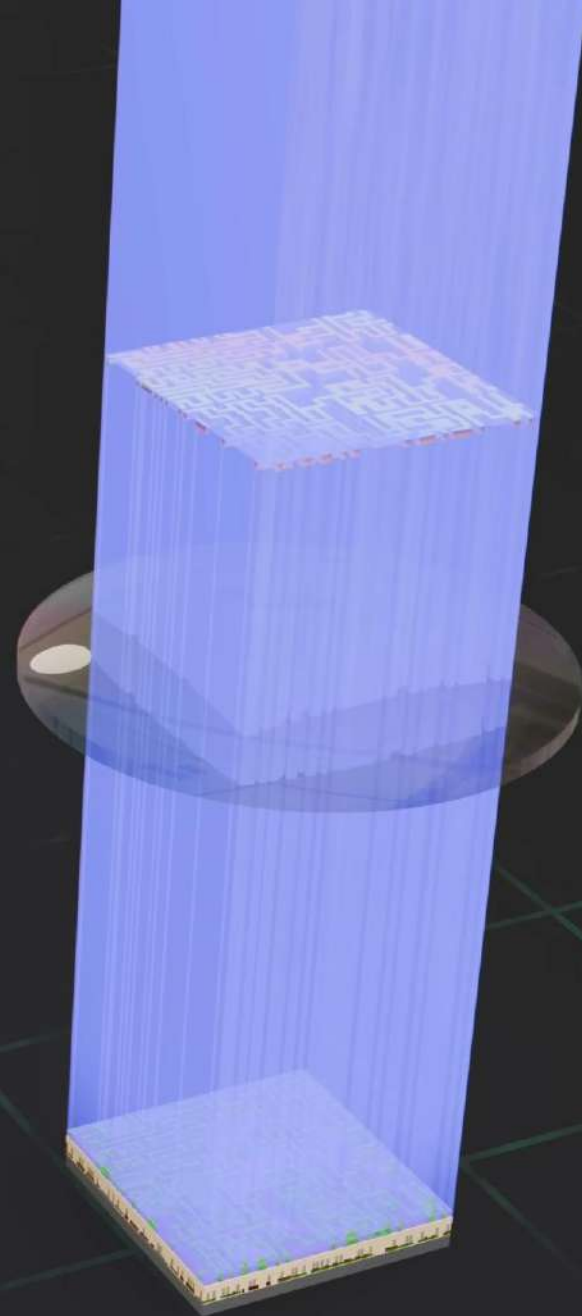
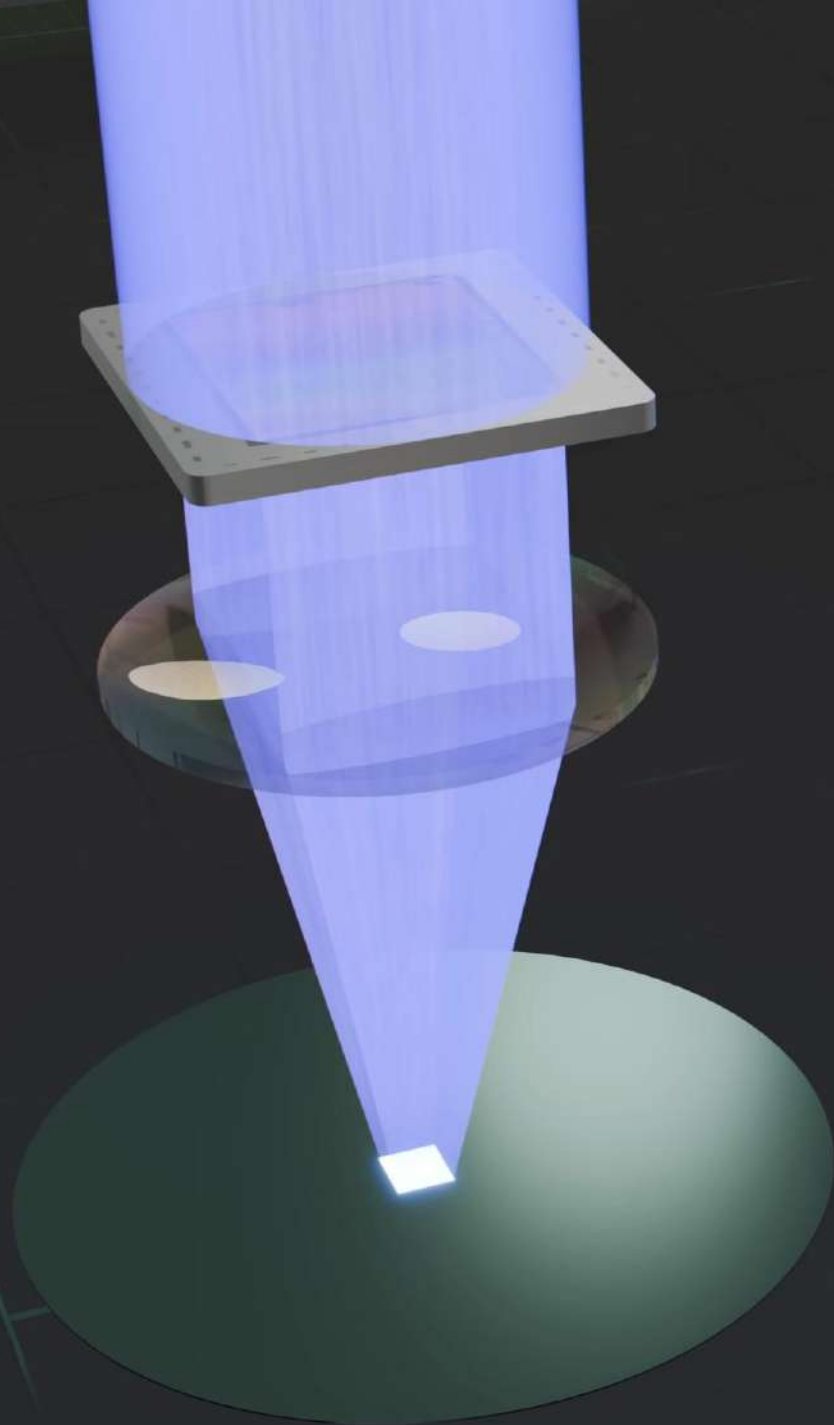
Photolithography

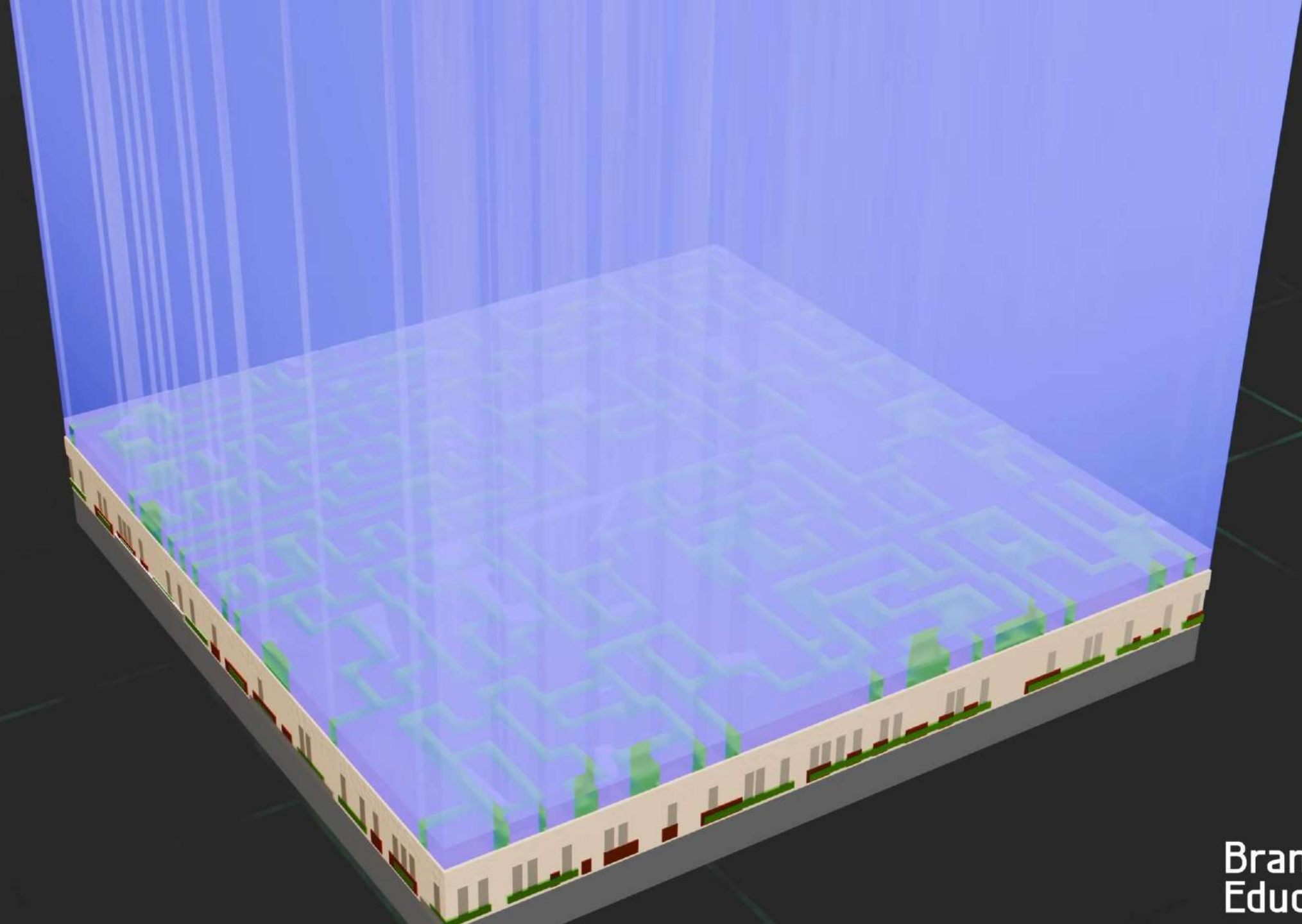
Photomask



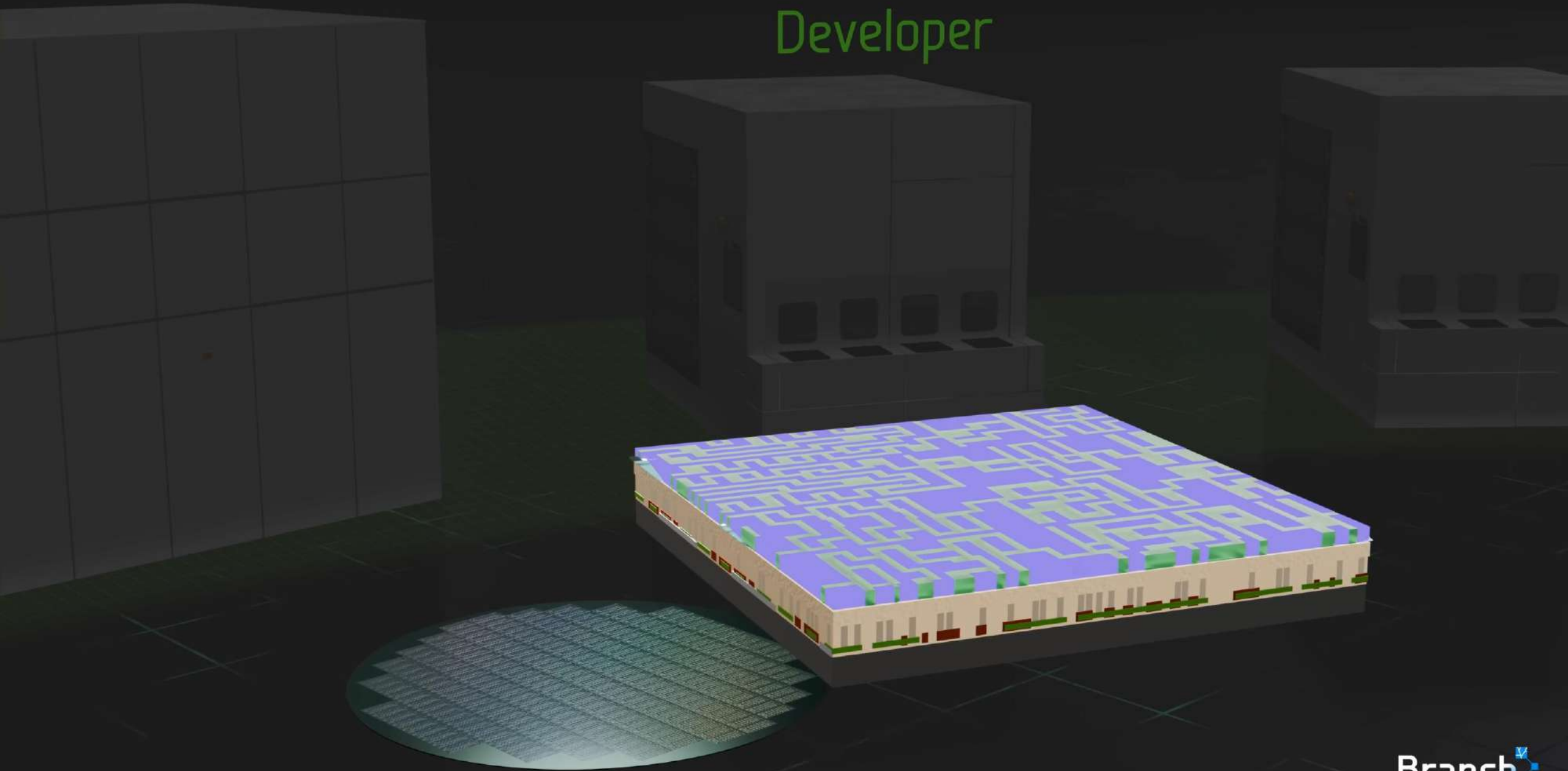




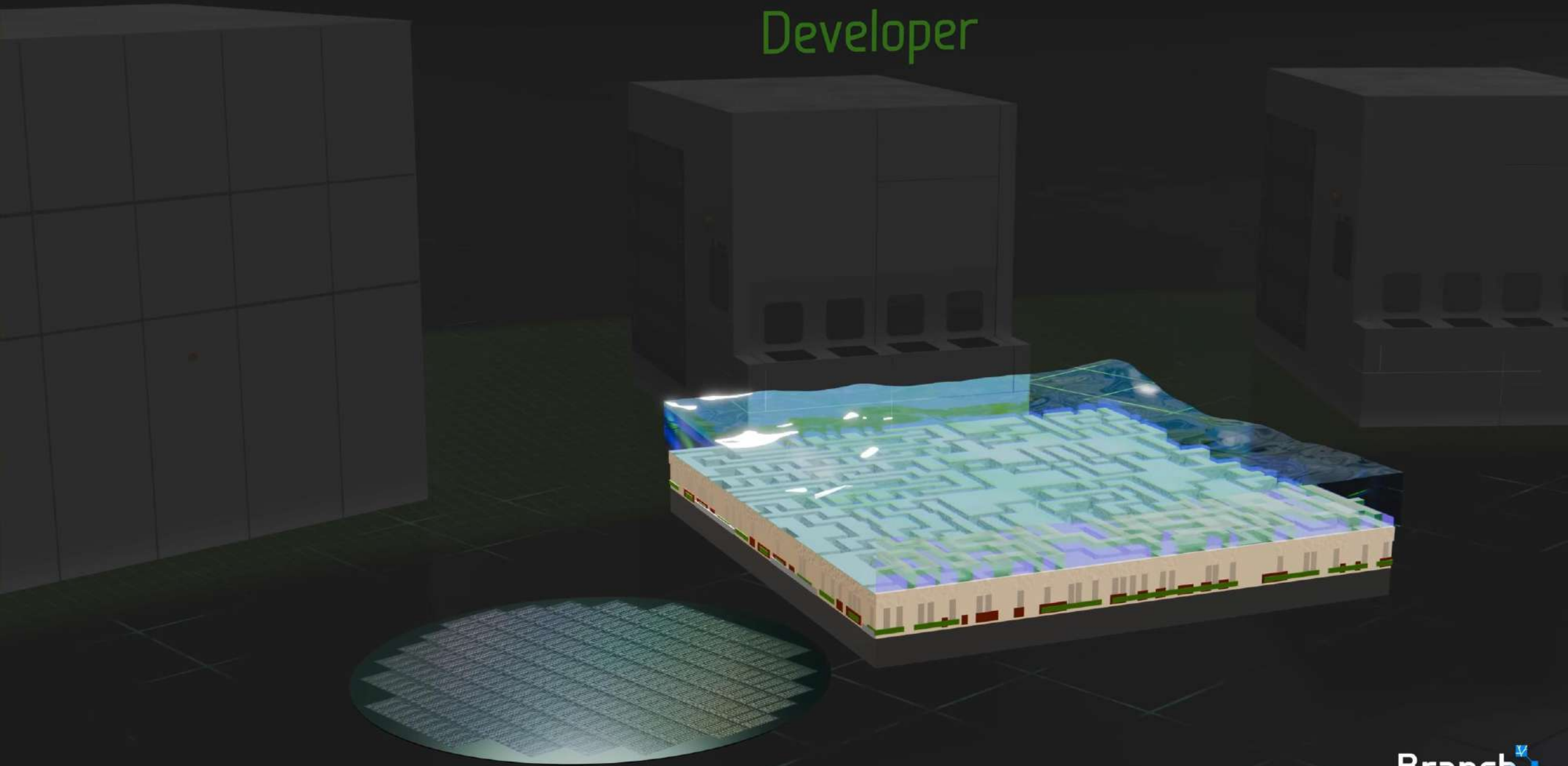




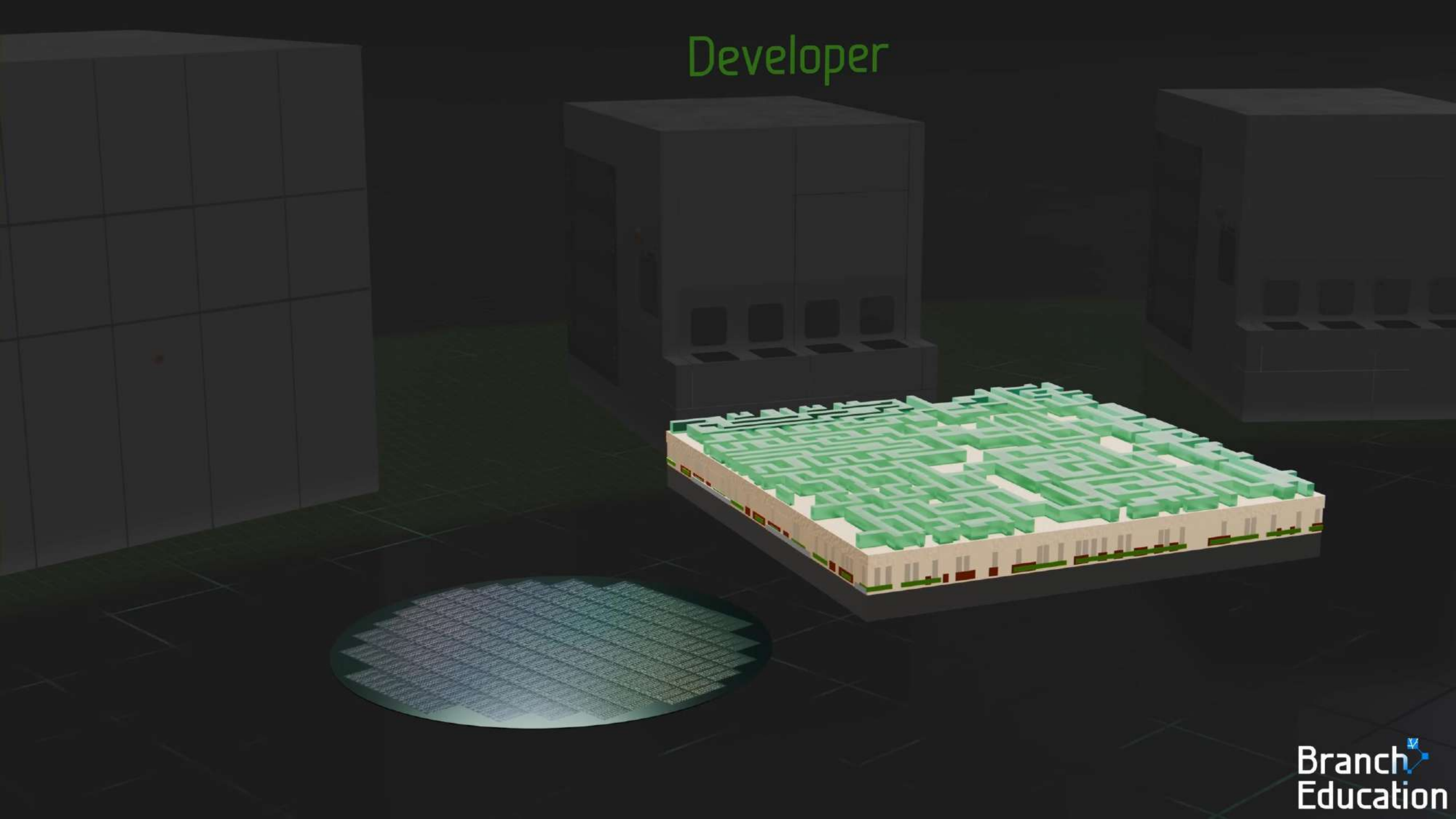
Developer

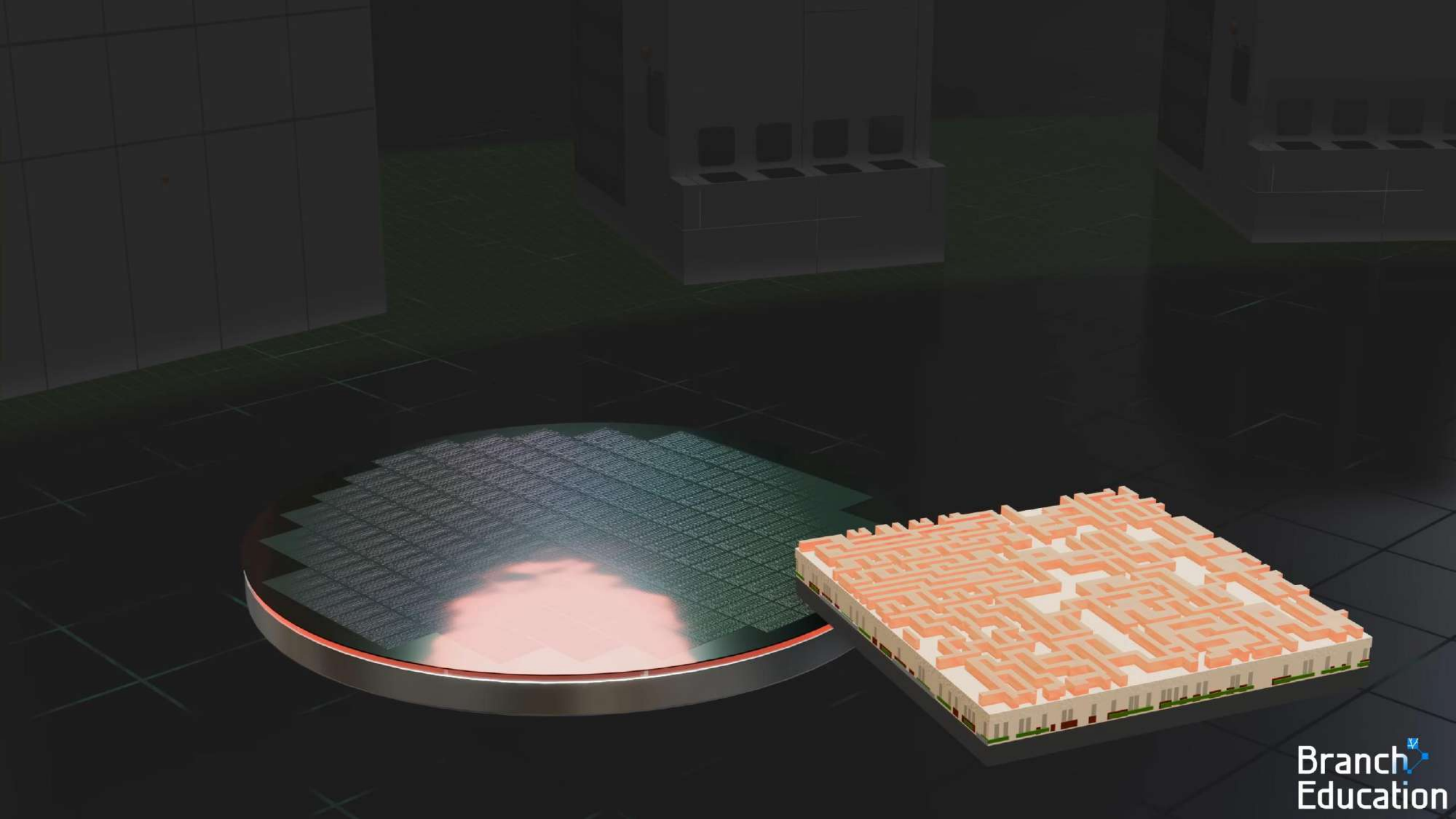


Developer

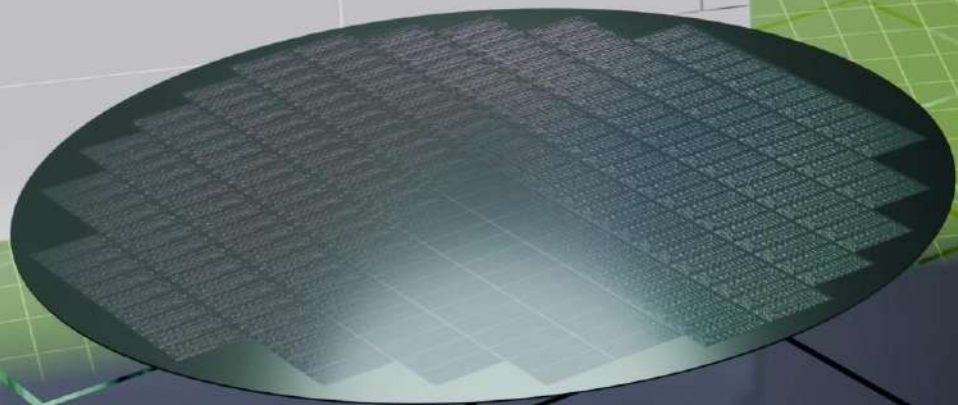


Developer





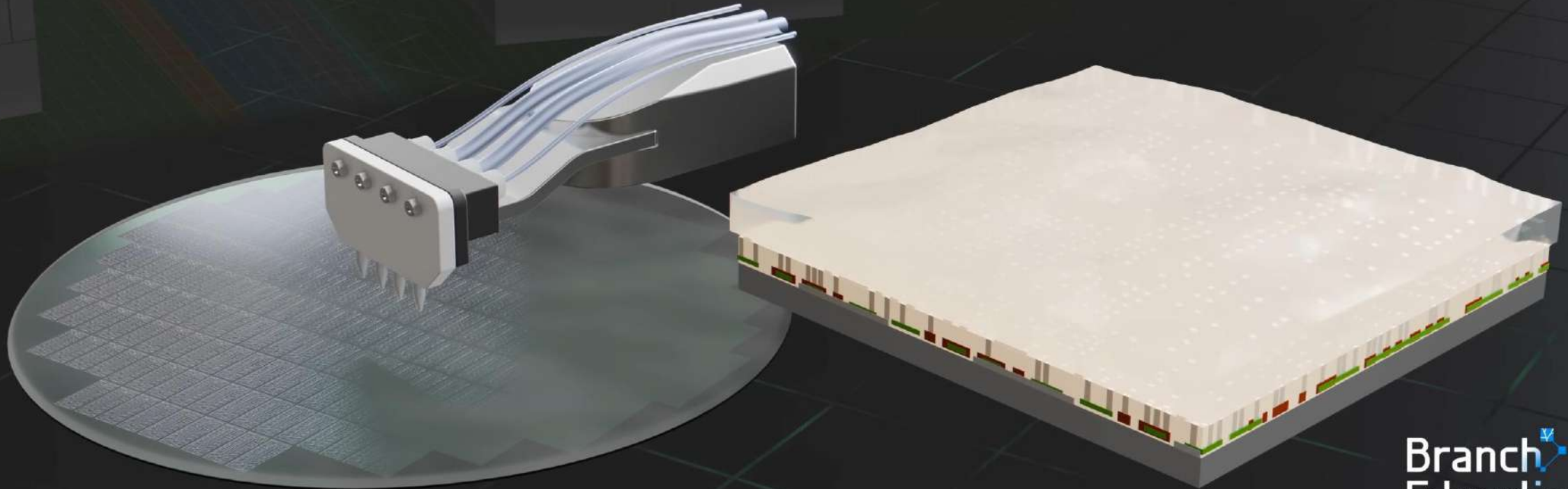
Wet/Dry Etching
Oxide, Metal Deposition | Silicon Growth
Ion Implantation Annealing
Wafer Inspection
Cleaning



Wet/Dry Etching
Oxide, Metal Deposition | Silicon Growth
Ion Implantation Annealing
Wafer Inspection
Cleaning

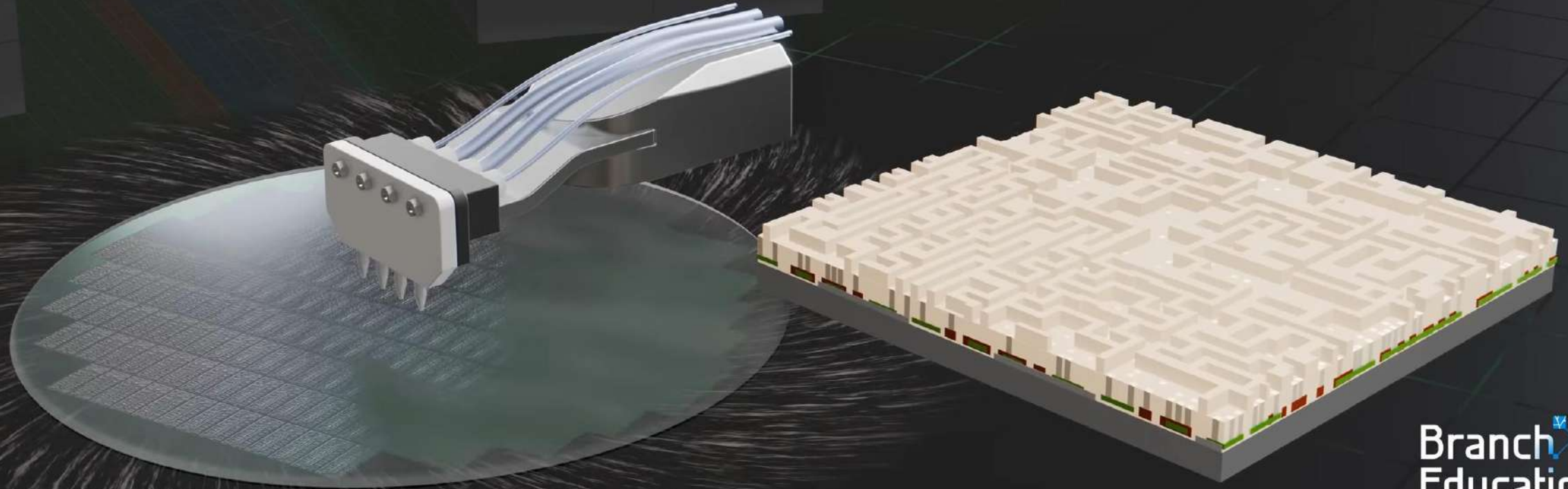
Plasma Dry Etching
Oxide, Metal Deposition | Silicon Growth
Ion Implantation Annealing
Wafer Inspection
Cleaning

Photoresist Stripper



Wet/Dry Etching
Oxide, Metal Deposition / Silicon Growth
Ion Implantation Annealing
Wafer Inspection
Cleaning

Photoresist Stripper

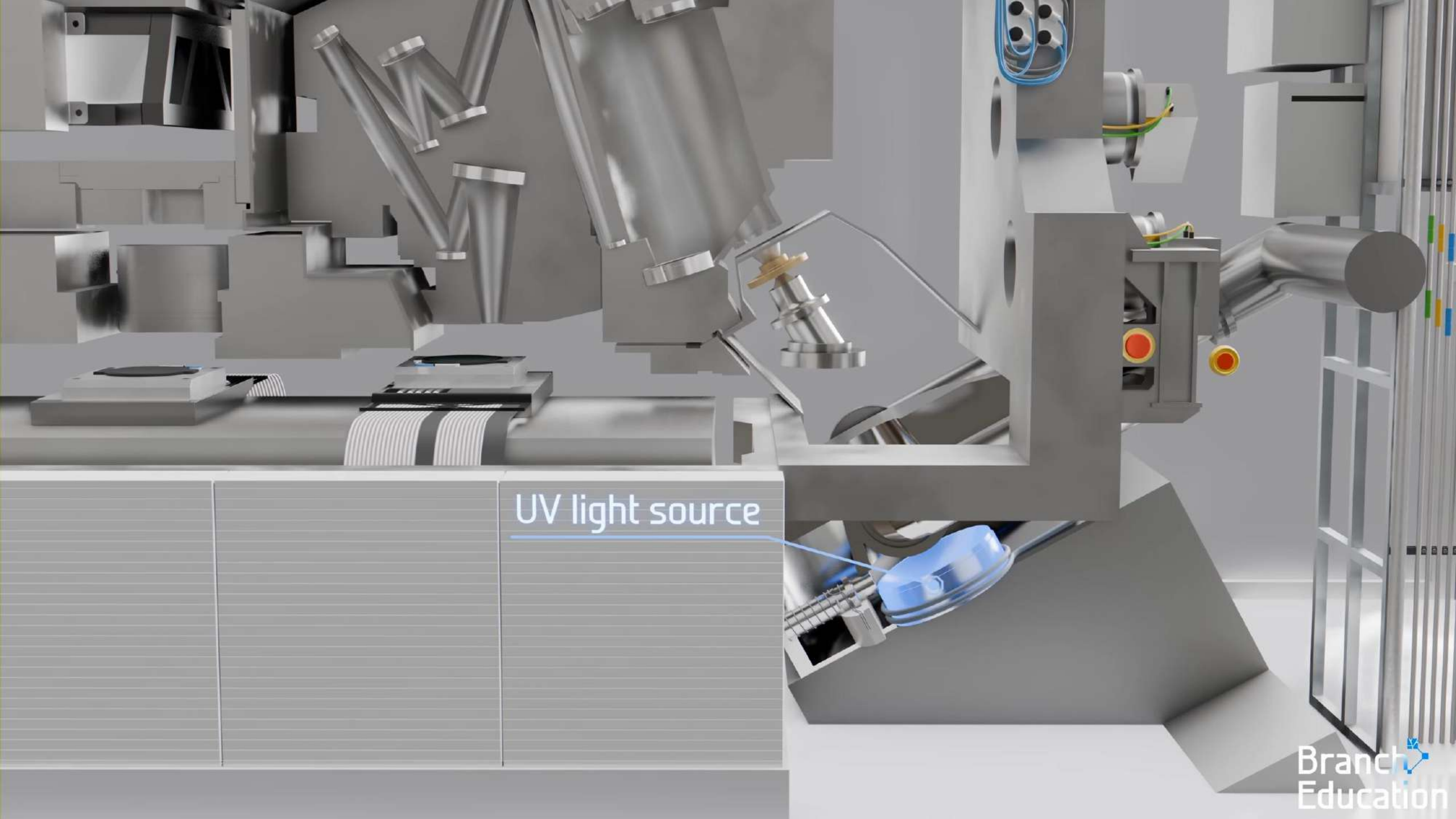


Photolithography Tool



Photolithography Tool

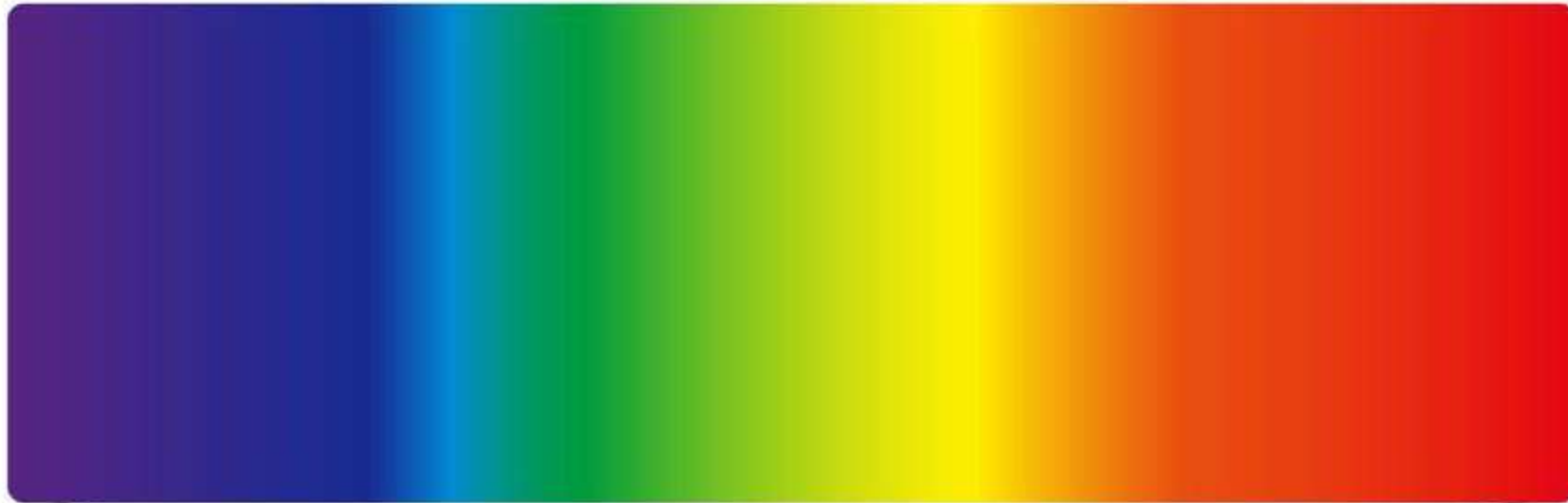


A detailed 3D cutaway diagram of a microscope, showing its internal components. The main body is grey, with various optical elements like lenses and mirrors visible. A blue light source is located at the base, with a blue line pointing to it from the label 'UV light source'. The base has a control panel with a red emergency stop button and a yellow button. The microscope is mounted on a stand with a safety cage on the right side.

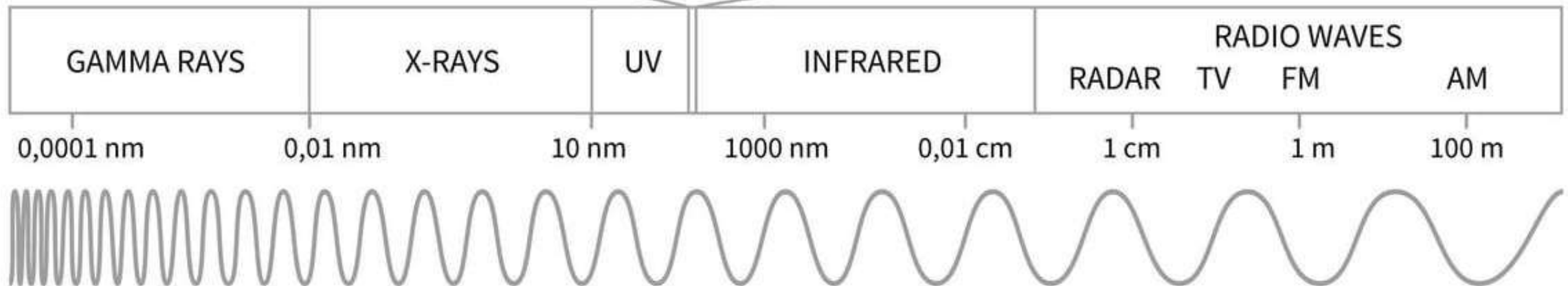
UV light source

빛을 어떻게 만들어 낼까?

VISIBLE SPECTRUM



VISIBLE LIGHT

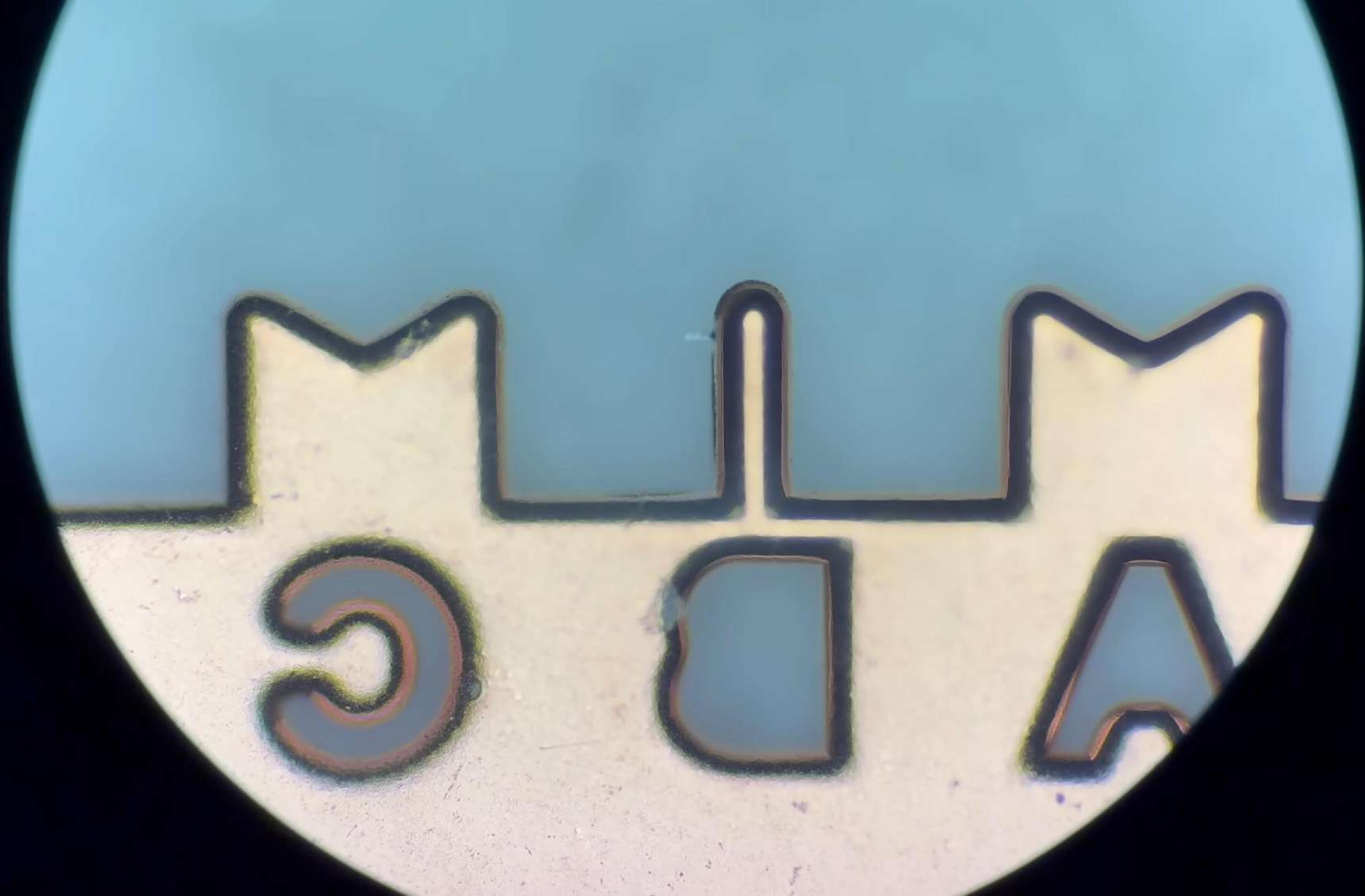






566/DFN25/D
N PLAN EPI
20x/0.40 BD

566
20x/0.40
12381







3 mm

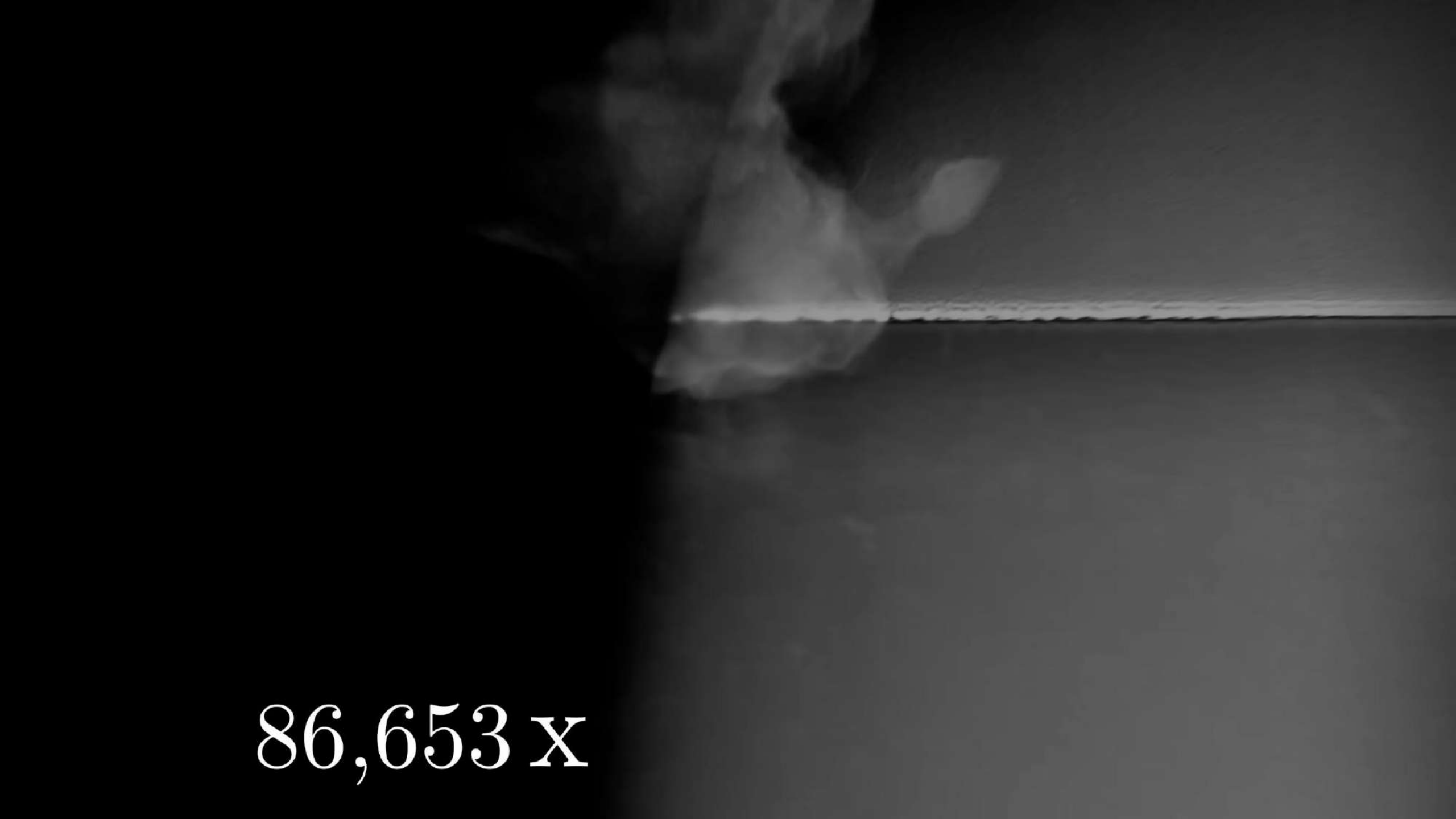
SDA

163 x

973 x



4,079 x



86,653 x

827,871 x

42,661,887 x

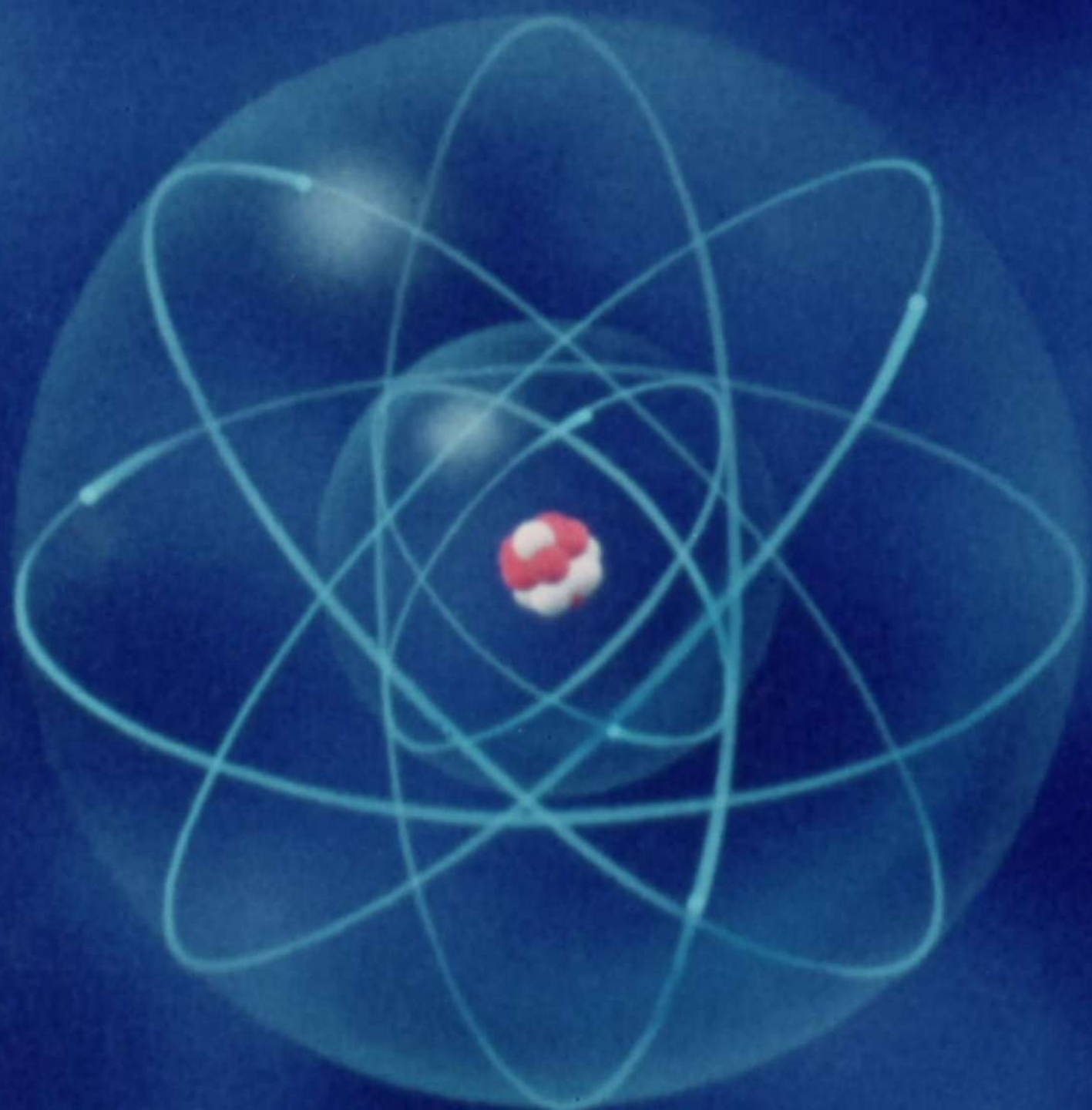
13,553,225 x

49,296,735 x

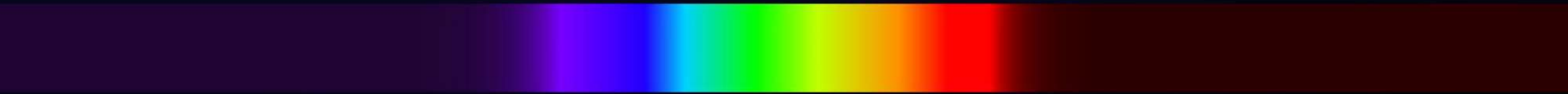
50,000,000 x

50,000,000 x



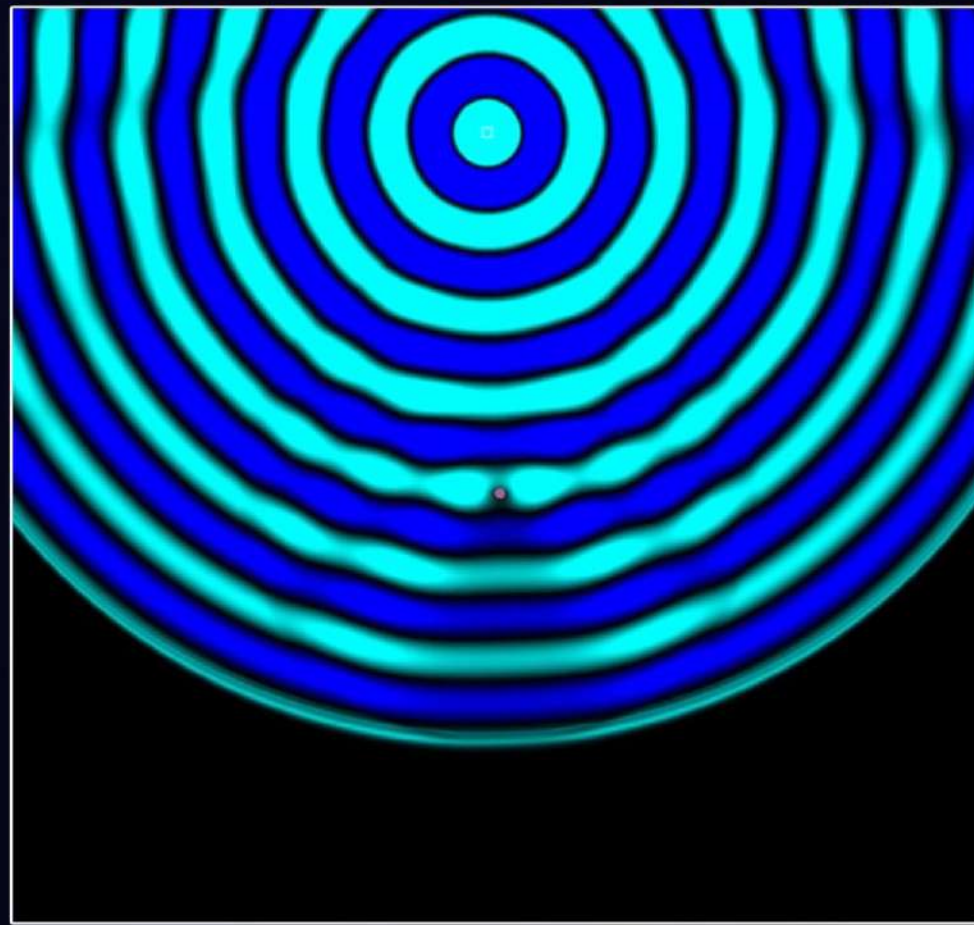
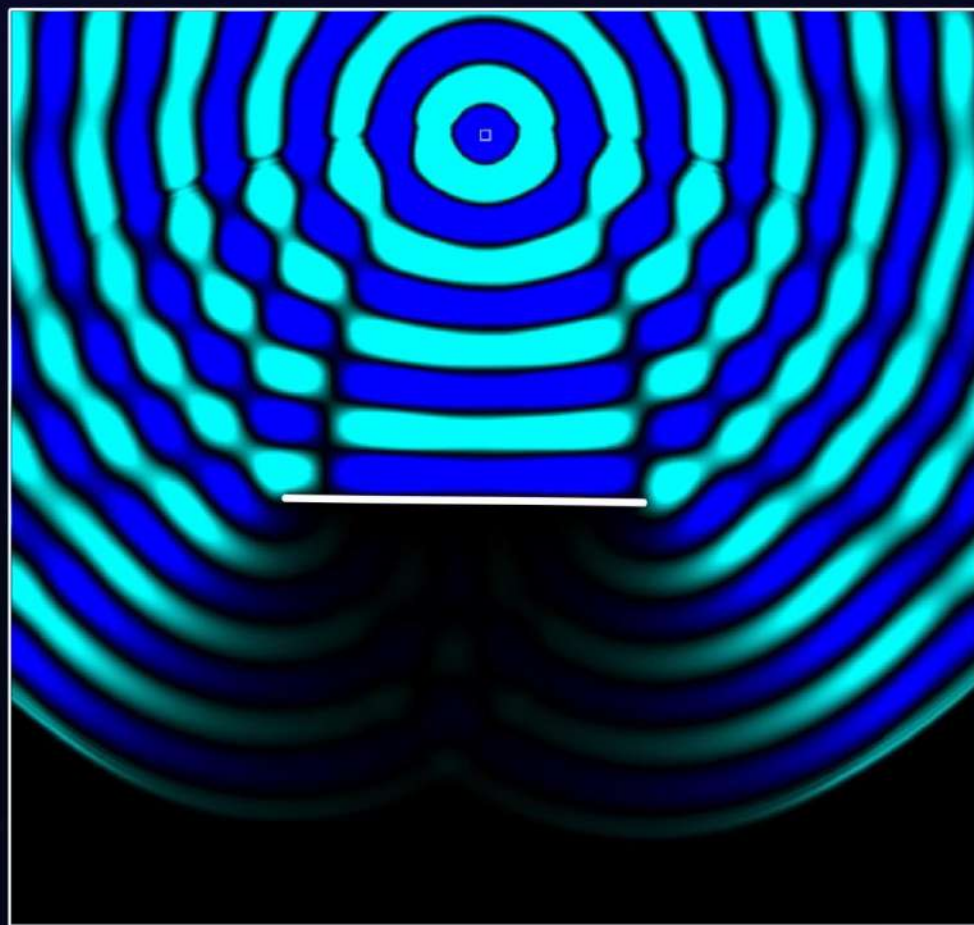


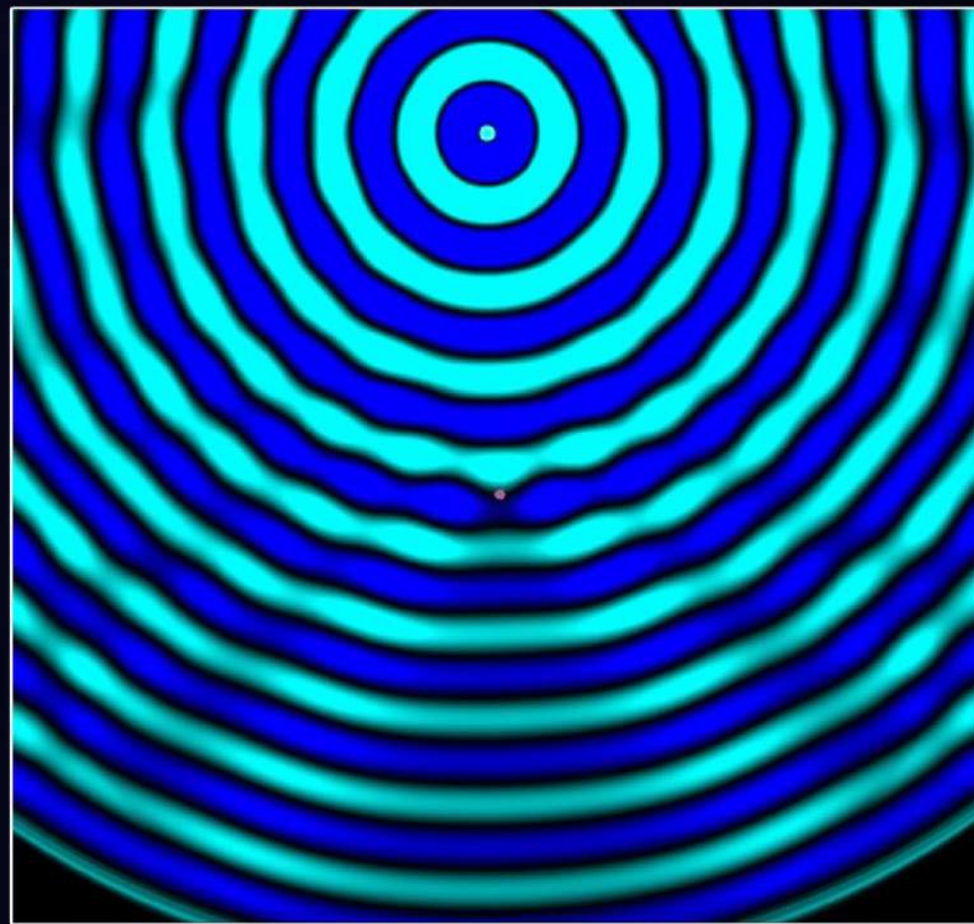
380 - 750 nm

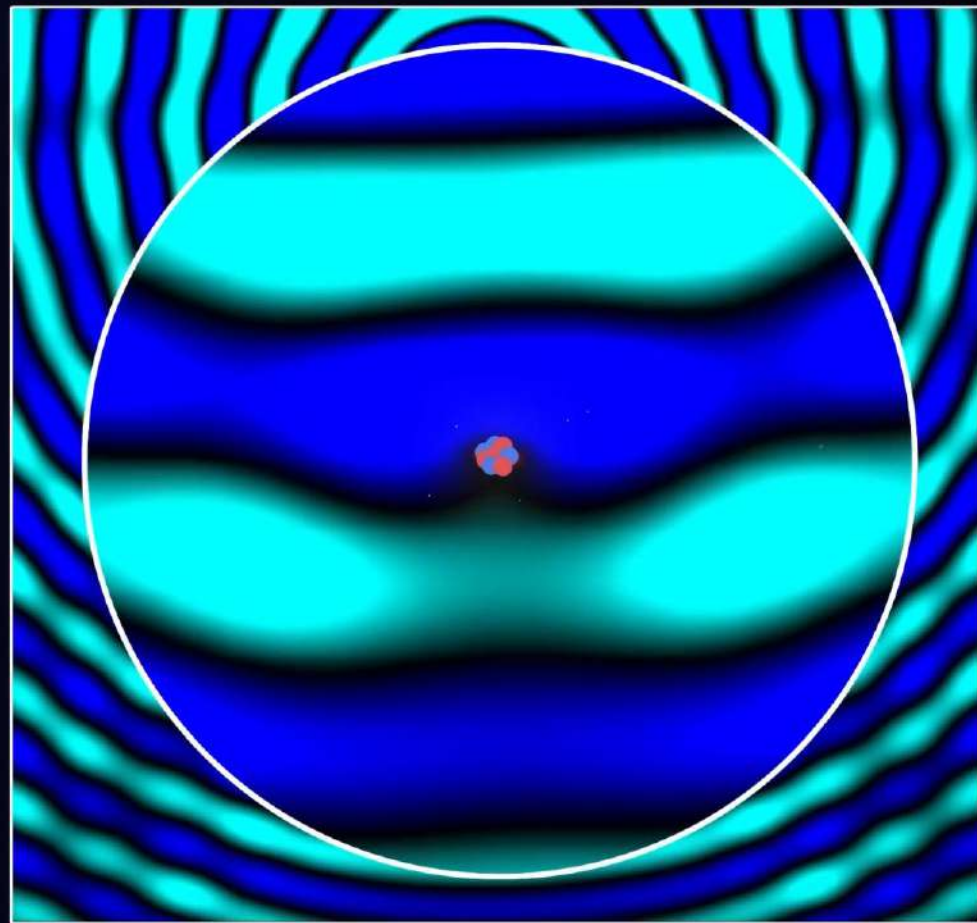
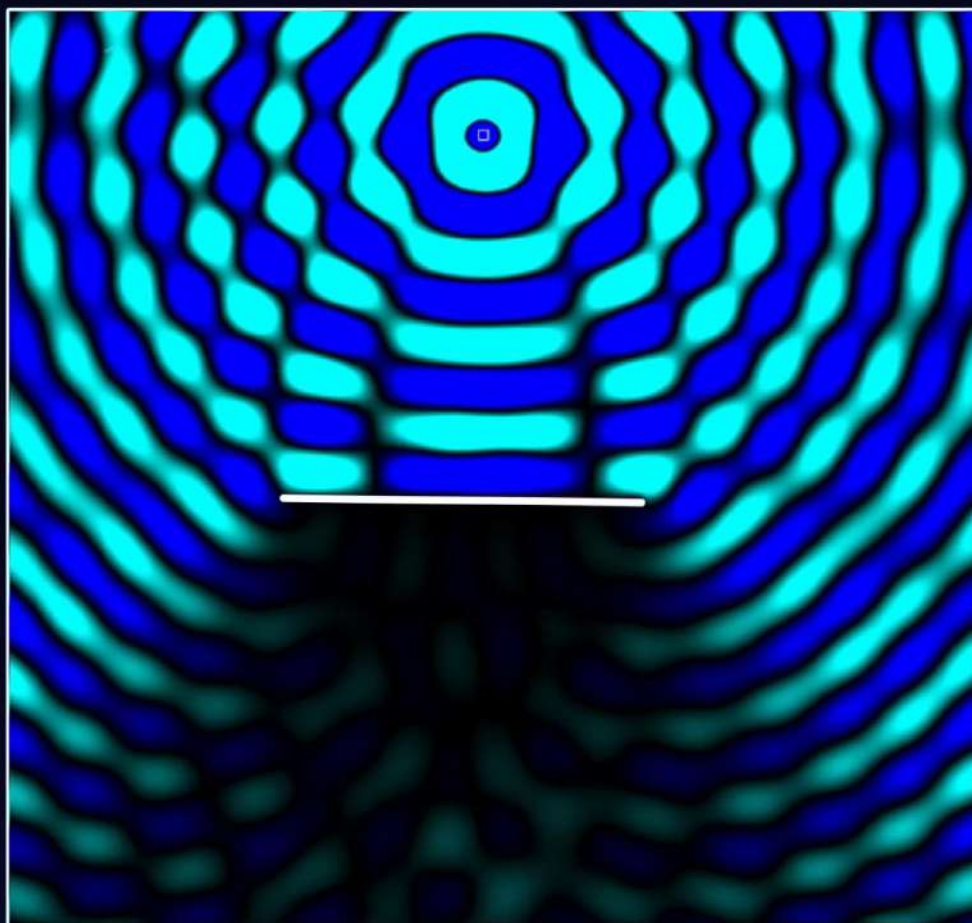


0.1 nm









EUV 광원의 원리 (Tin Droplet Plasma)

- 주석(Tin, Sn) 방울(Droplet) 생성

주석 방울을 매우 정교하게 (초당 수만 개 수준) 고속으로 분사
이 주석 방울은 지름이 수십 마이크로미터(μm) 정도로 매우 작음

- 두 단계 레이저 타격 (Pre-pulse & Main-pulse)

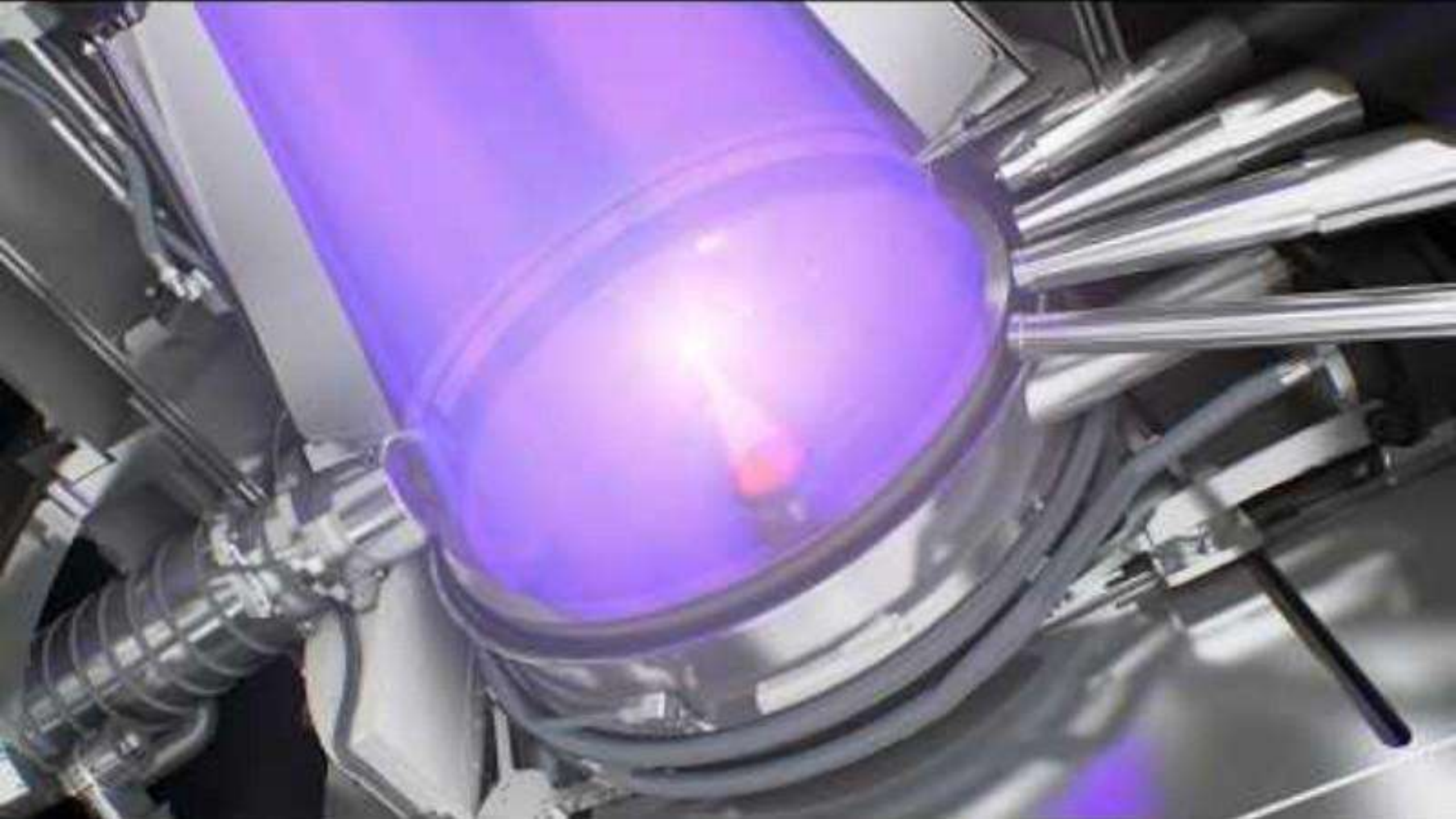
- 예비 타격(Pre-pulse)

비교적 약한 강도의 레이저를 방울에 미리 쏘아서 주석 방울을 납작한 형태로 펴줌
이 과정은 플라즈마 발생 효율을 높이기 위한 준비 과정

방울을 납작하게 만들면, 표면적이 넓어져 다음 단계에서
고출력 레이저가 주석을 기화시킬 때 더욱 효율적이고 균일한 플라즈마 생성이 가능
표면적이 늘어나면 레이저 에너지가 더 고르게 분포되어 효율과 밝기(출력)가 크게 향상

- 본 타격(Main-pulse)

본격적으로 초고강도 CO_2 레이저를 쏘서
방울을 순간적으로 고온 플라즈마 상태(약 $20\text{만}^\circ\text{C}$ 이상)로 기화
이 과정에서 극자외선(13.5nm 파장) EUV 광선이 방출



더 짧은 파장을 방출하는 광석으로 빛을 만들면
더 작게 그릴 수 있는 거 아닌가?

더 짧은 파장의 빛 사용의 어려움

- 흡수 문제

EUV보다 더 짧은 파장 영역(예: X-ray 영역)에서는 물질이 빛을 쉽게 흡수해버림
빛이 쉽게 흡수되면 투과와 반사가 어려워지고, 렌즈나 거울로 빛을 정밀하게 제어하기 어려움

- 광학 기술의 한계

EUV는 이미 렌즈가 아닌 특수한 다층 거울(반사경)을 사용해 빛을 모음.
하지만 X-ray 영역은 거울의 효율이 떨어지고, 정밀 제어가 어려움
더 짧은 파장으로 갈수록 정밀한 광학 설계가 매우 어려워짐

- 에너지·장비의 복잡성

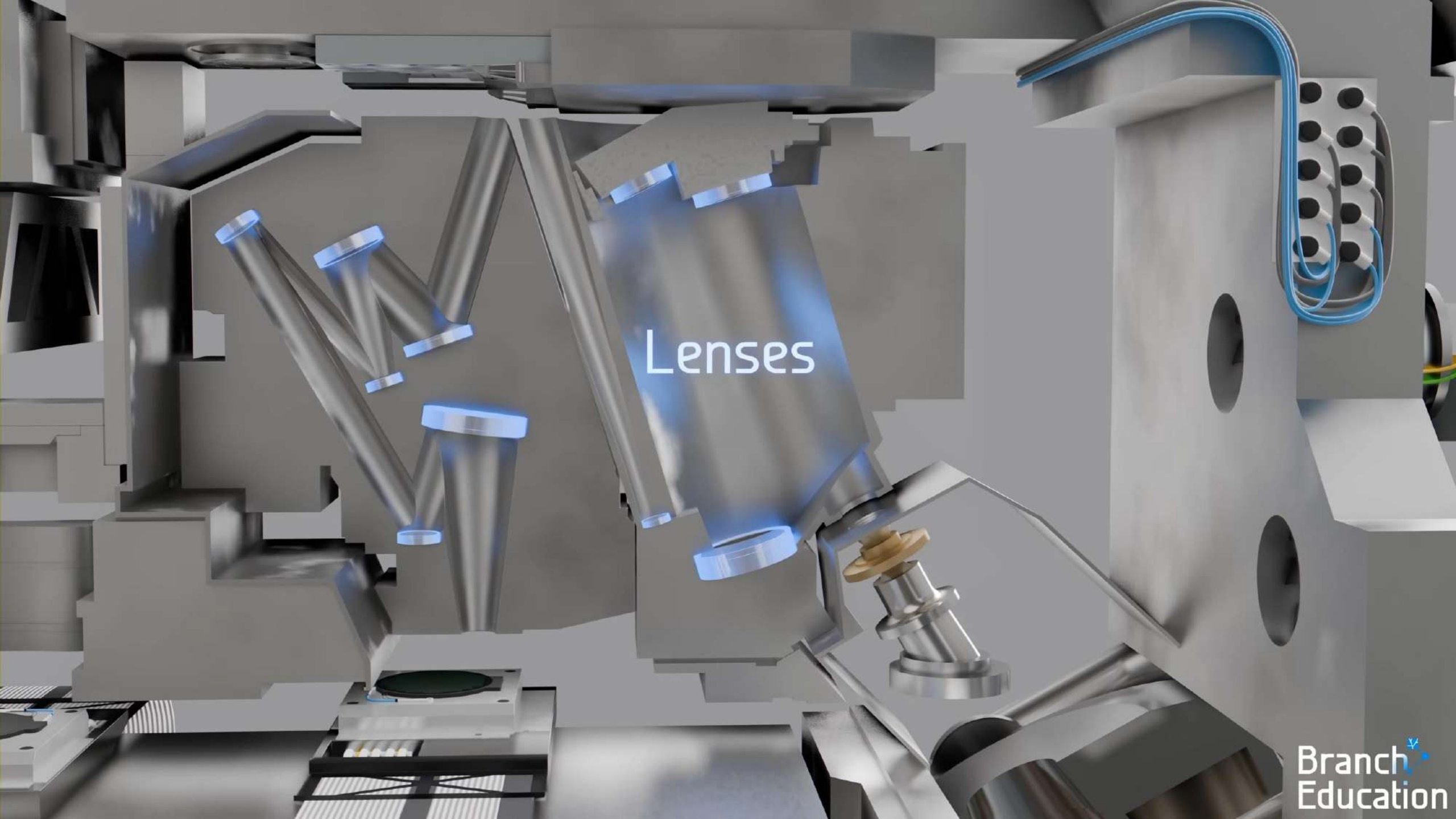
더 짧은 파장을 내기 위해선 더 높은 에너지 상태의 플라즈마를 만들어야 하는데,
이것은 장비와 제어 기술에 엄청난 부담을 줌
열, 방사선, 장비의 수명 문제가 매우 심각해짐

이론적으로는 더 짧은 파장이 더 미세한 패턴을 만들 수 있지만,
기술적, 물리적, 경제적 이유로 인해
현재는 13.5nm EUV가 현실적인 최적점으로 자리 잡고 있는 것

이보다 더 짧은 파장 영역의 활용은 아직 연구 단계이며,
장기적으로는 X-ray 리소그래피와 같은 더 진보된 기술이 나타날 가능성도 있음

포토리소그래피(Photolithography)

빛을 이용해 마스크(mask)에 그려진 회로 패턴을 실리콘 웨이퍼 위에 정밀하게 전사하는 공정
E U V Exposure

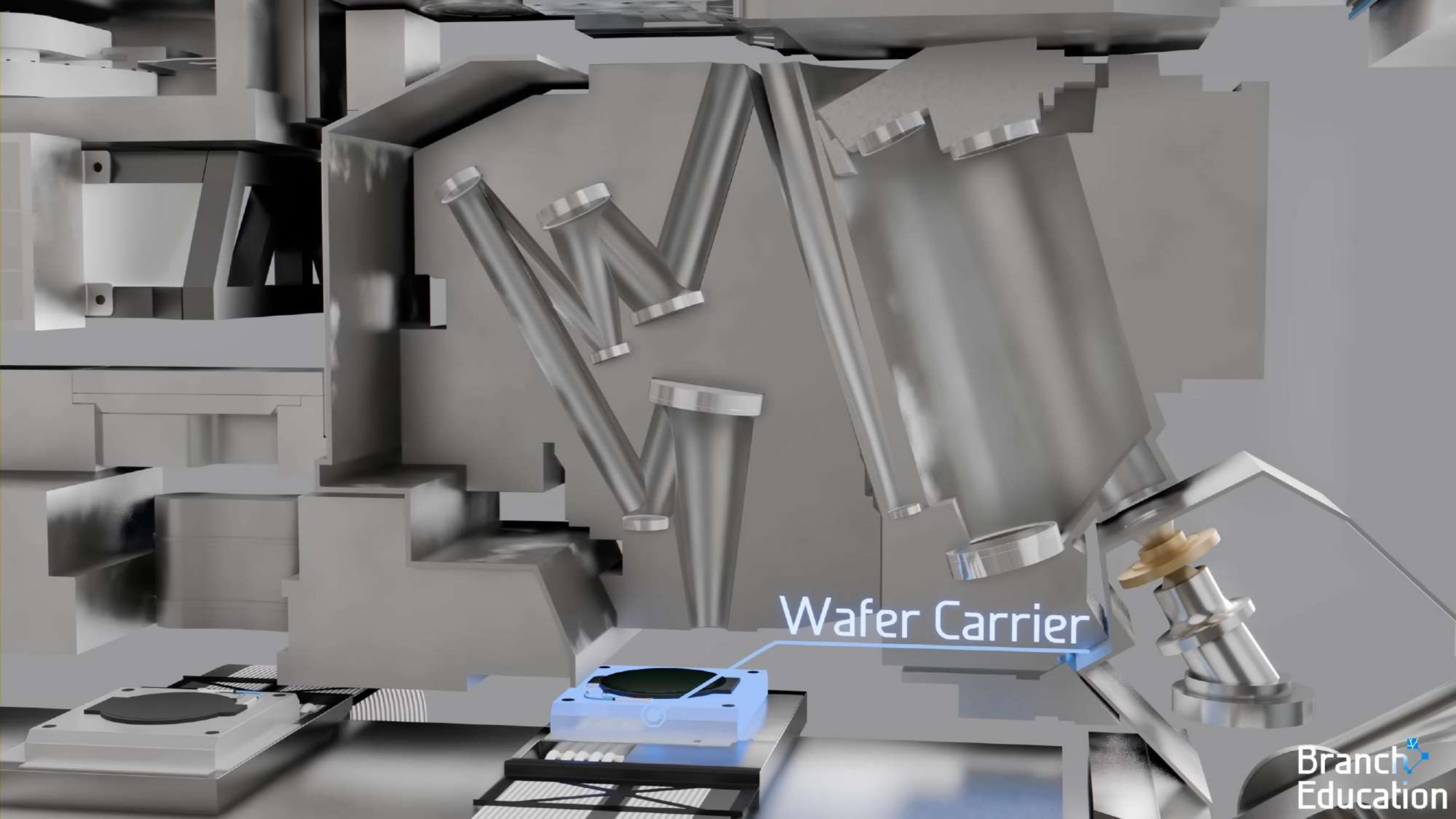


A detailed 3D cutaway illustration of a microscope. The central component, the eyepiece, is highlighted in a vibrant blue and labeled "Lenses". To its left, the objective lenses are also highlighted in blue. The surrounding mechanical parts, including the base, stage, and housing, are rendered in a realistic grey metal finish. A blue cable is visible on the right side, and a yellow adjustment knob is located near the base of the eyepiece.

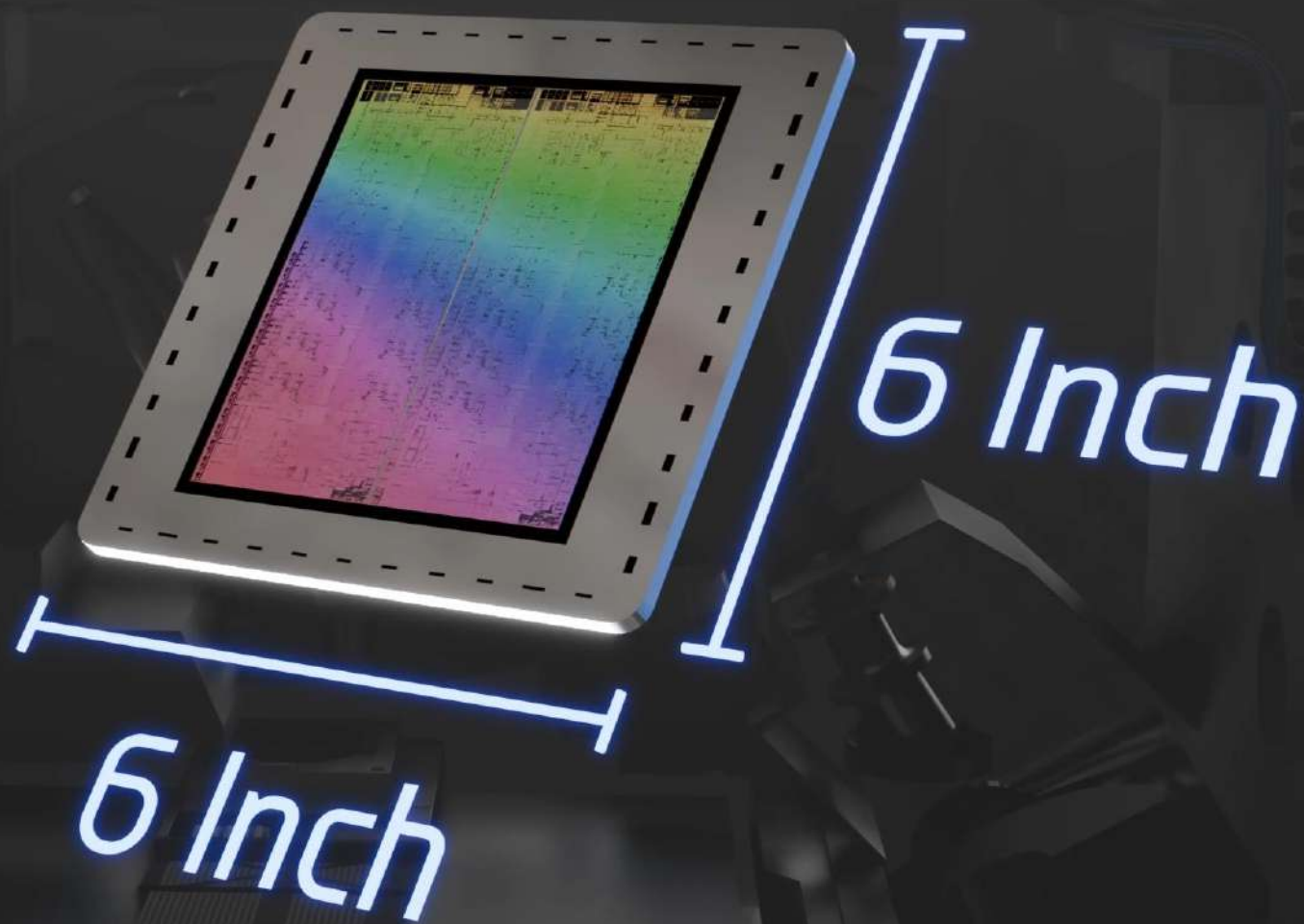
Lenses

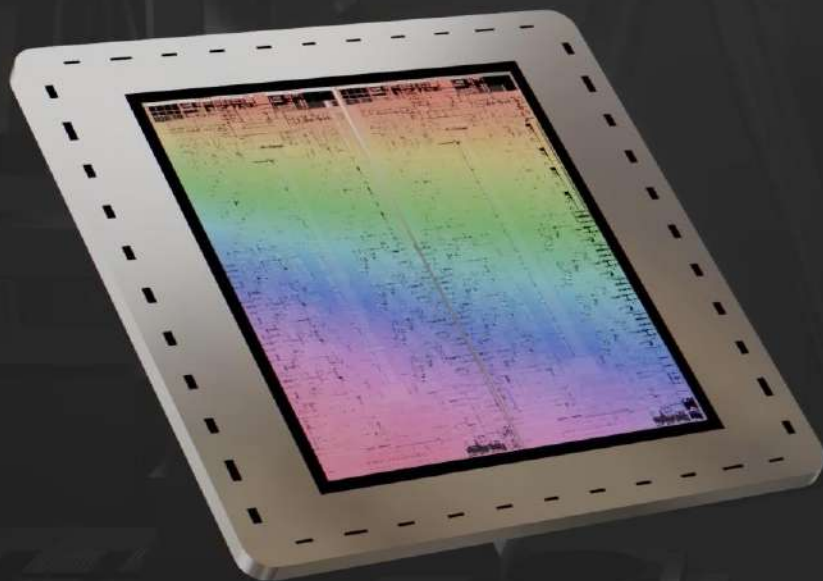
A detailed 3D rendering of a photolithography system. At the top, a complex arrangement of cables (blue, yellow, grey) is connected to a control unit. Below this, a horizontal stage holds a square photomask, which is highlighted by a blue circle and a line pointing to the label 'Photomask'. The mask is positioned above a series of vertical, tapered optical components that funnel light downwards. The entire setup is housed within a grey, industrial-looking frame.

Photomask

A detailed 3D cutaway rendering of a semiconductor furnace. The image shows the internal structure, including a large central cylindrical chamber, several smaller angled tubes, and a complex support system. At the bottom, a wafer carrier is shown holding multiple silicon wafers. A blue line points from the text 'Wafer Carrier' to the carrier assembly.

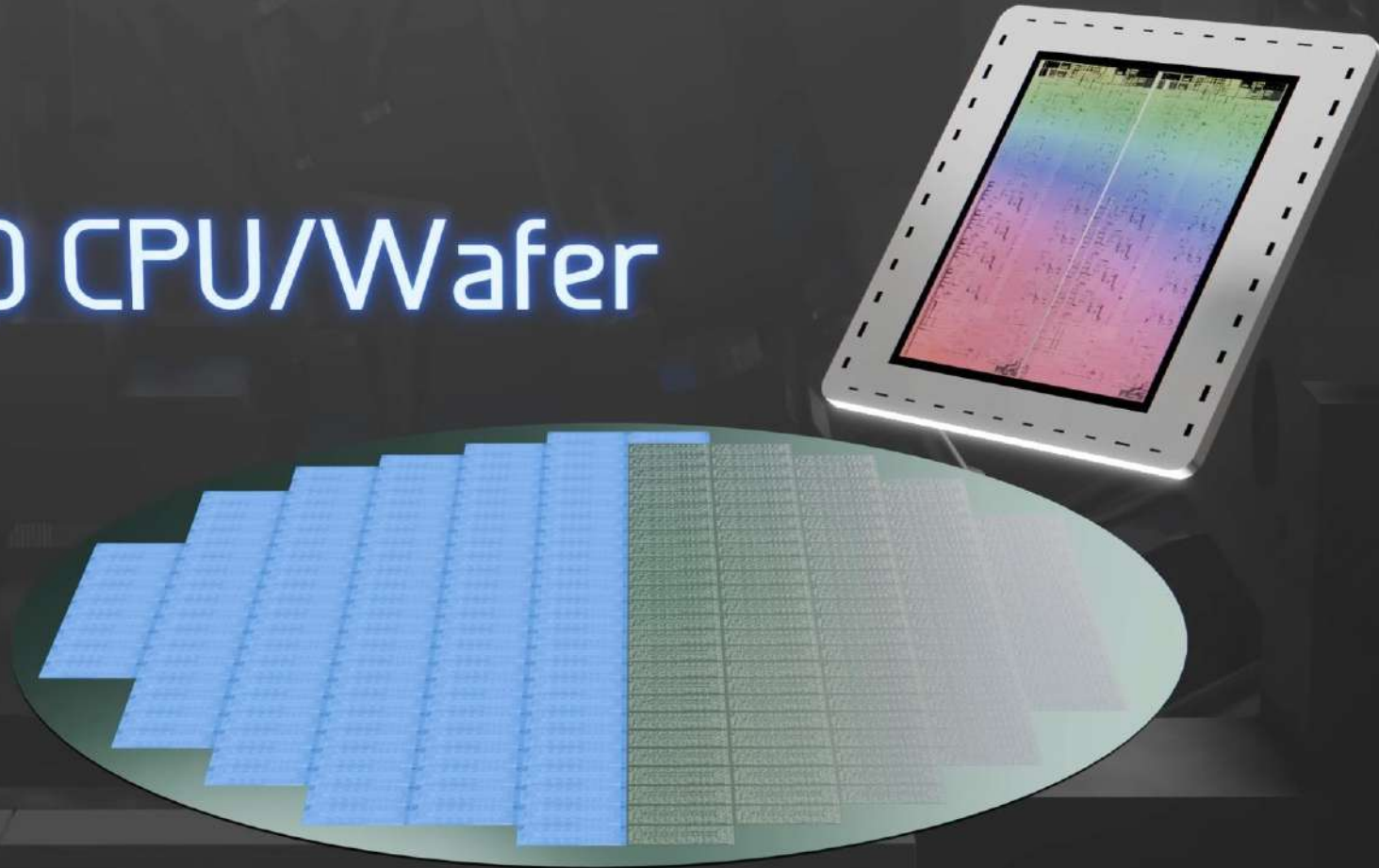
Wafer Carrier



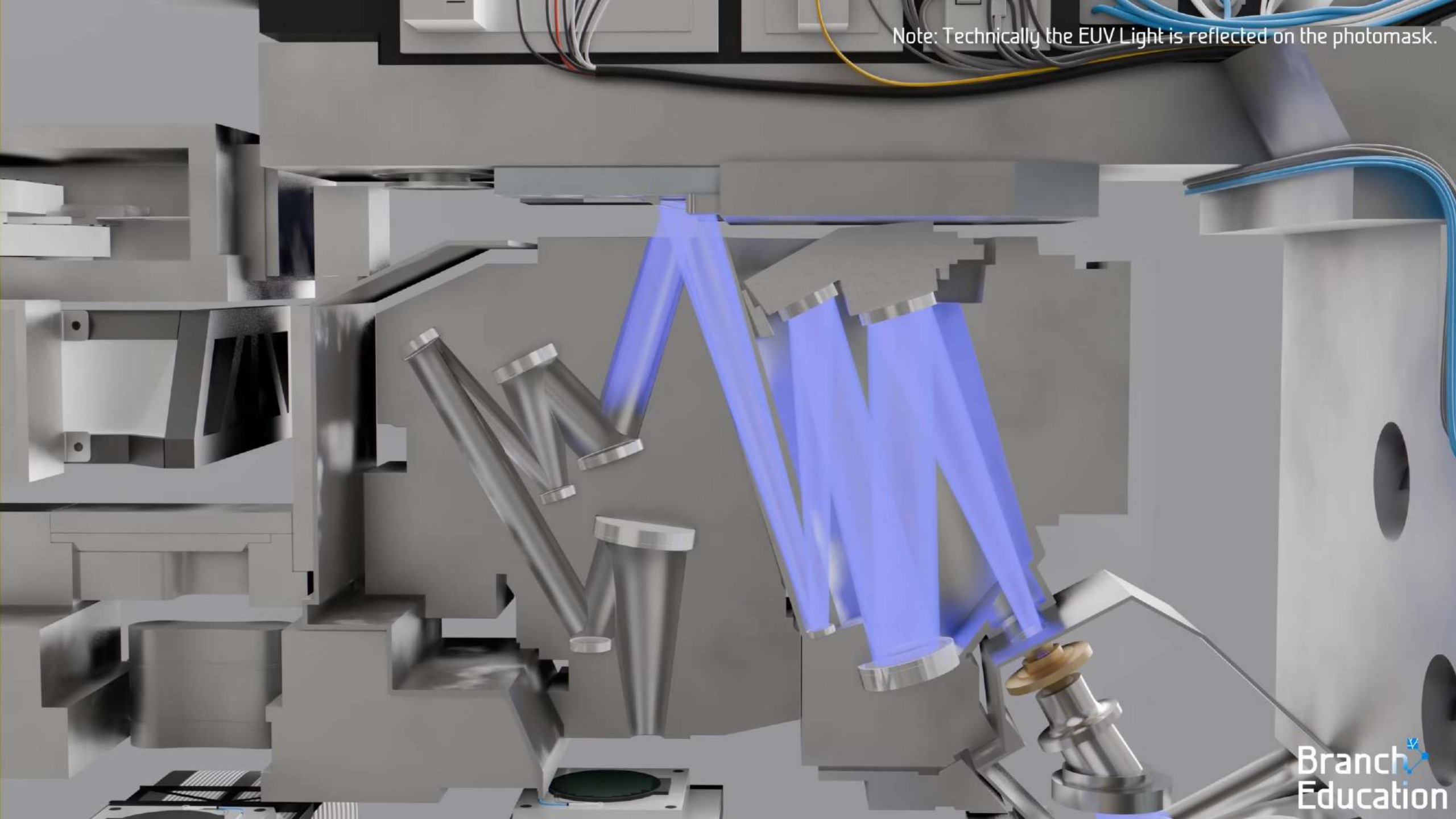


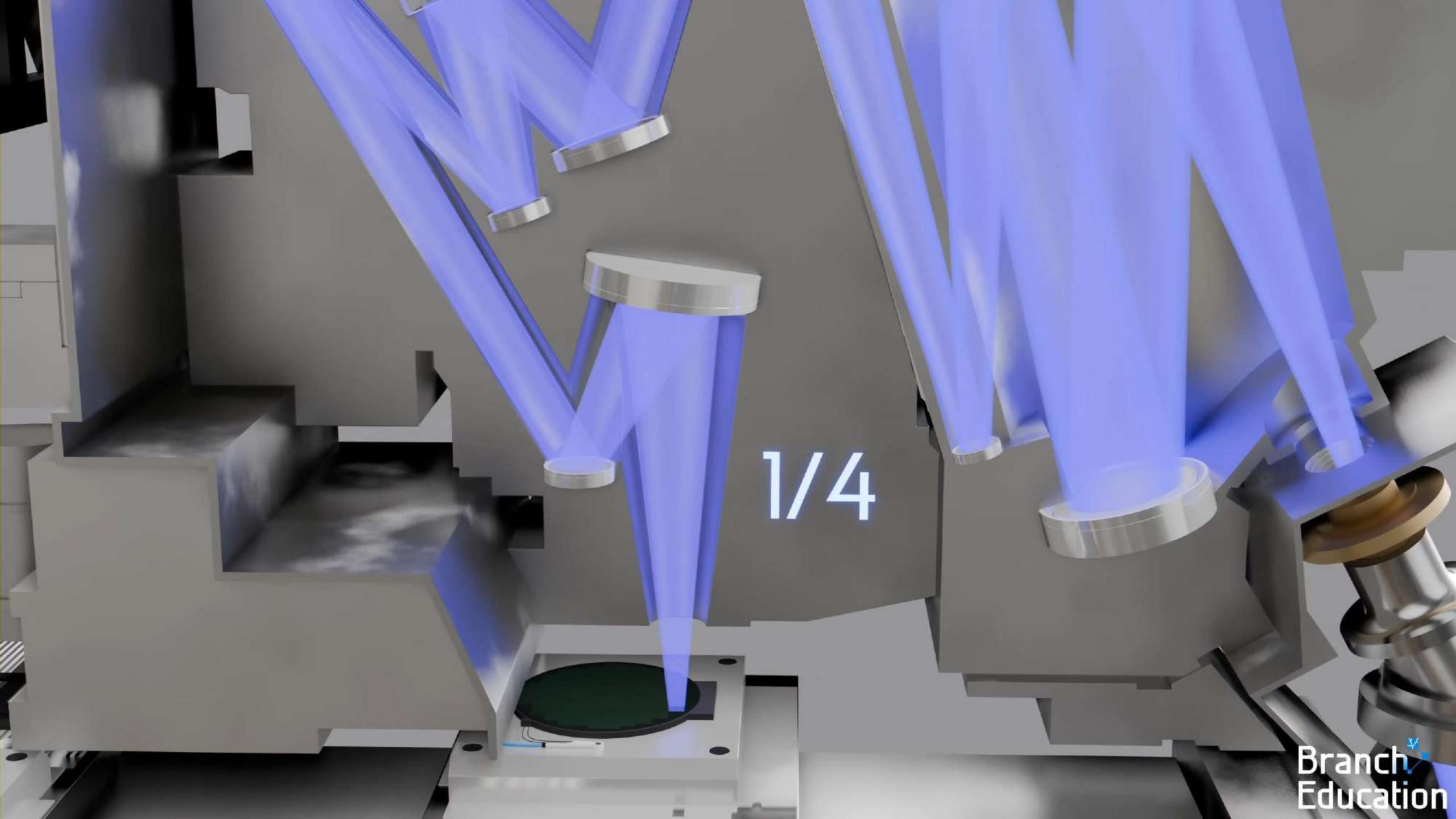
CPU Dimensions
Width = 10.5mm
Height = 23.9mm

230 CPU/Wafer



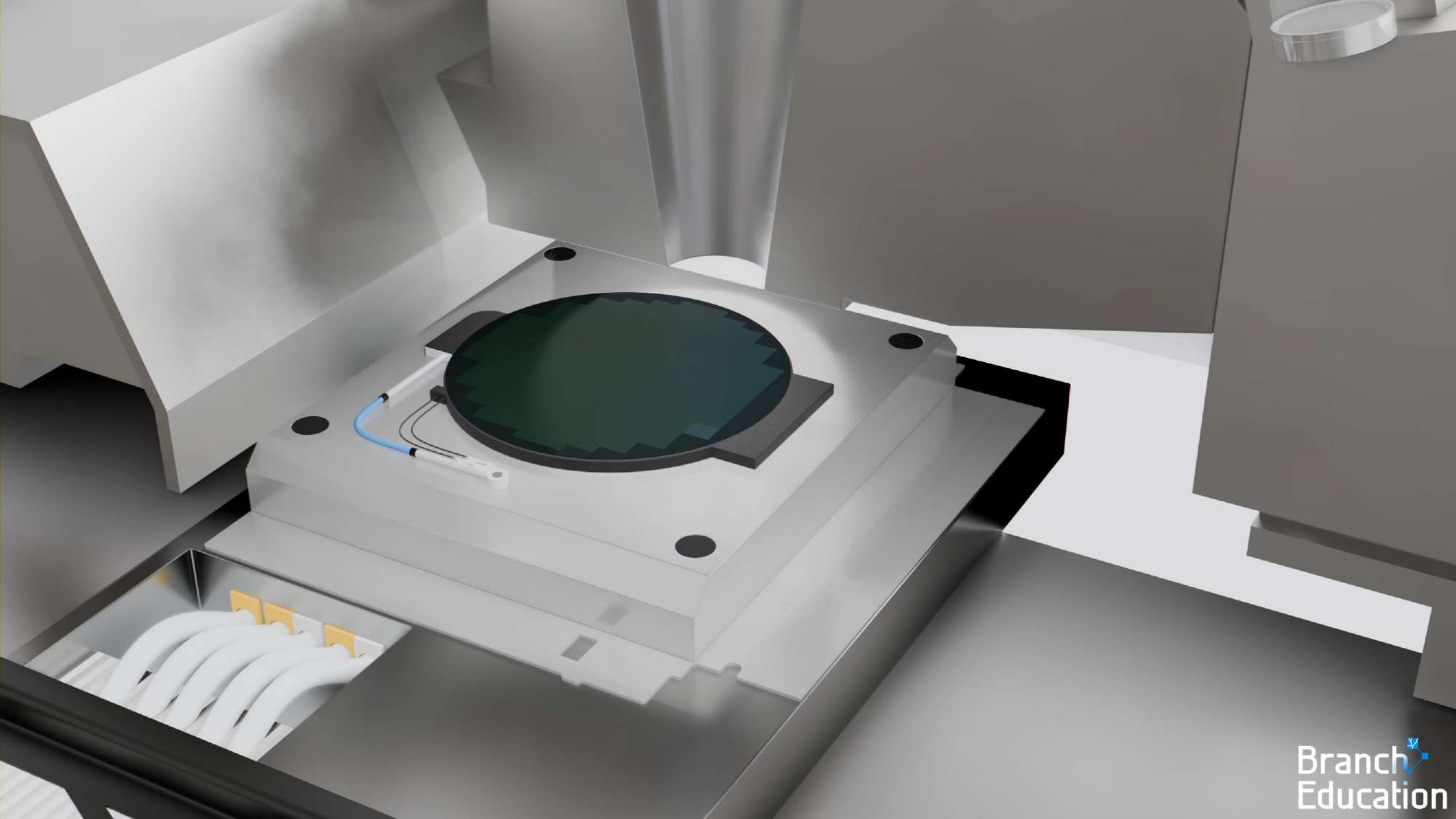
Note: Technically the EUV Light is reflected on the photomask.

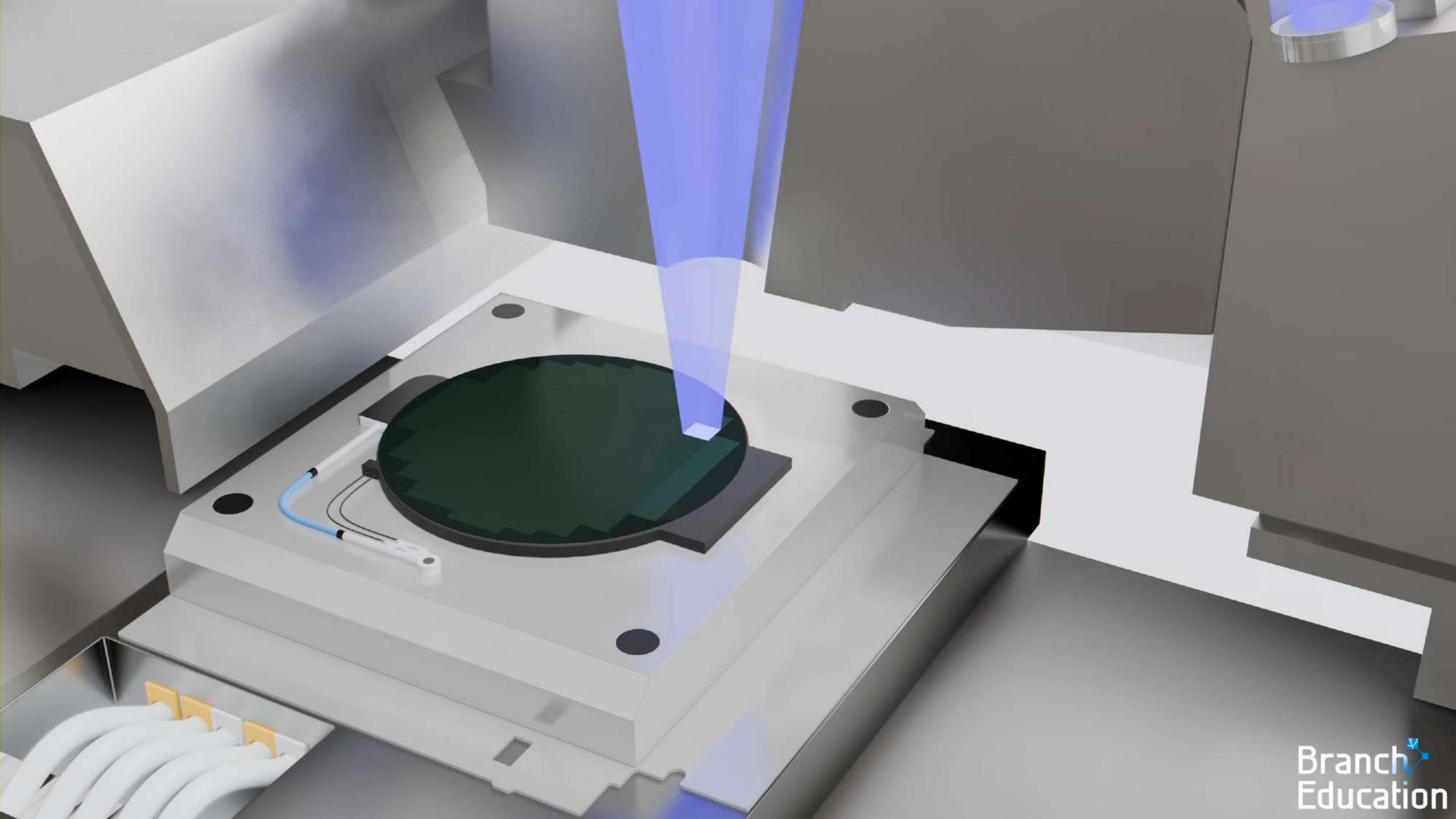




A 3D schematic diagram of a multi-beam laser system. The diagram shows a complex arrangement of mirrors and lenses that direct multiple blue laser beams. One beam is focused onto a green circular target on a rotating stage. The entire setup is housed within a grey mechanical structure. The fraction 1/4 is displayed in the center.

$1/4$

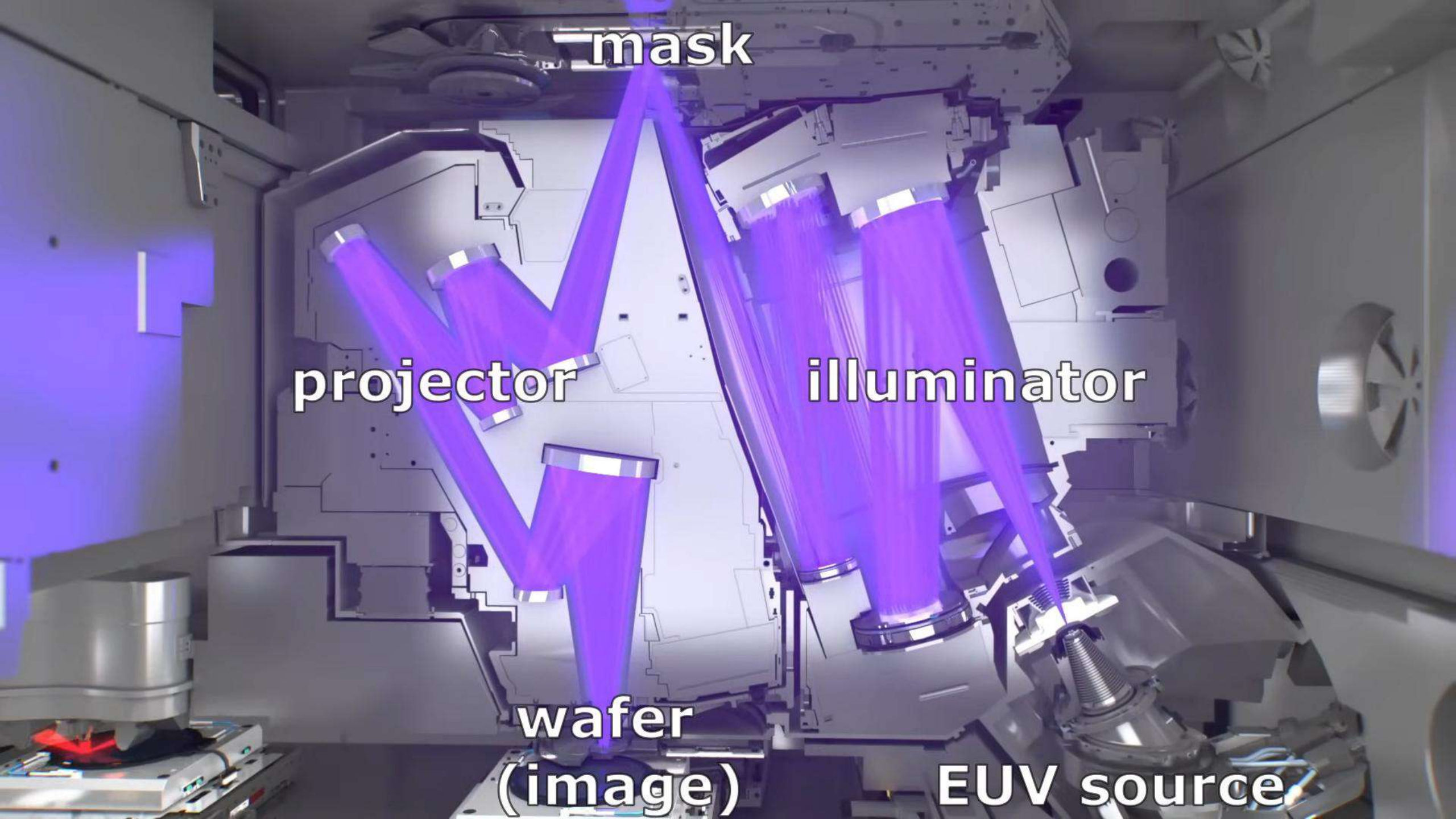




The background of the image is a close-up, low-angle shot of an EUV lithography system. Several bright, parallel beams of light are visible, emanating from the top and shining down onto a complex, dark structure. The scene is dimly lit, with the primary light source being the beams themselves, creating a high-contrast, industrial atmosphere. The ASML logo is positioned in the upper right corner.

ASML

A fly-through of
High NA
EUV



mask

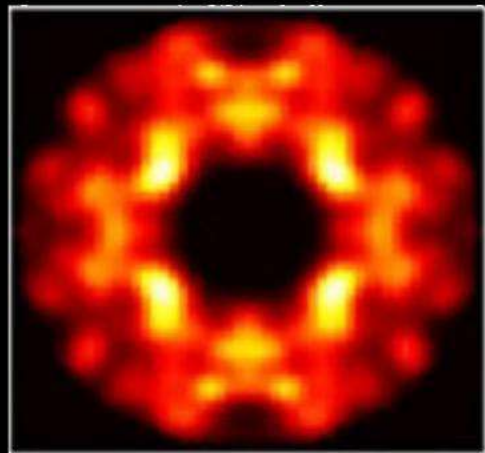
projector

illuminator

wafer
(image)

EUV source

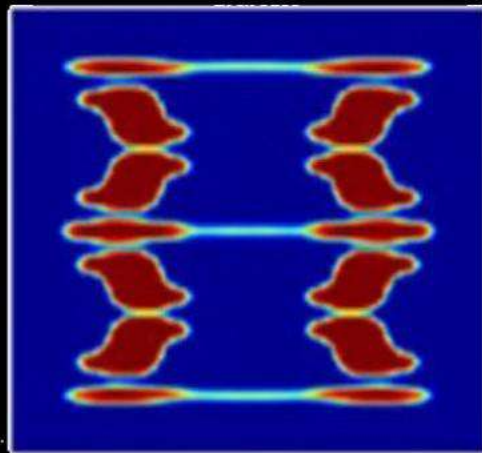
Illumination
field



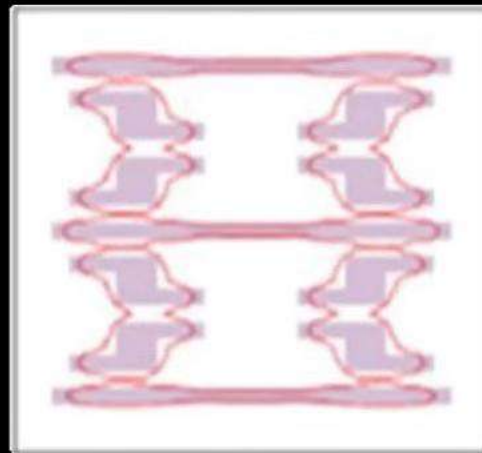
Mask



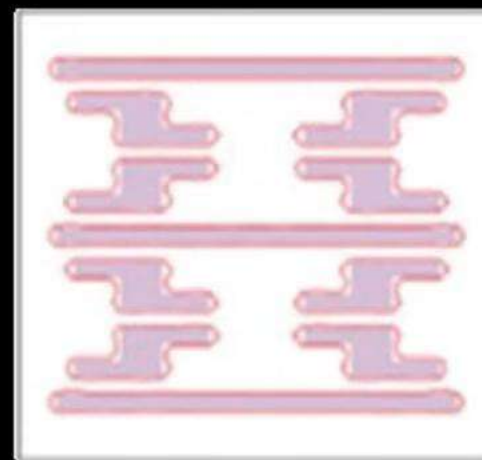
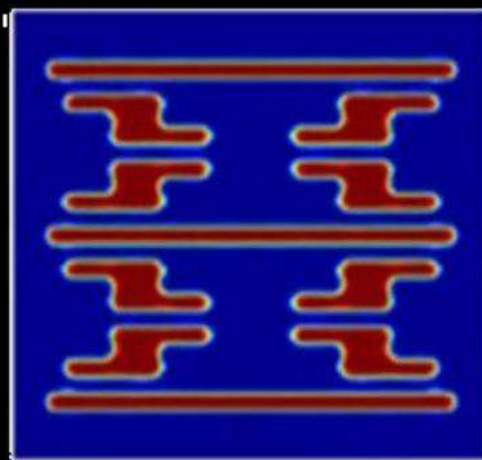
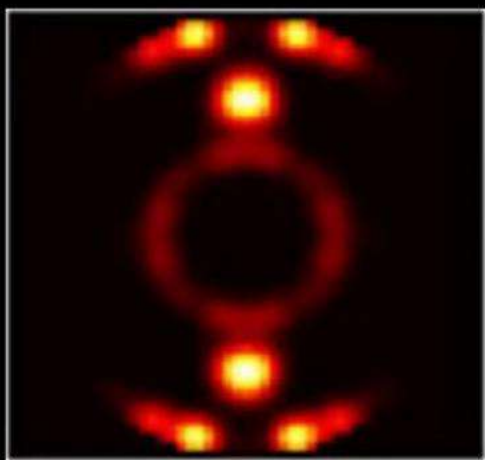
Image

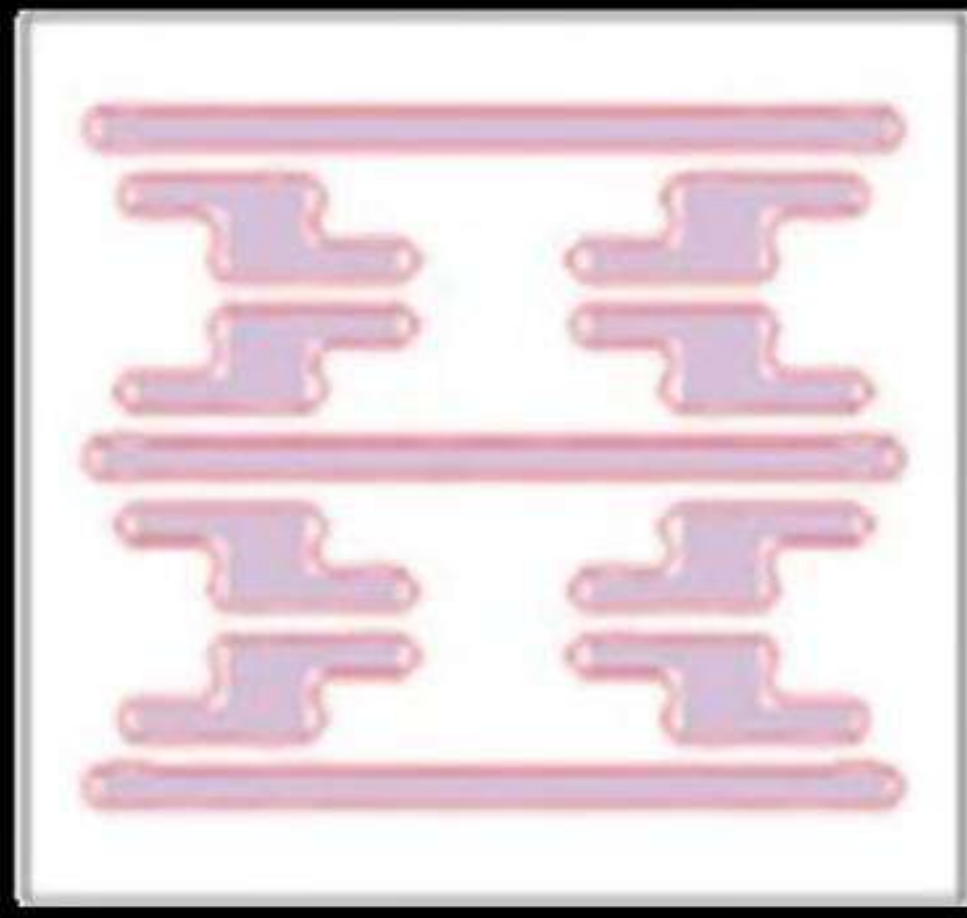
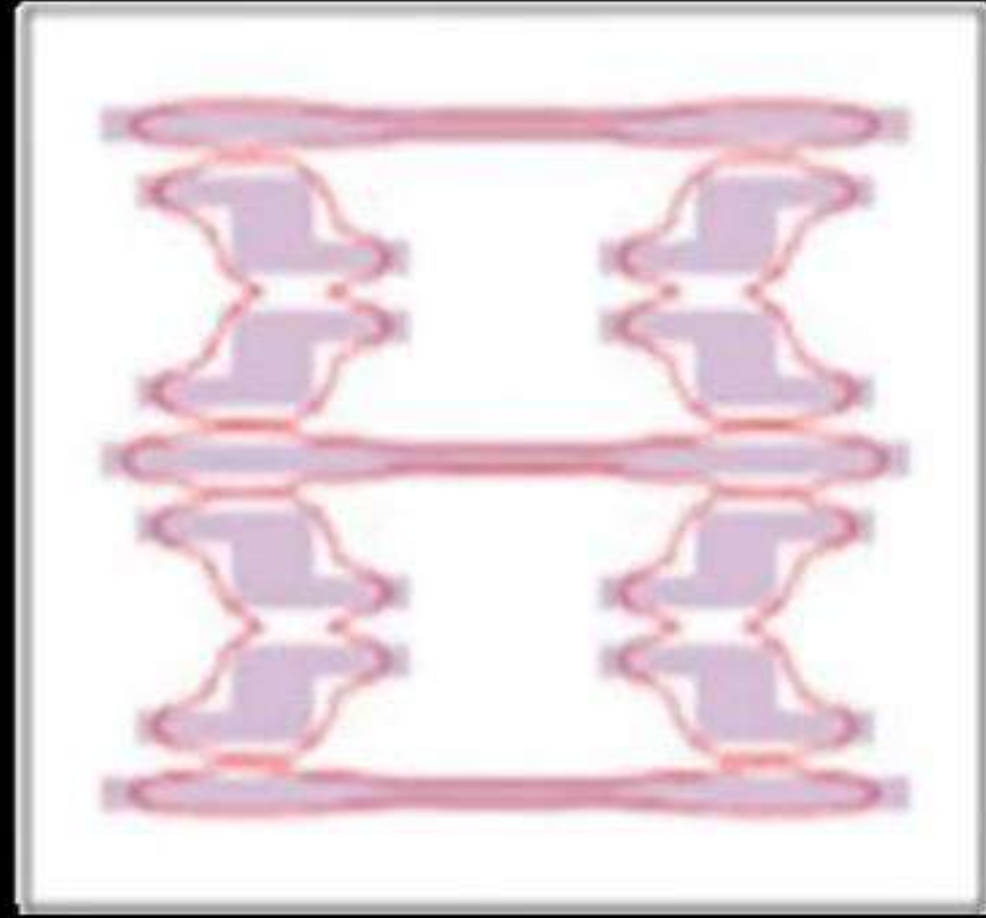


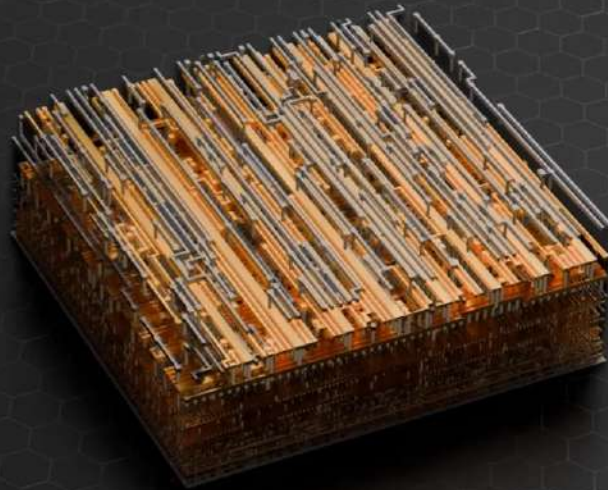
Error

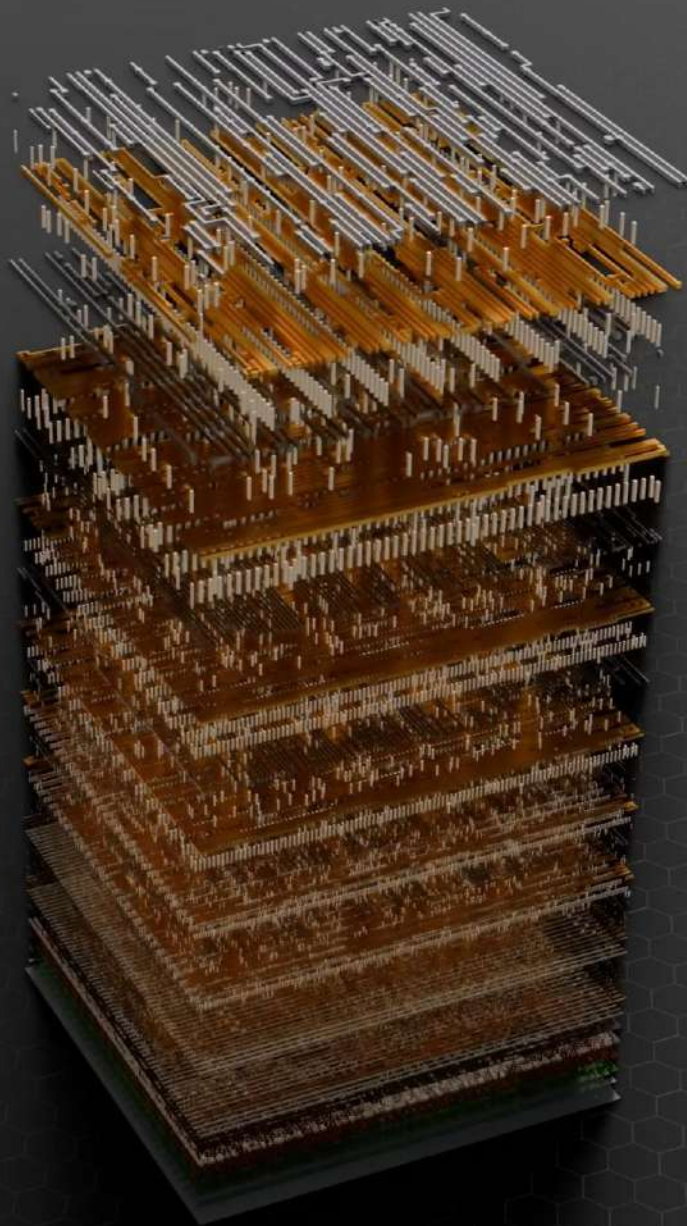
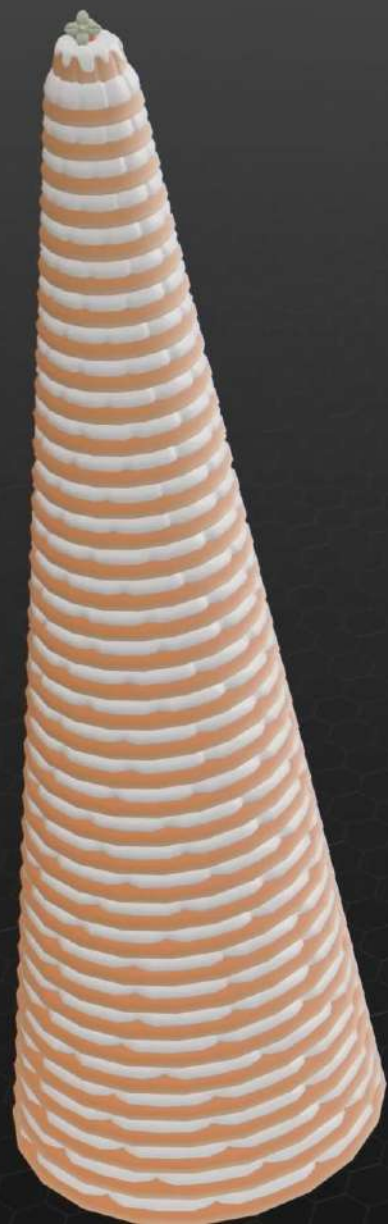


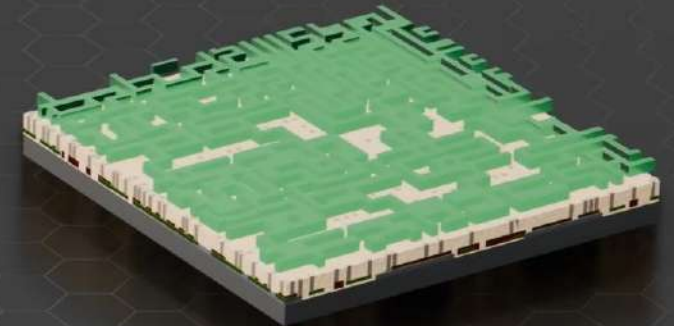
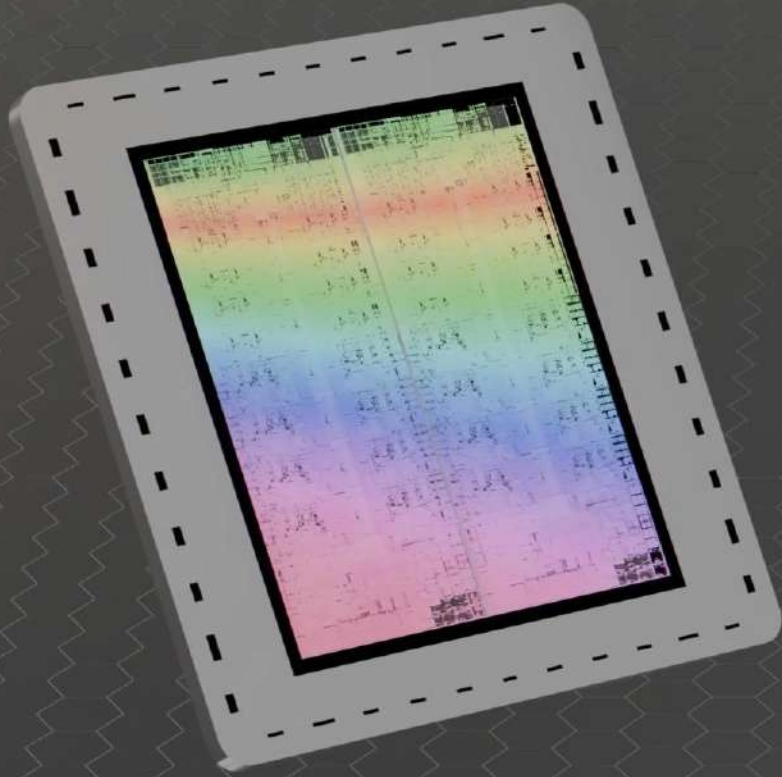
(Source-mask optimization)

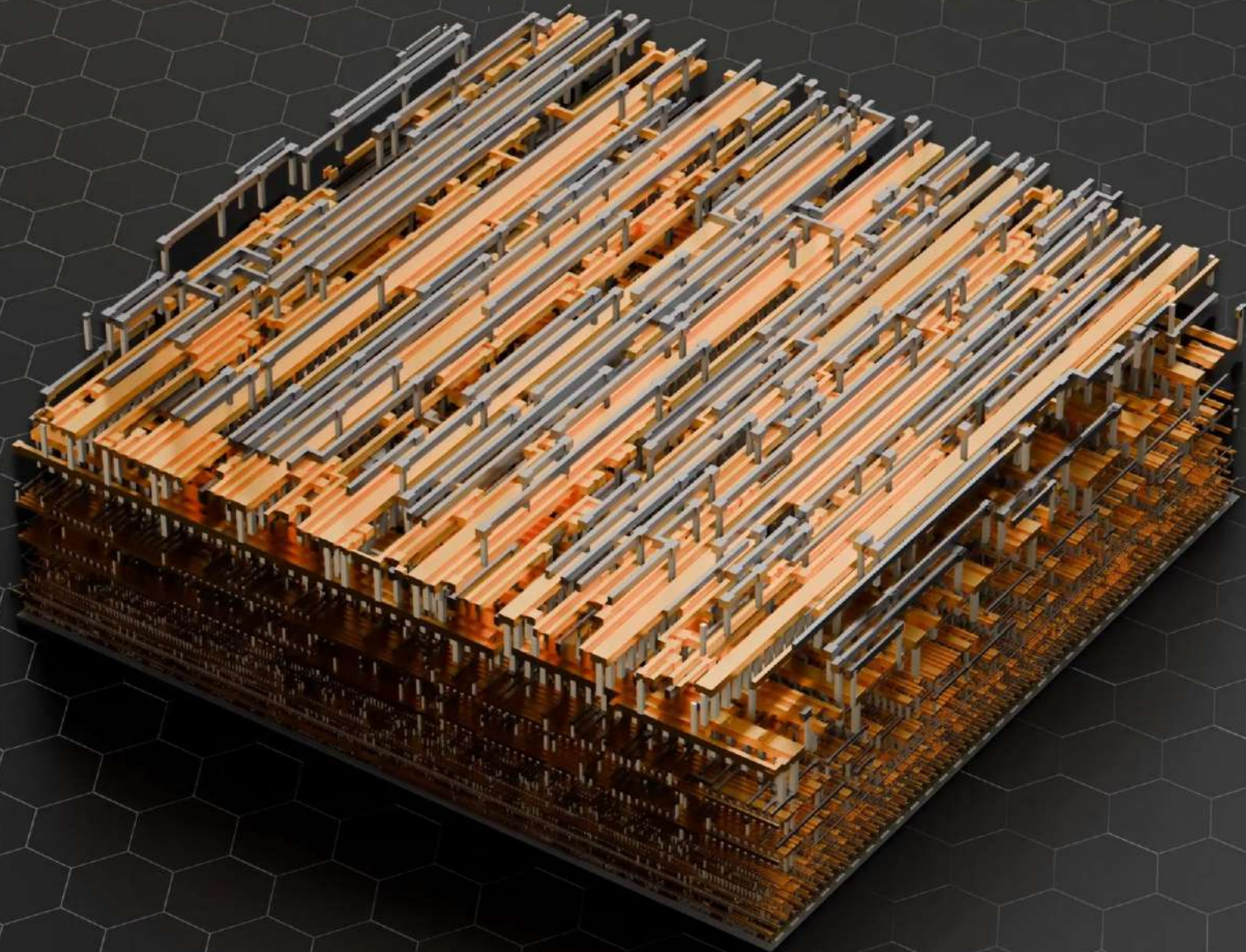




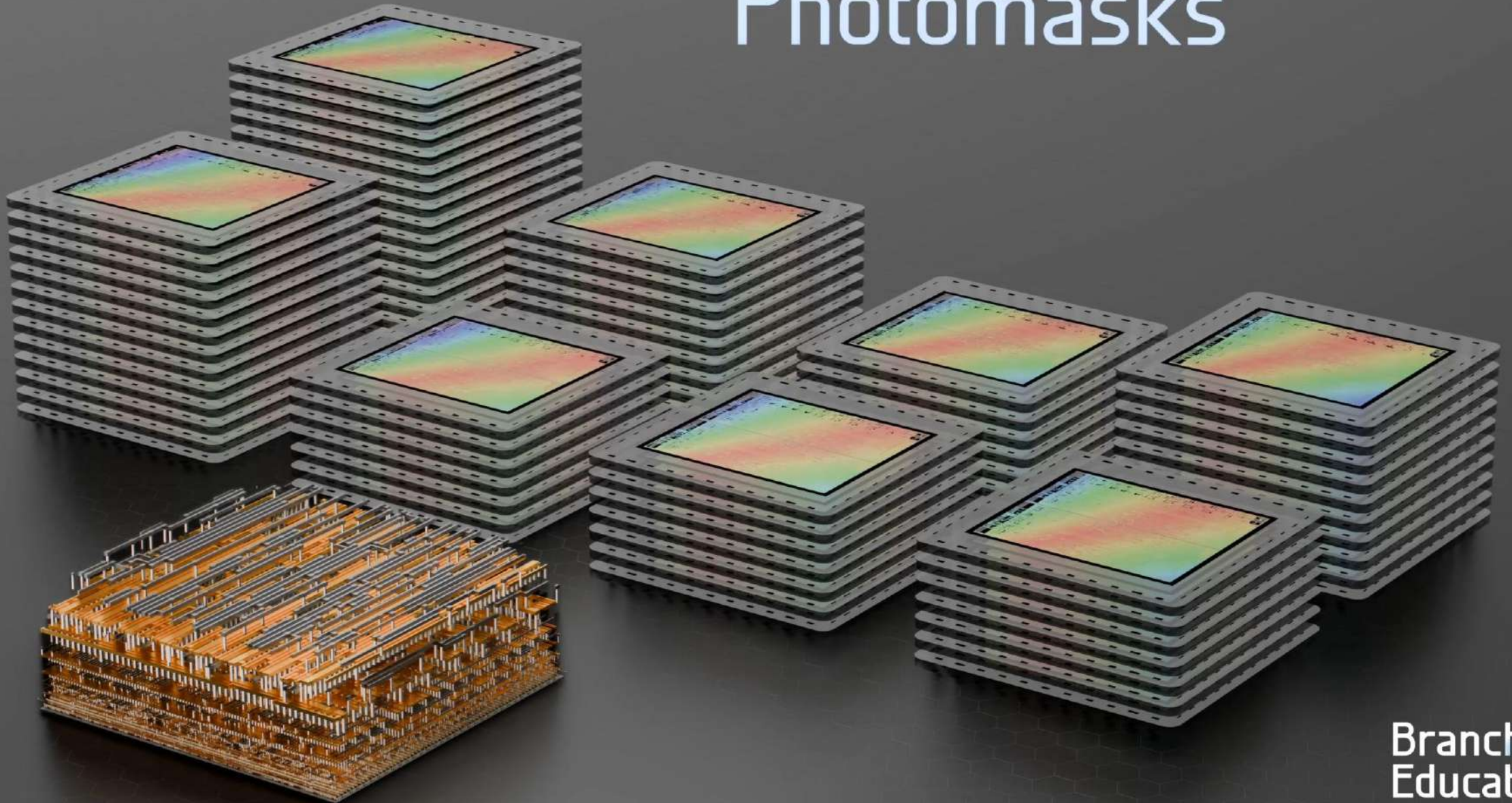


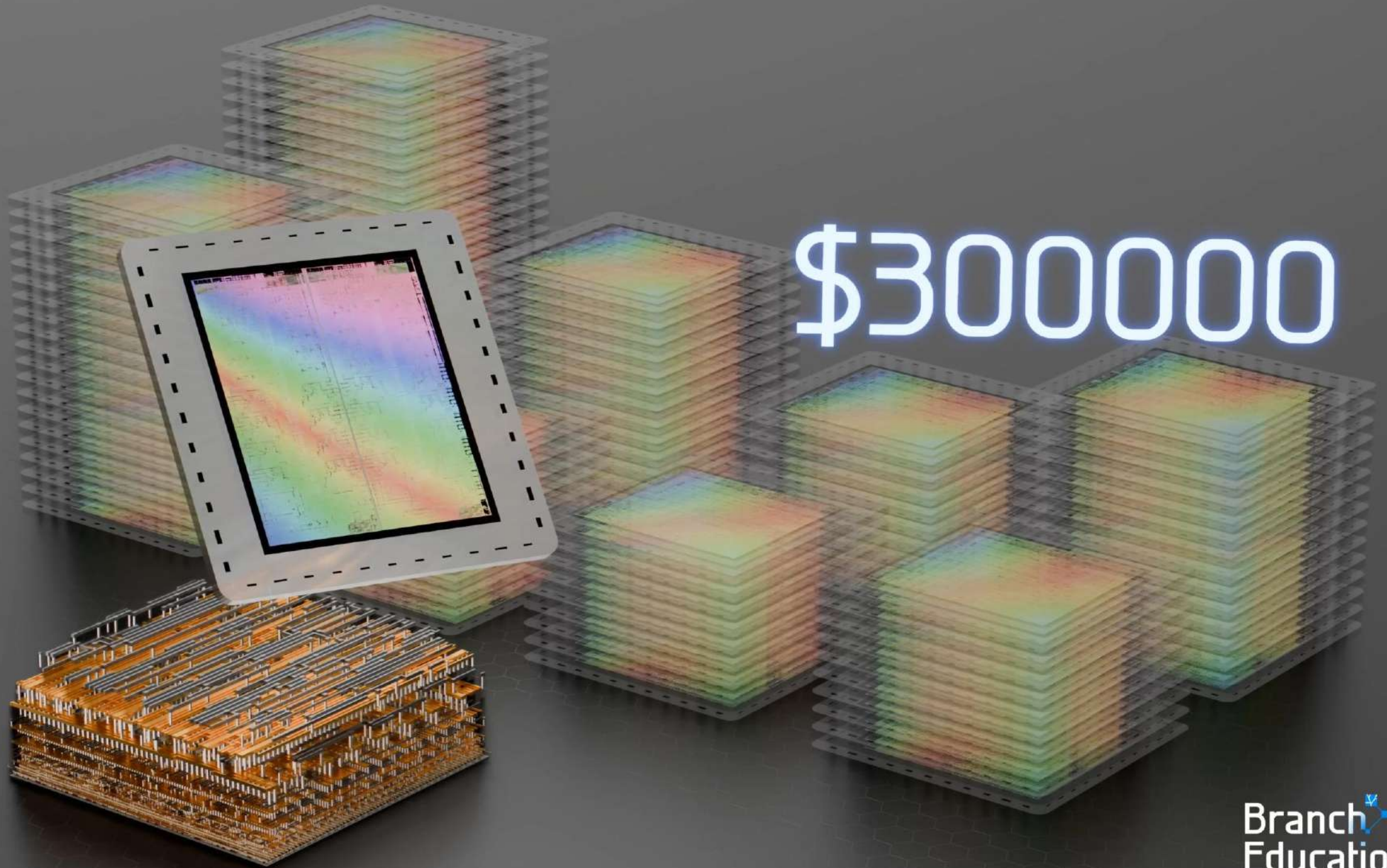






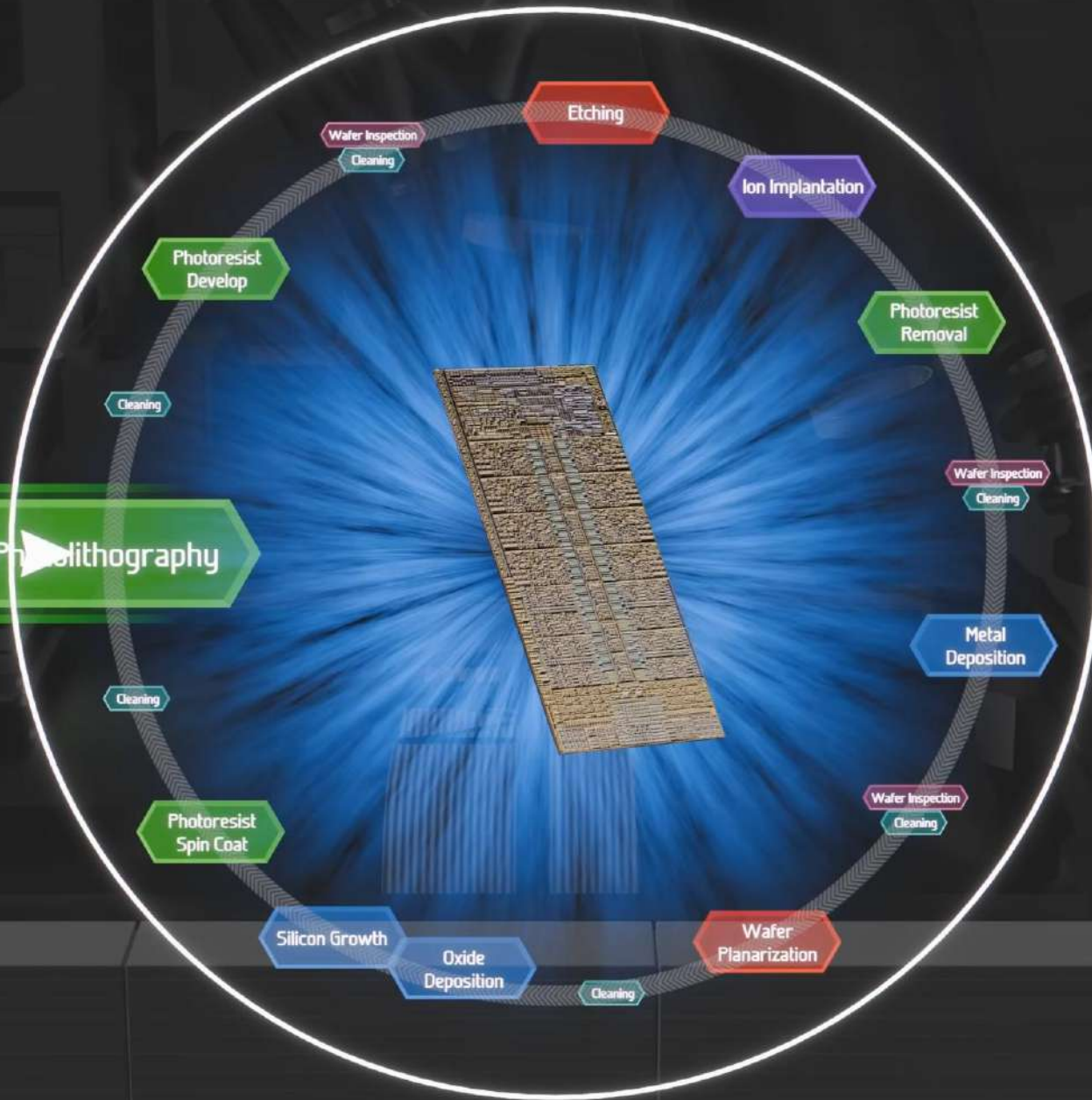
80 Different Photomasks

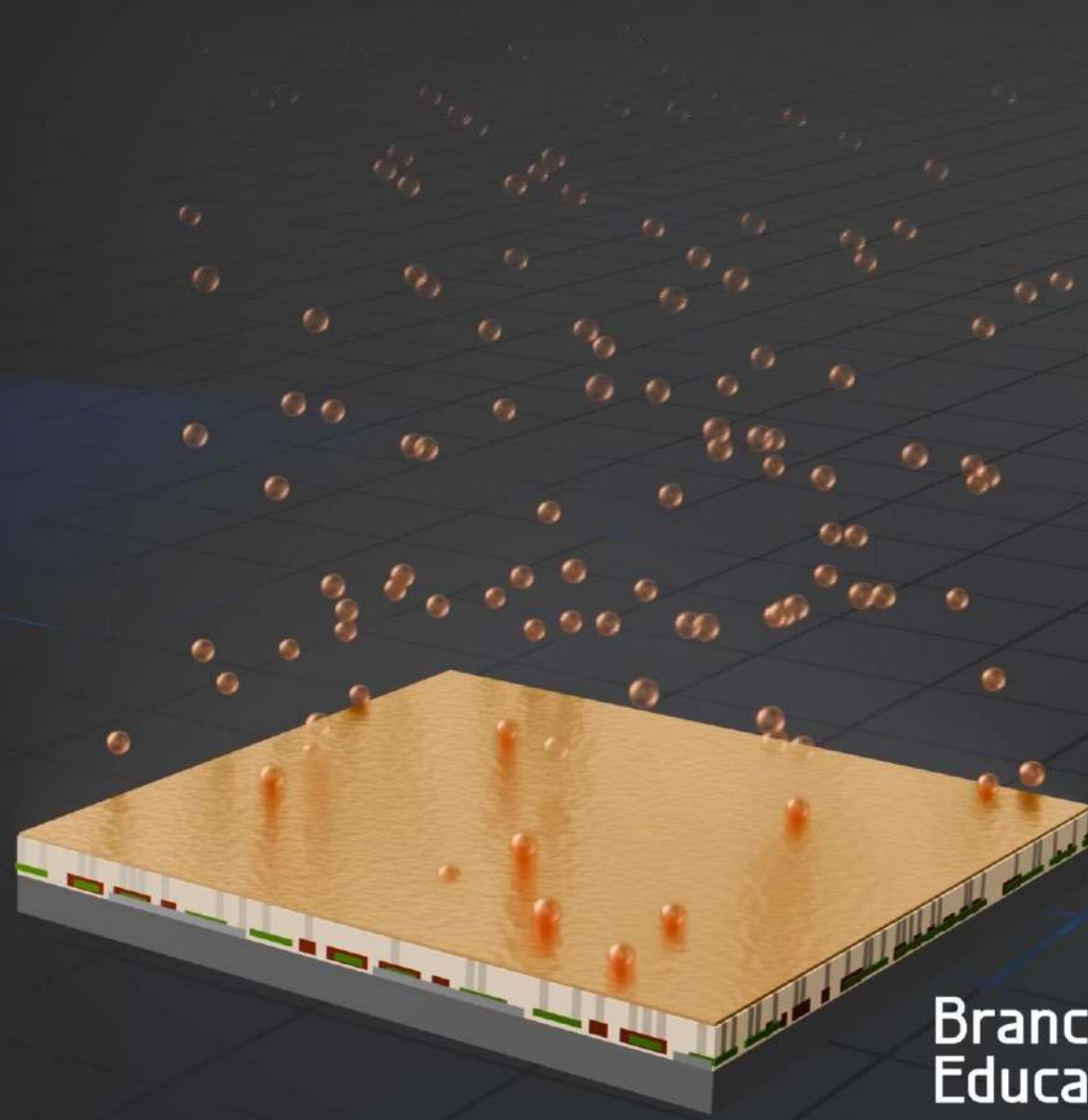
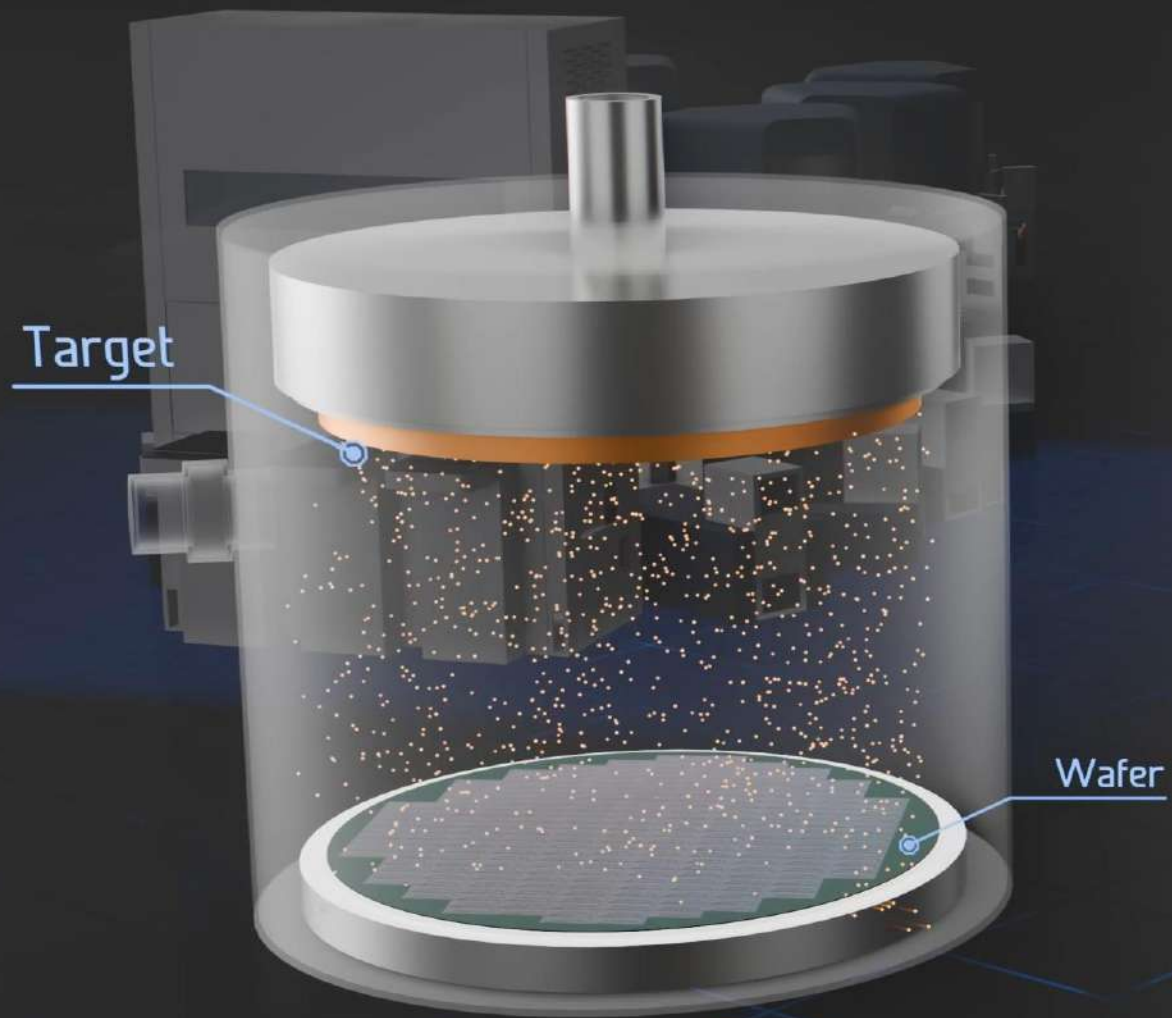


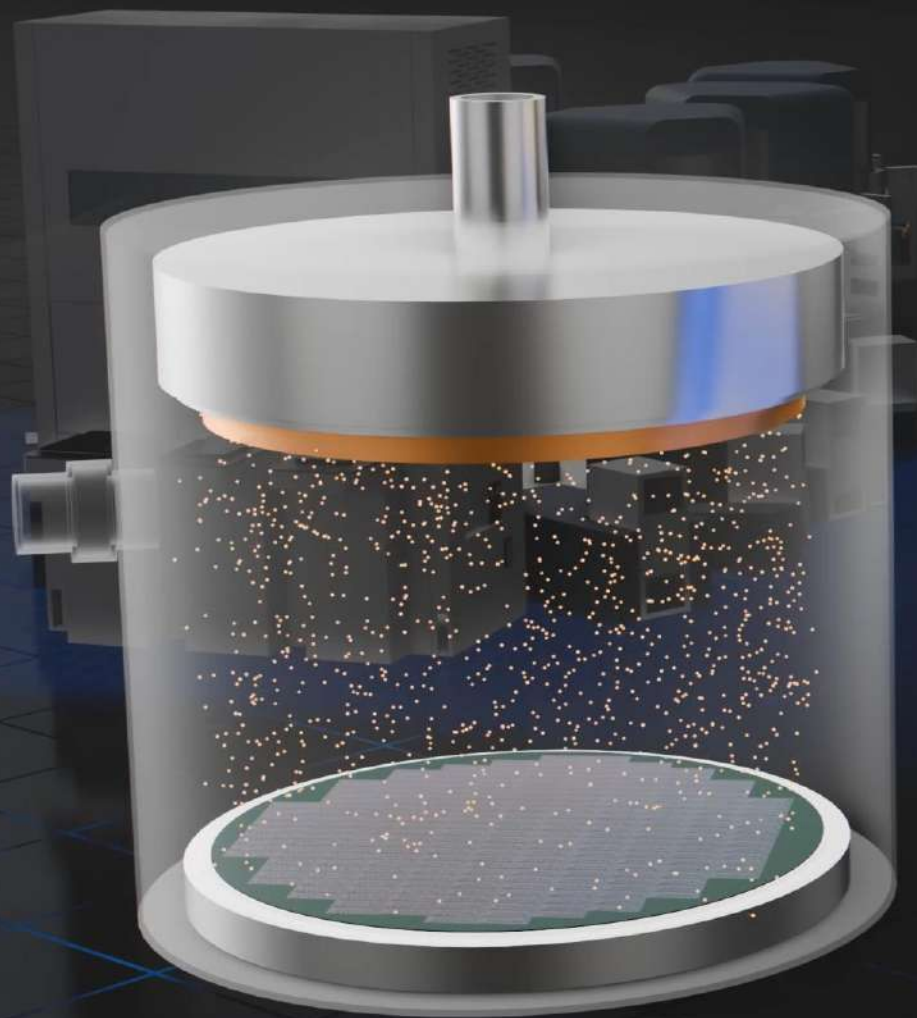


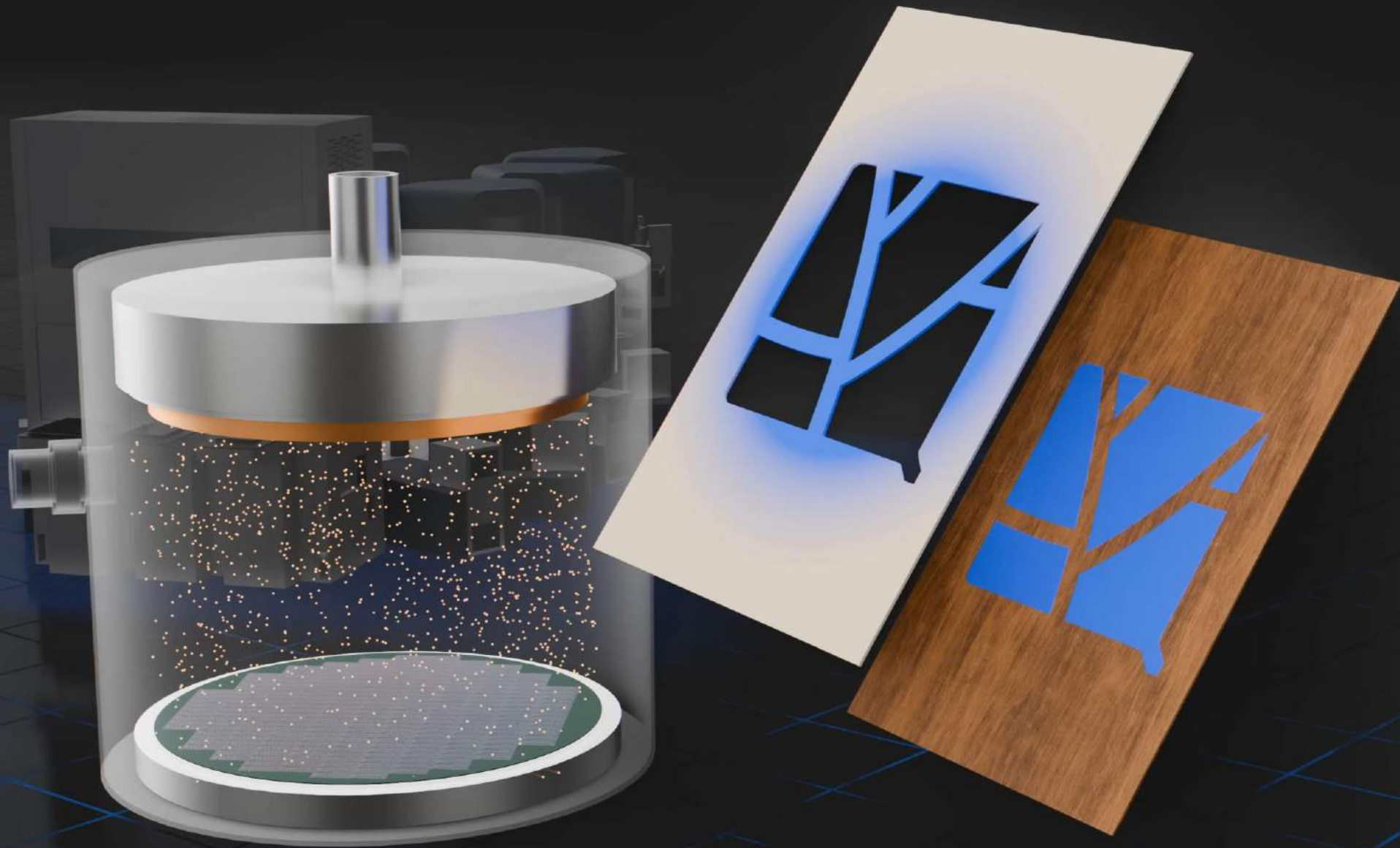
\$300000

80x Photolithography







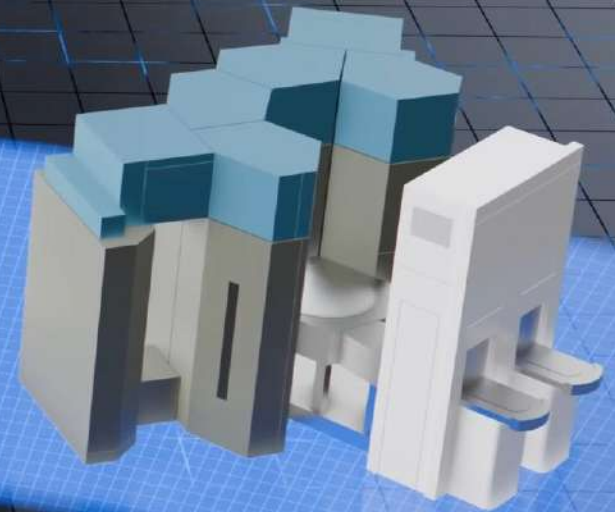
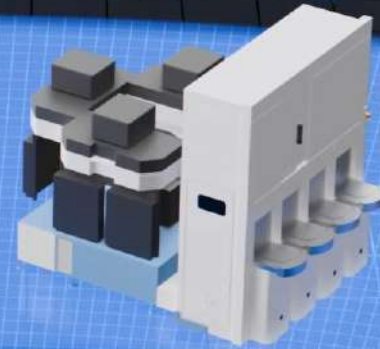




Physical Vapour Deposition



Chemical Vapor Deposition



Epitaxial Growth

Etching

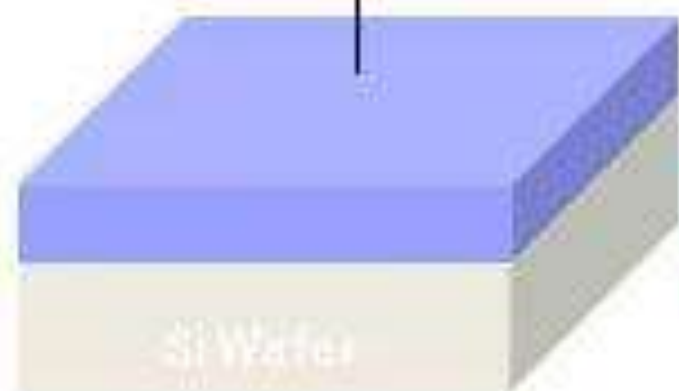
에칭(Etching)

노광된 포토레지스트(PR) 패턴을 마스크로 활용해 반도체 웨이퍼 표면에서 불필요한 재료를 제거하는 공정이다.

습식 에칭(산성·알칼리성 화학 용액)과 건식 에칭(플라즈마·DRIE 등 물리·화학적 방법) 방식이 있으며,

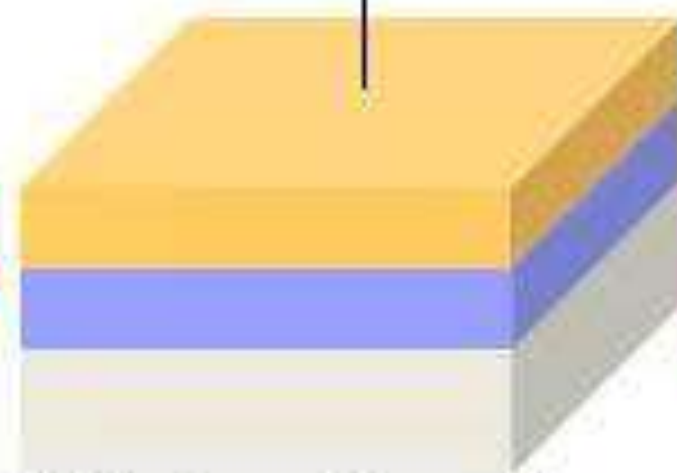
포토레지스트로 보호되지 않은 노출 영역만 선택적으로 식각해 회로 패턴을 웨이퍼에 새긴다.

The etched material
Al, Poly-Si, SiO_2

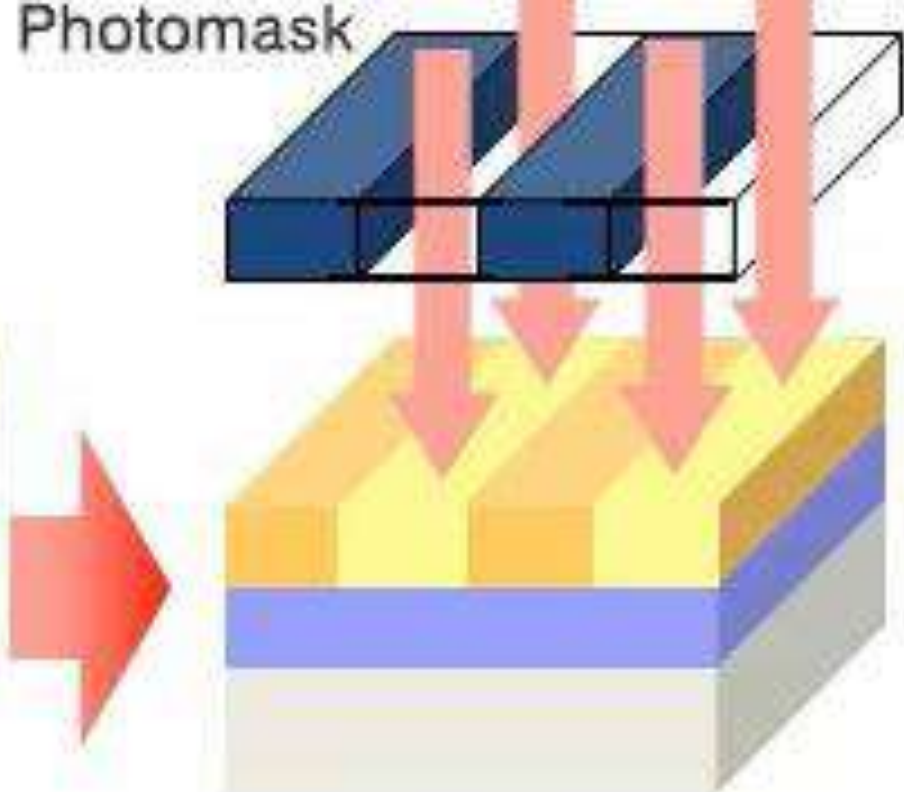


(1) Deposition

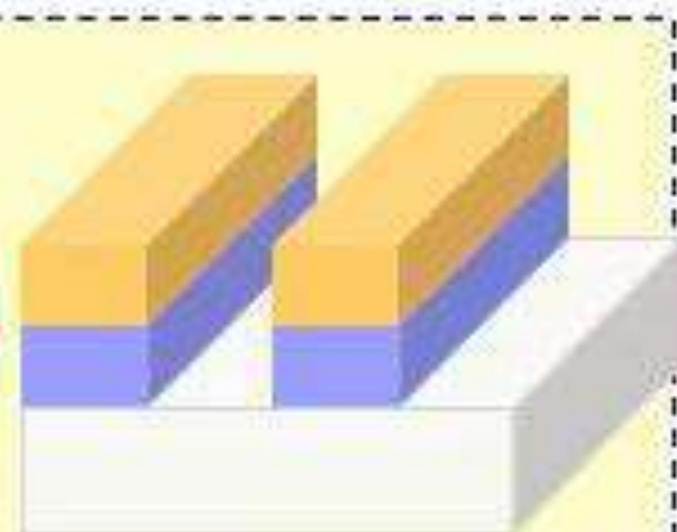
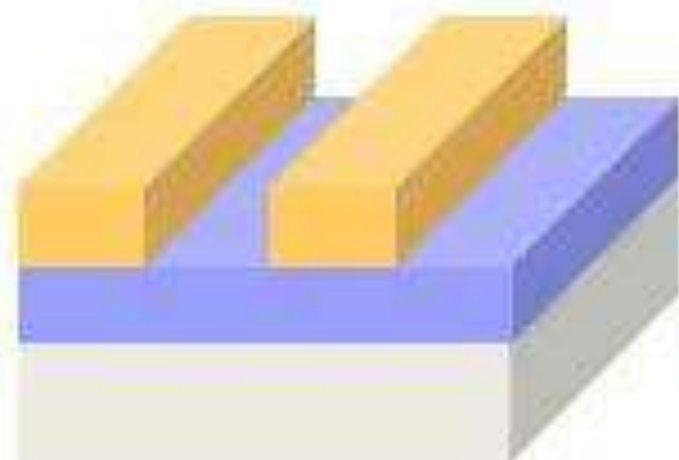
Photoresist
(positive photoresist)

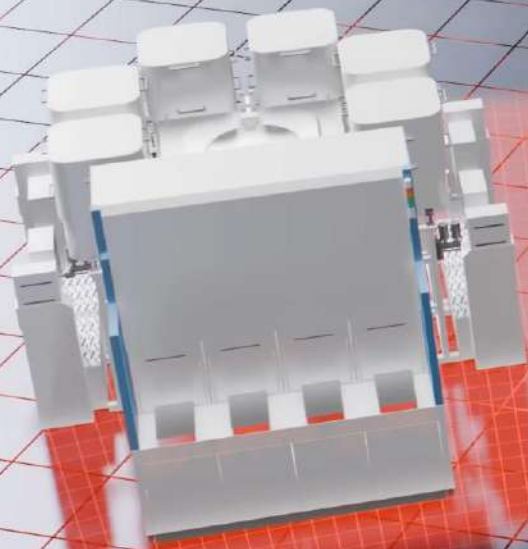


(2) Photoresist coating



(3) Exposure

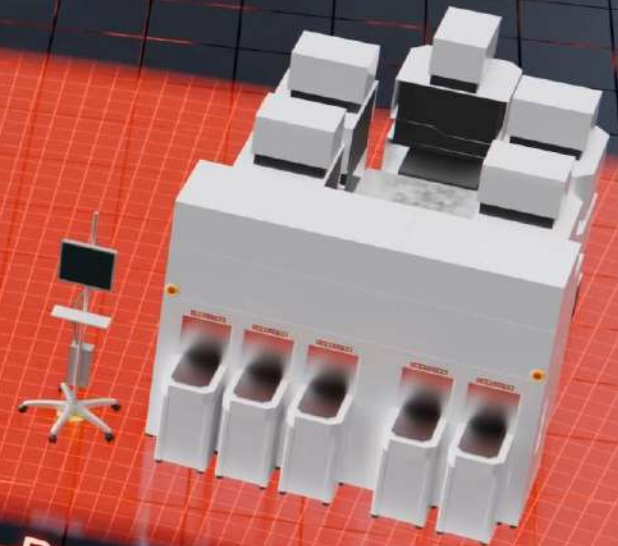




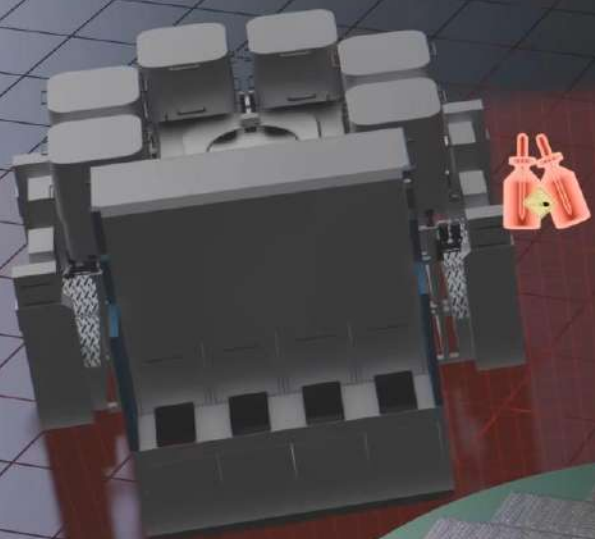
Etching Tool



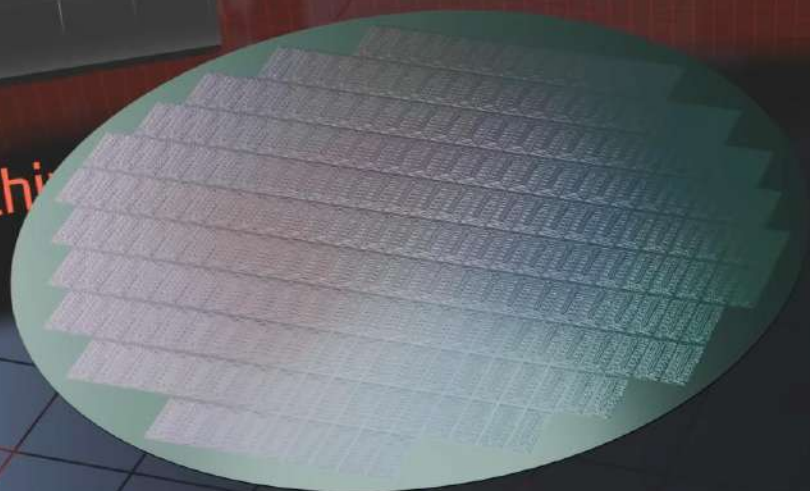
Etching Tool



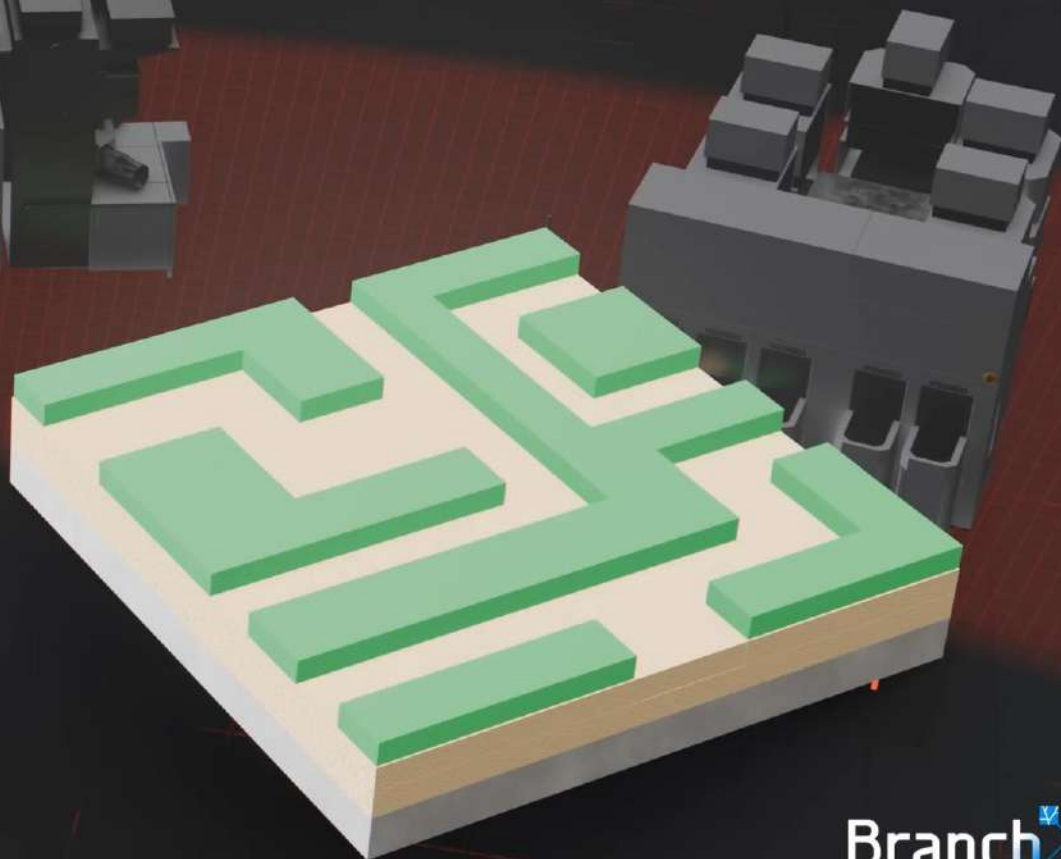
Dry Etching Tool

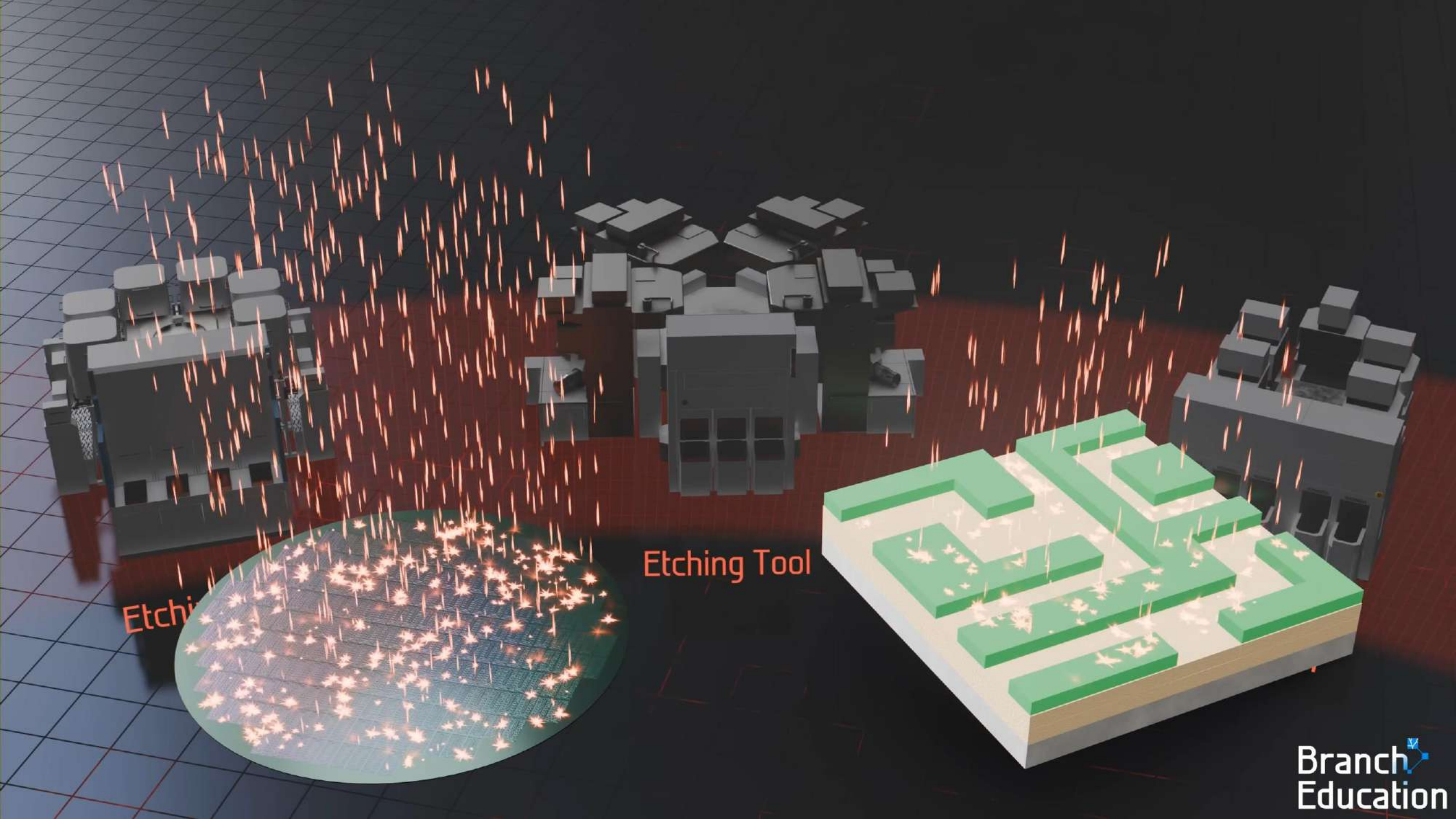


Etching



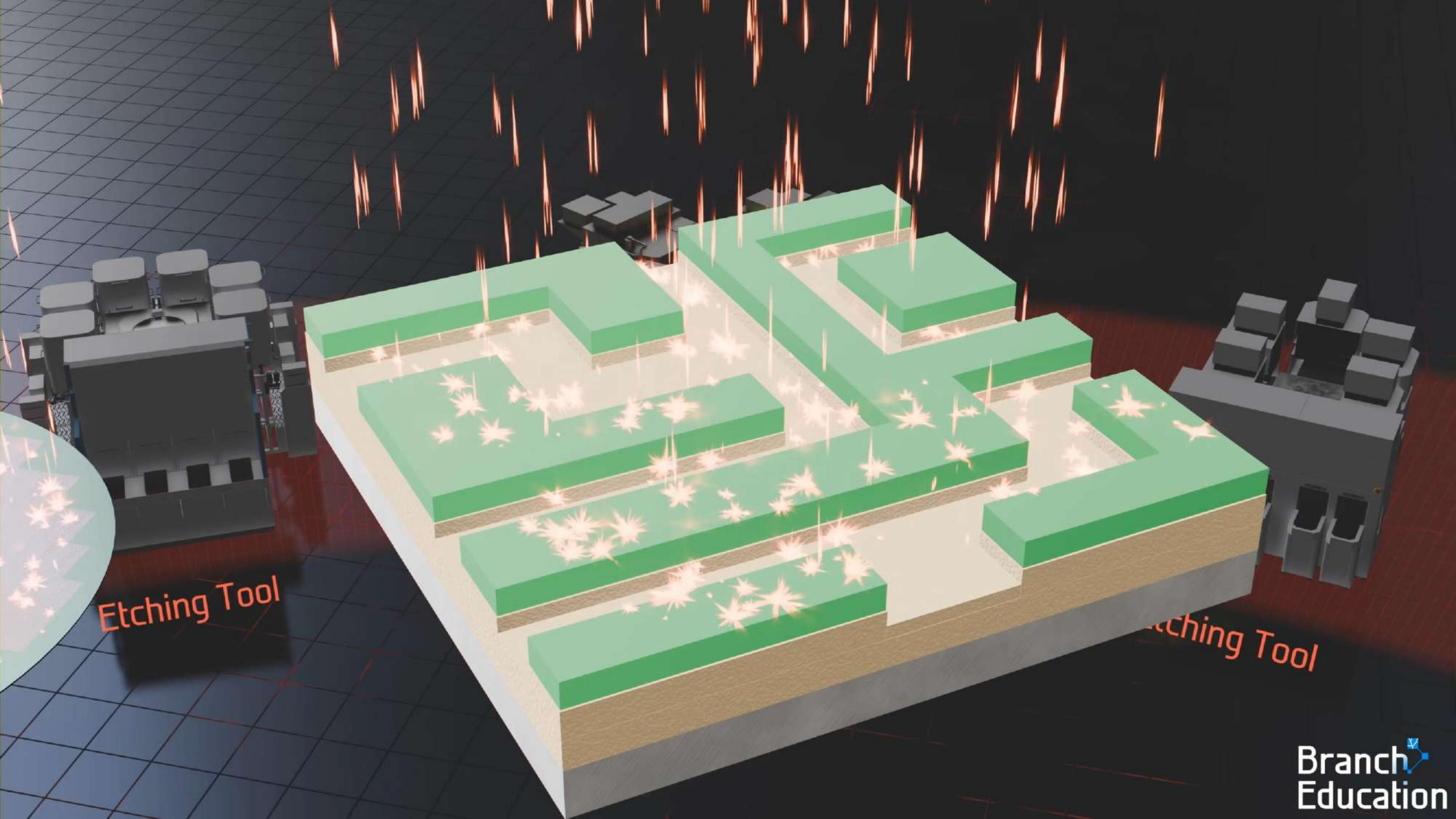
Etching Tool





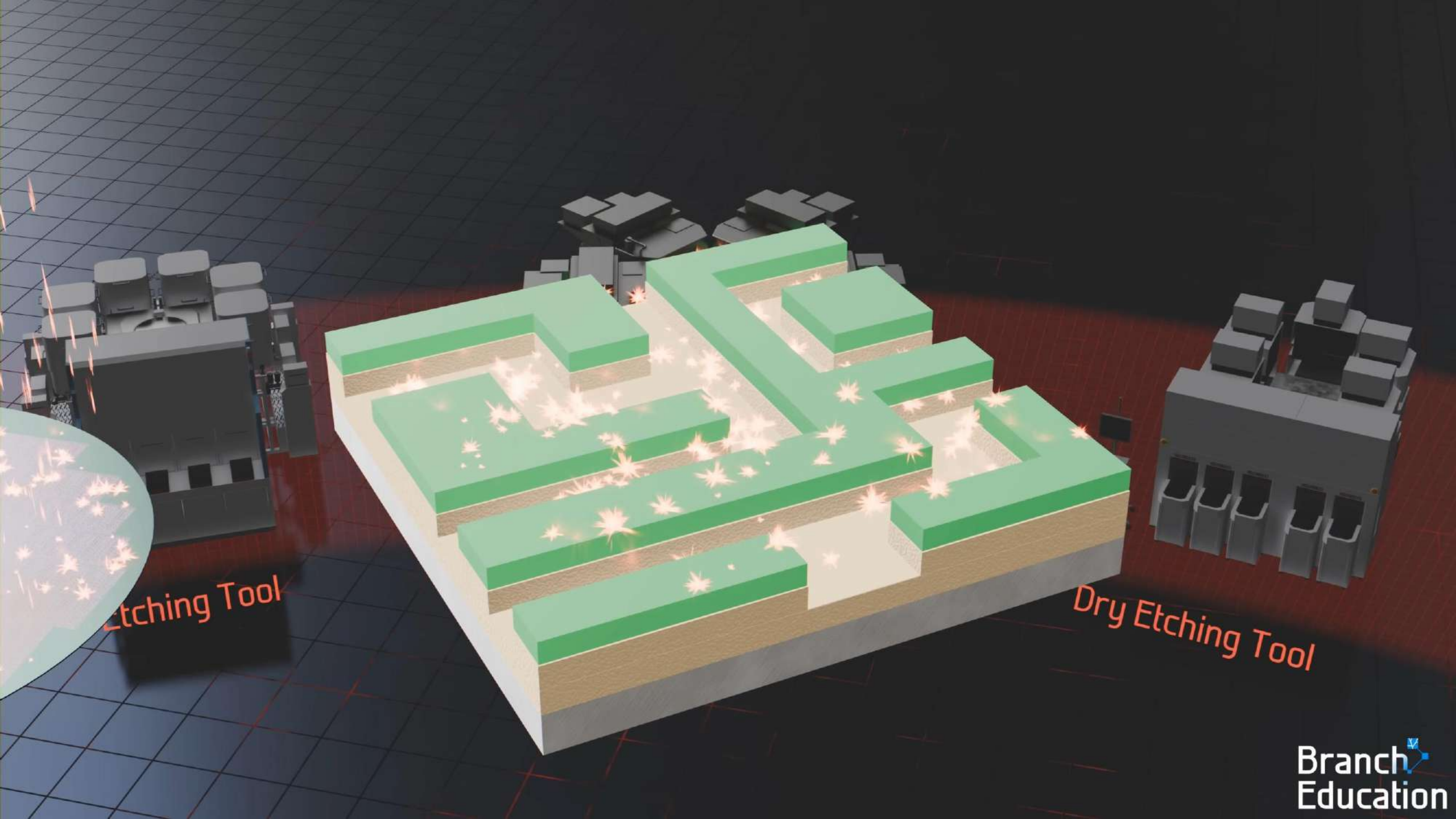
Etching

Etching Tool



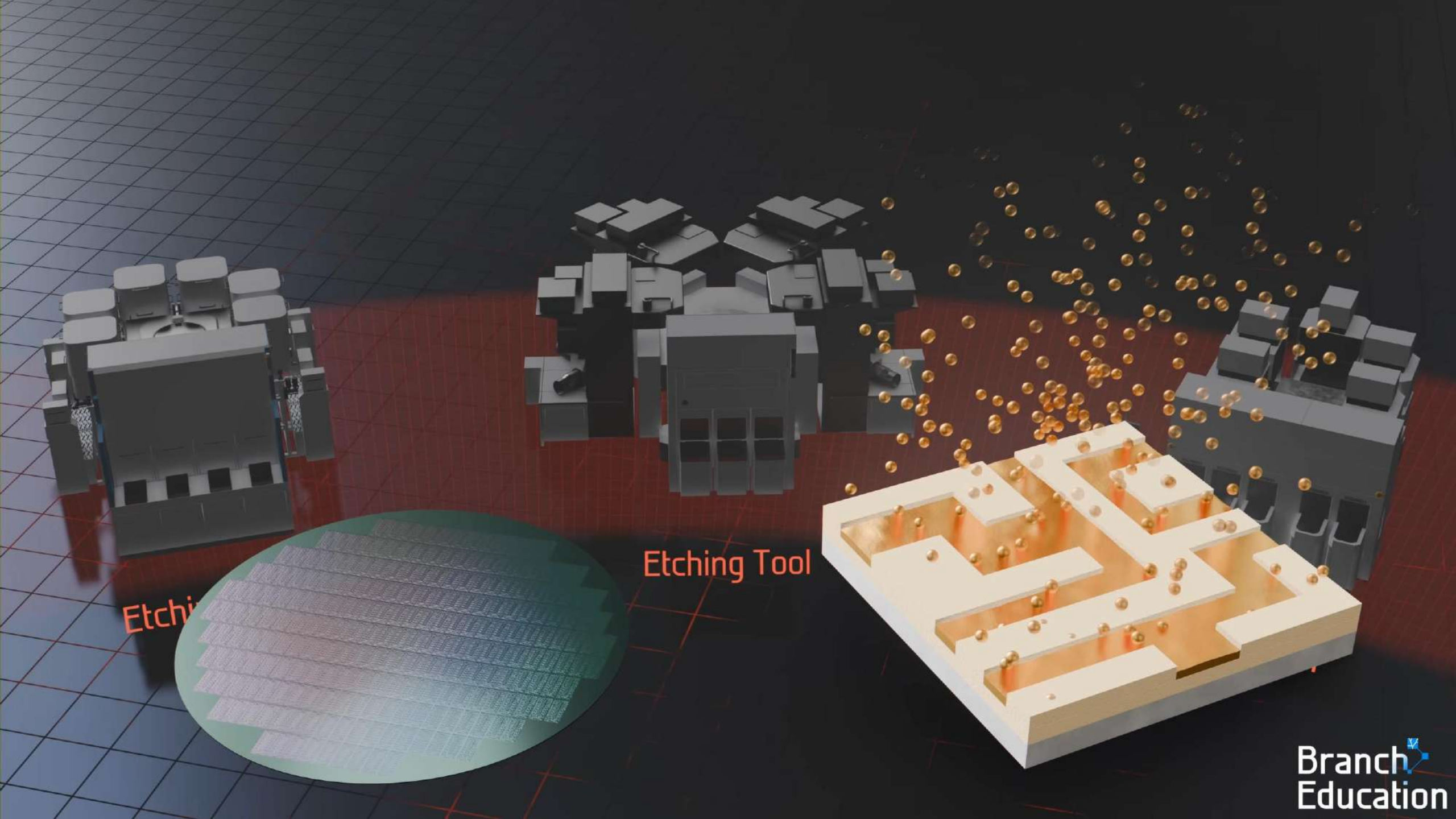
Etching Tool

Etching Tool



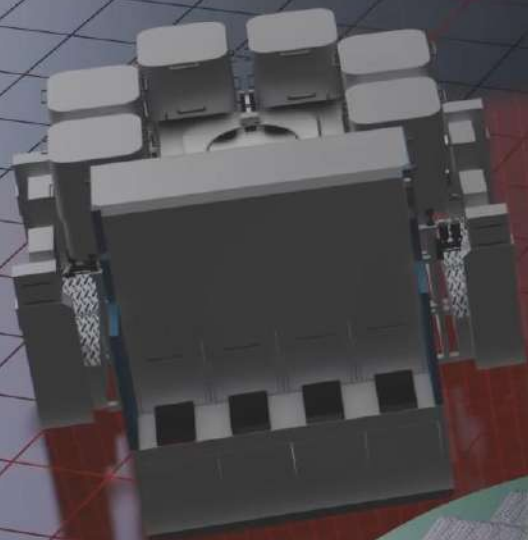
Etching Tool

Dry Etching Tool

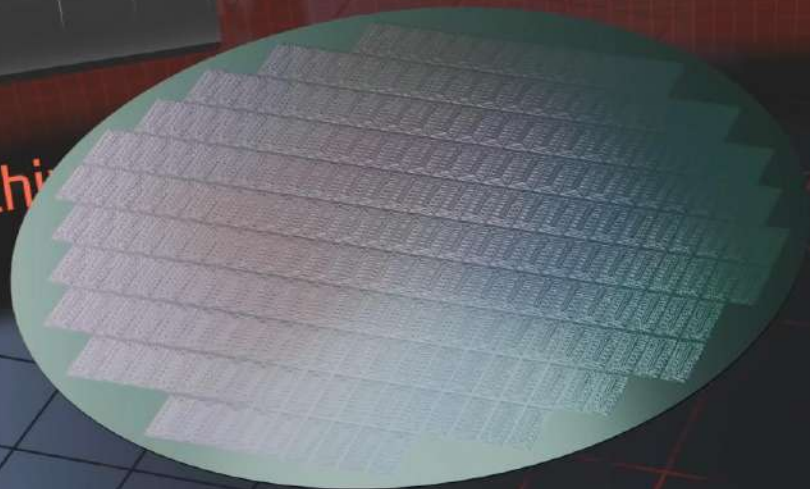


Etching Tool

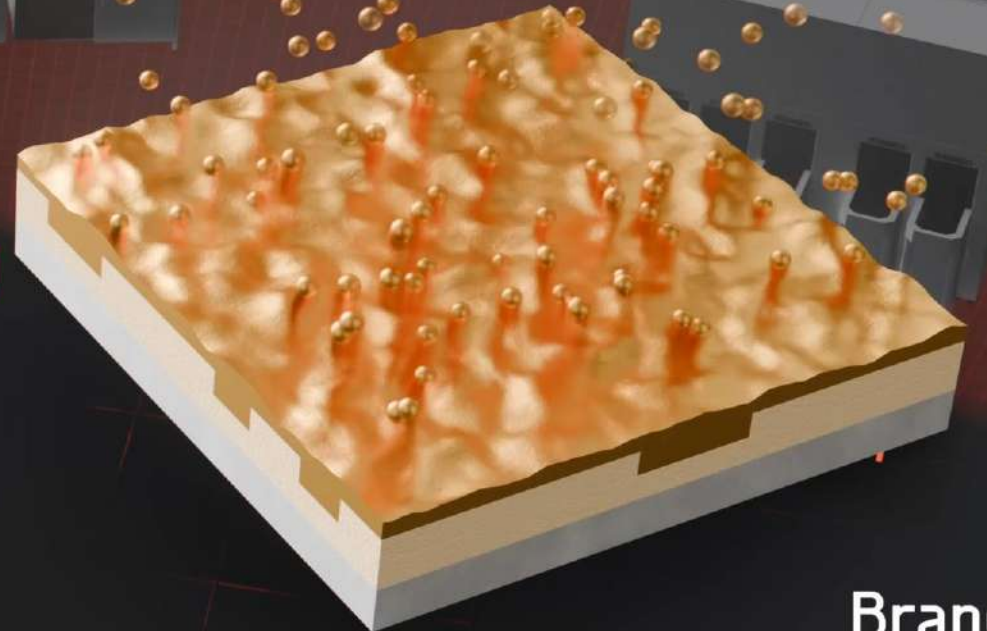
Etching

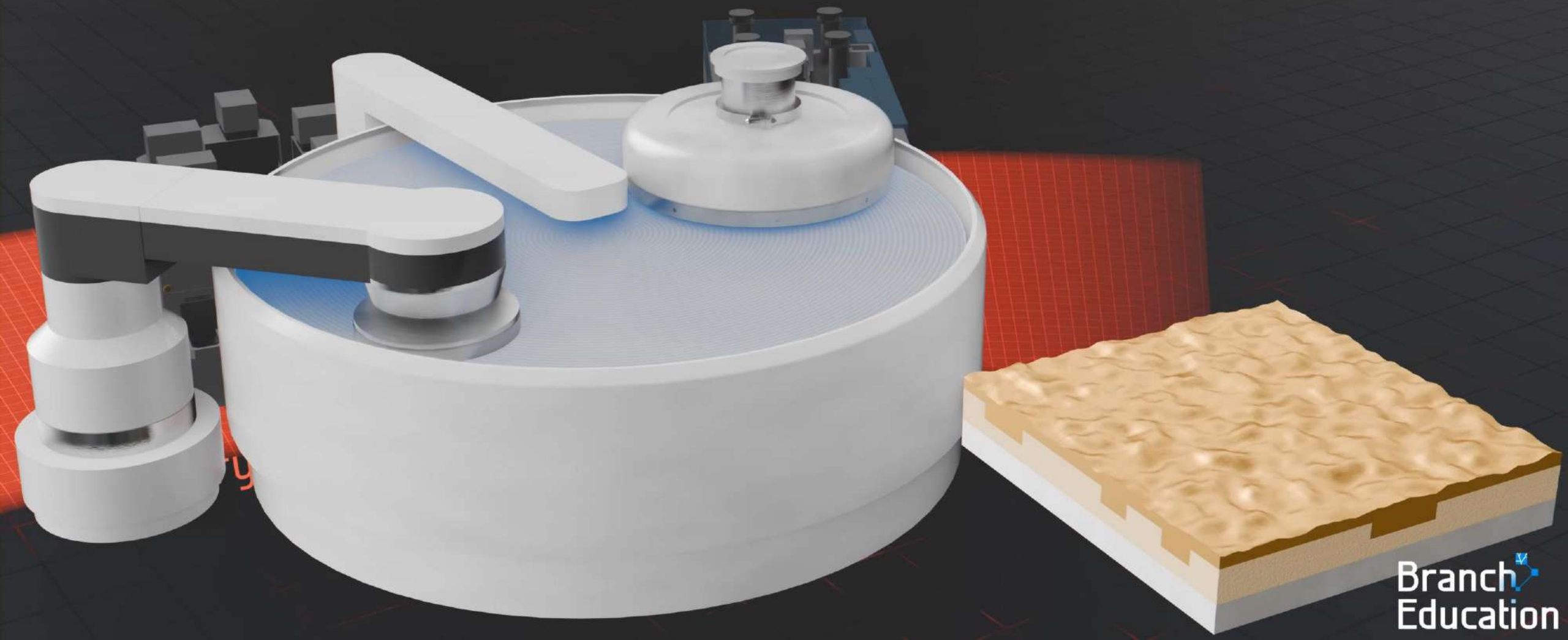


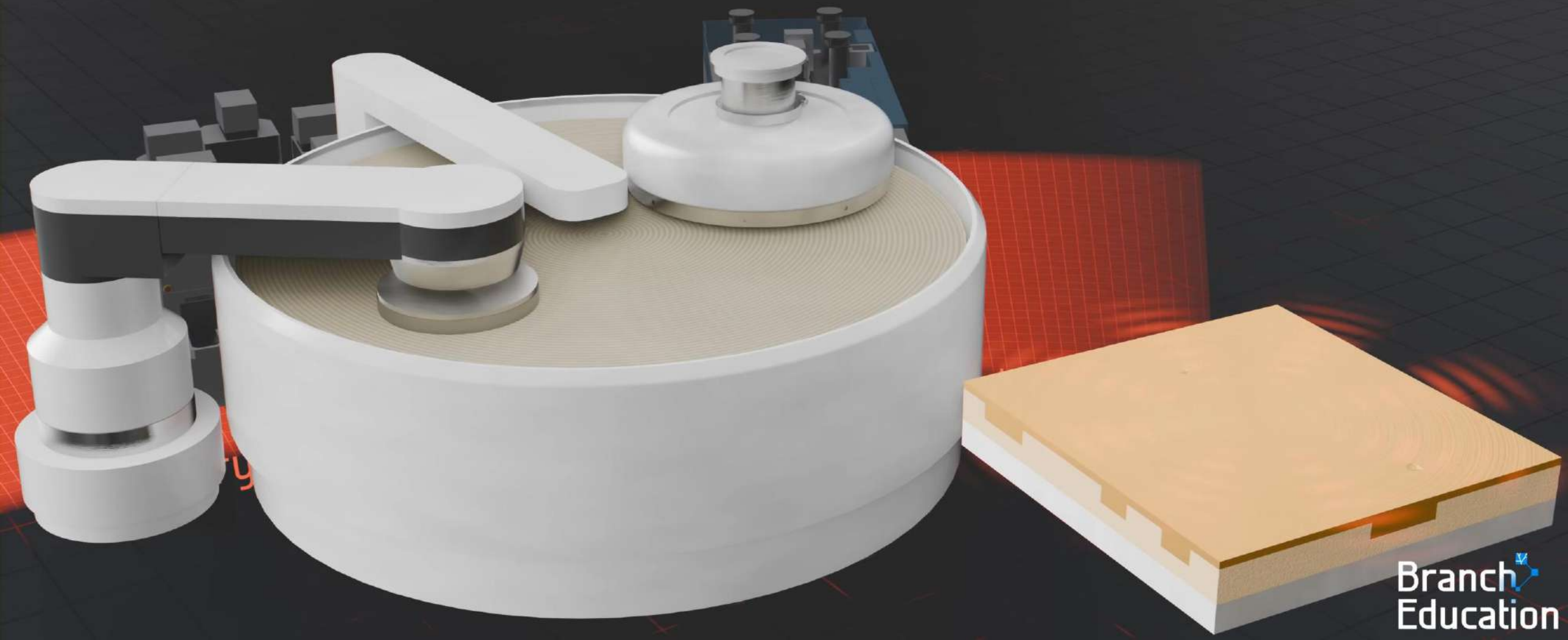
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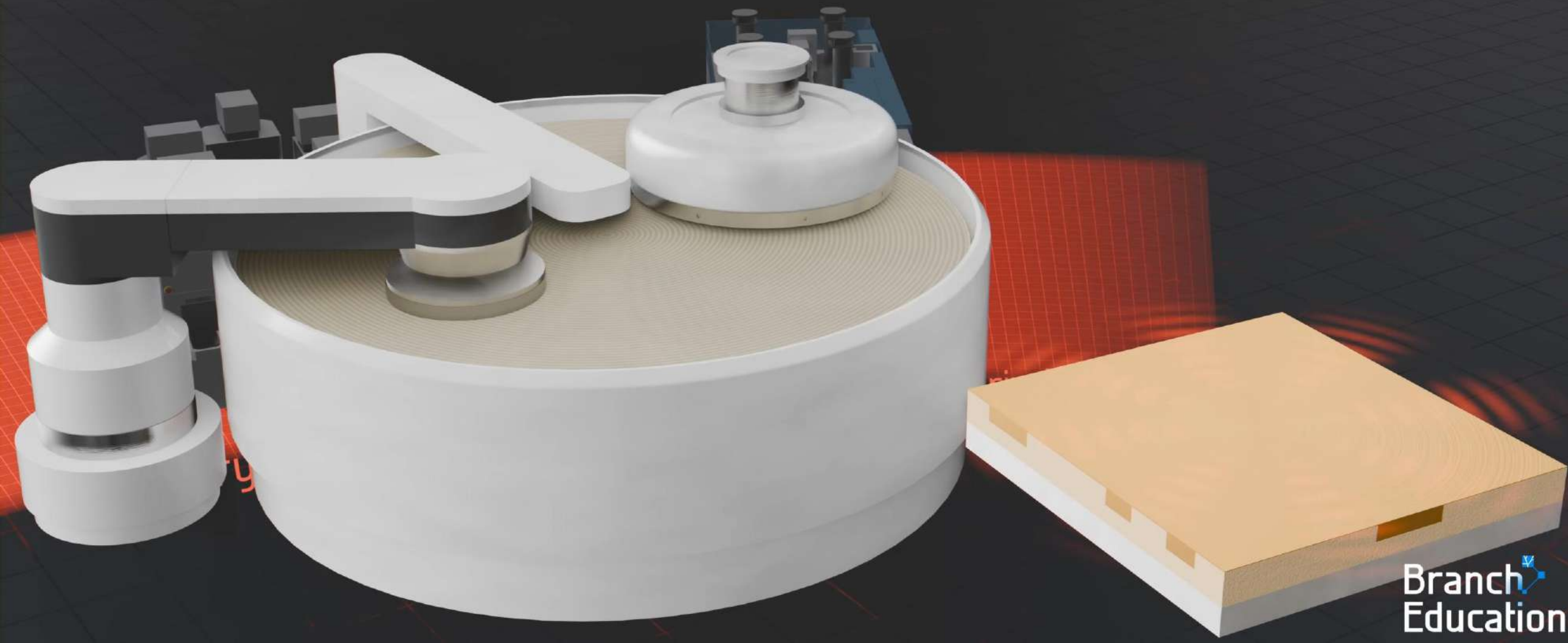


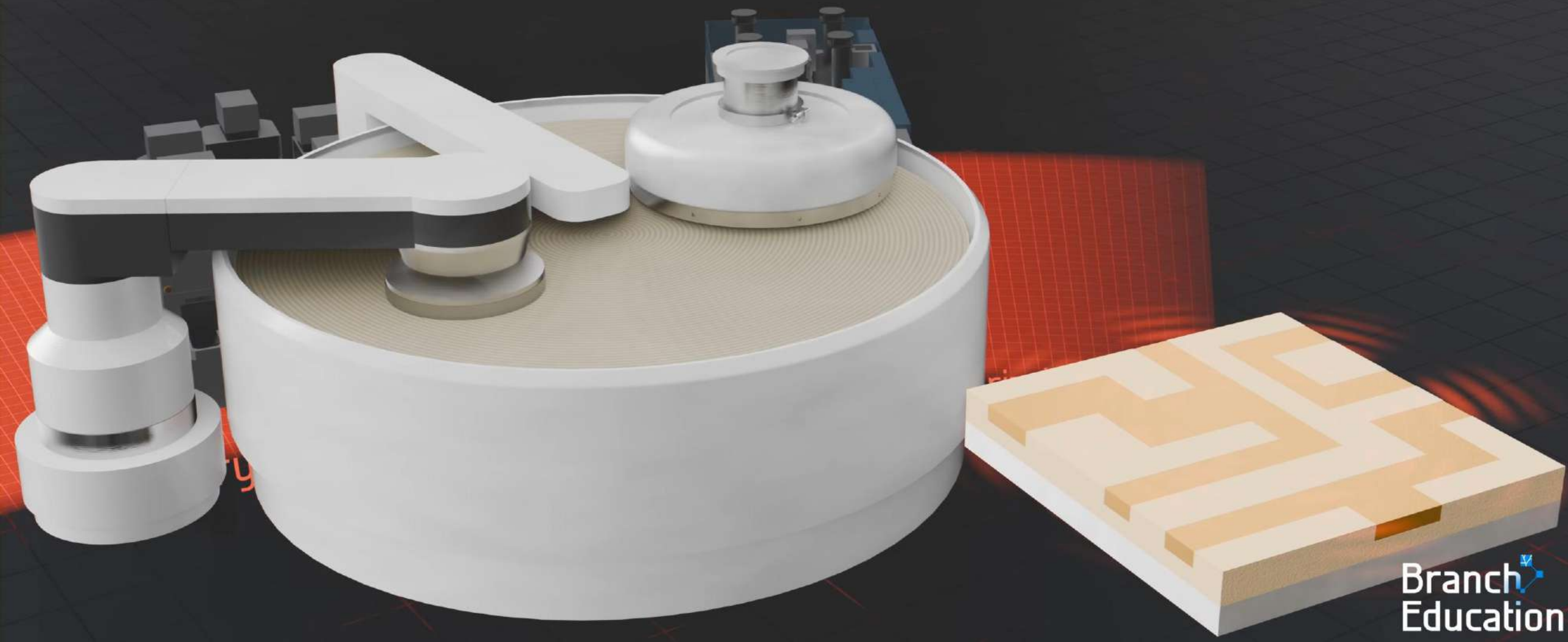
Etching Tool











Photoresist
Removal

Ion Implantation

Etching

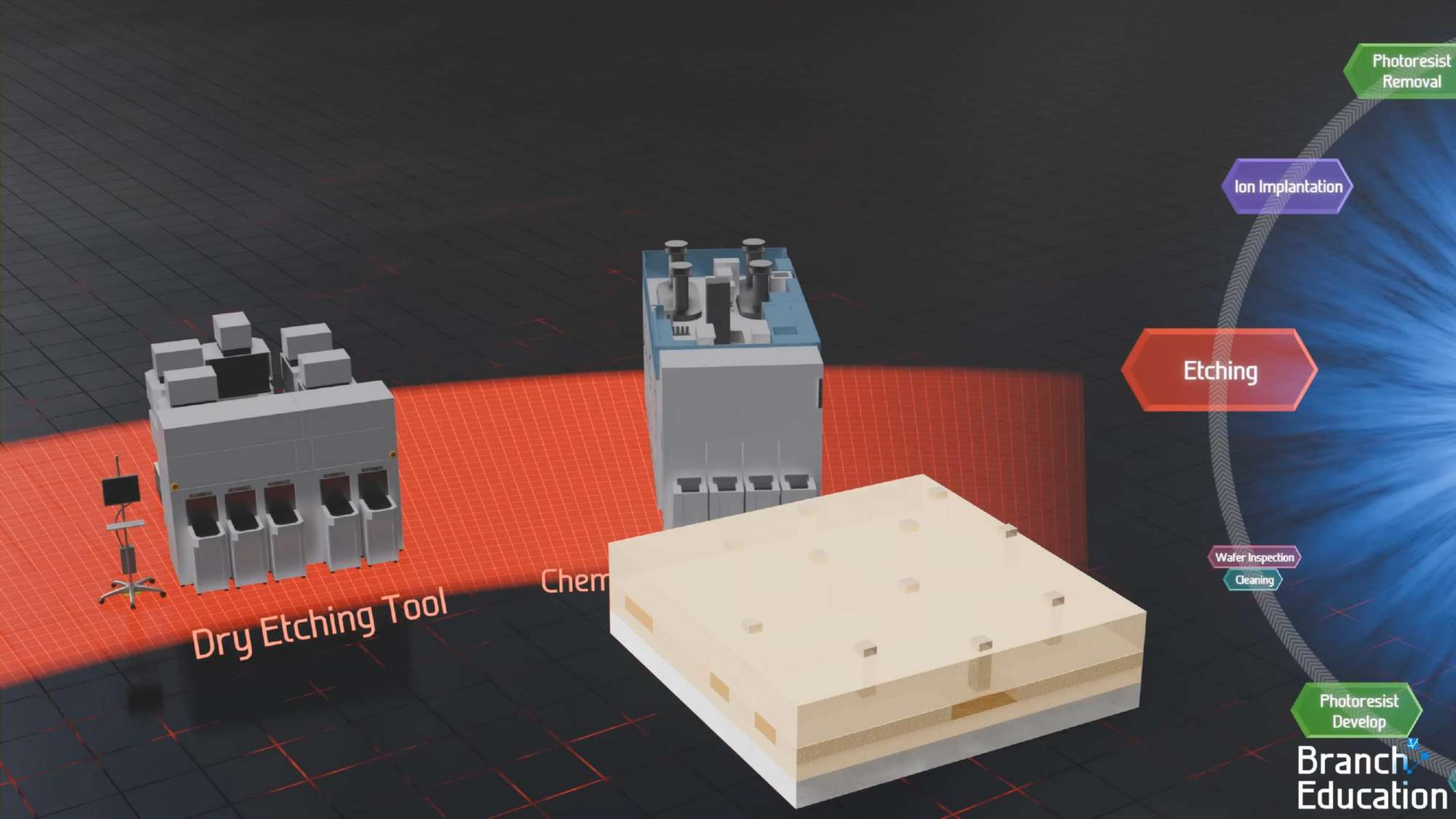
Wafer Inspection

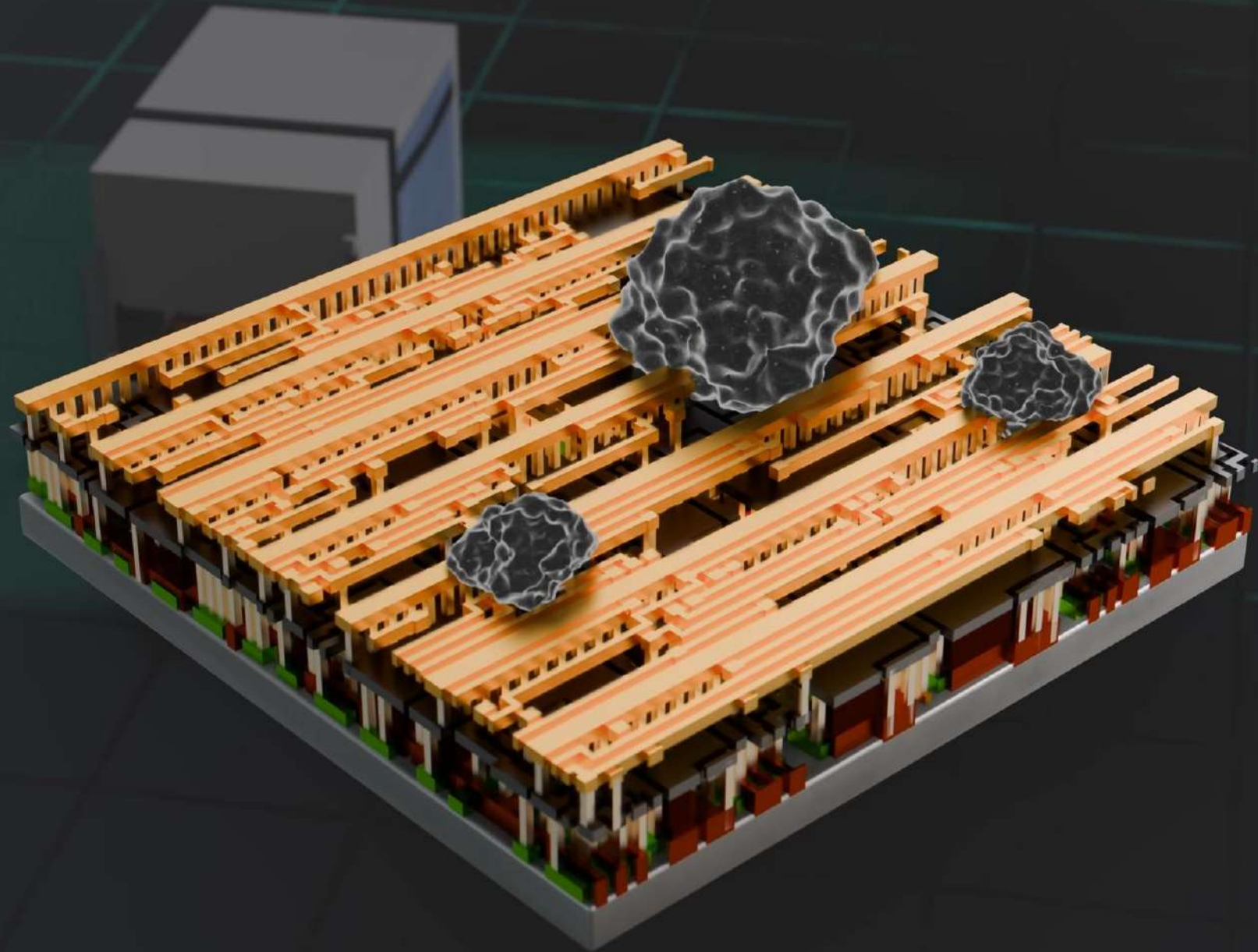
Cleaning

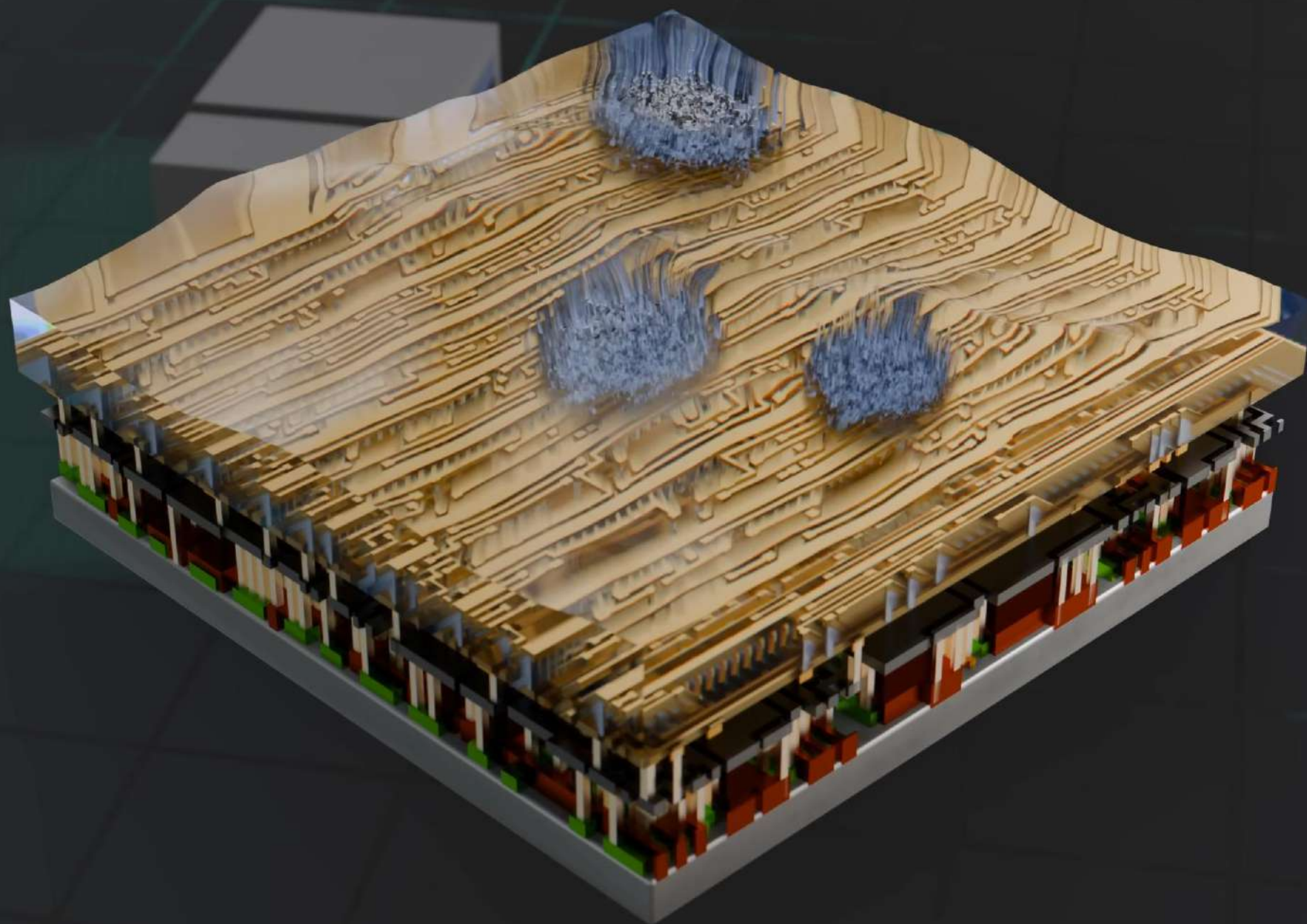
Photoresist
Develop

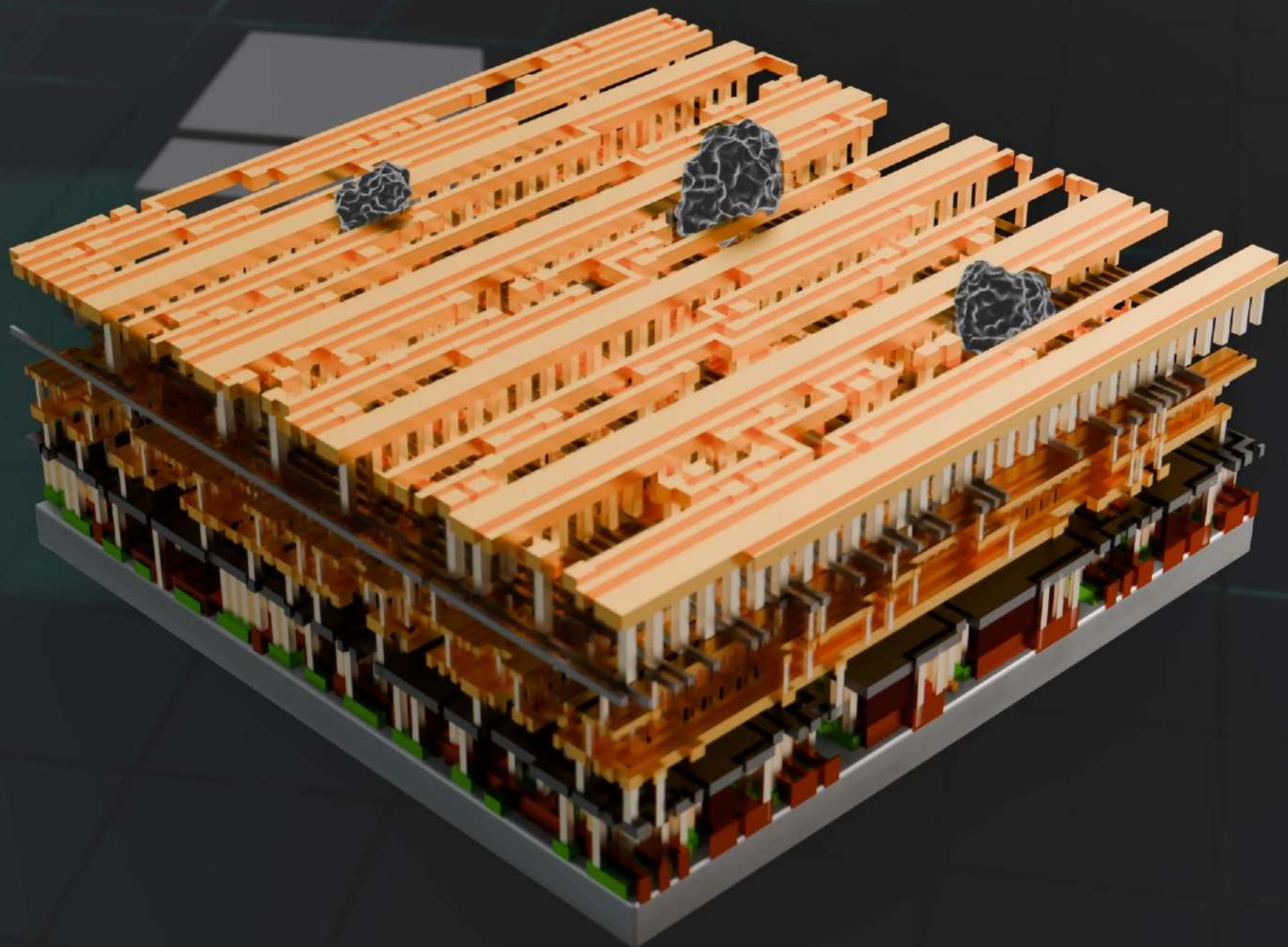
Dry Etching Tool

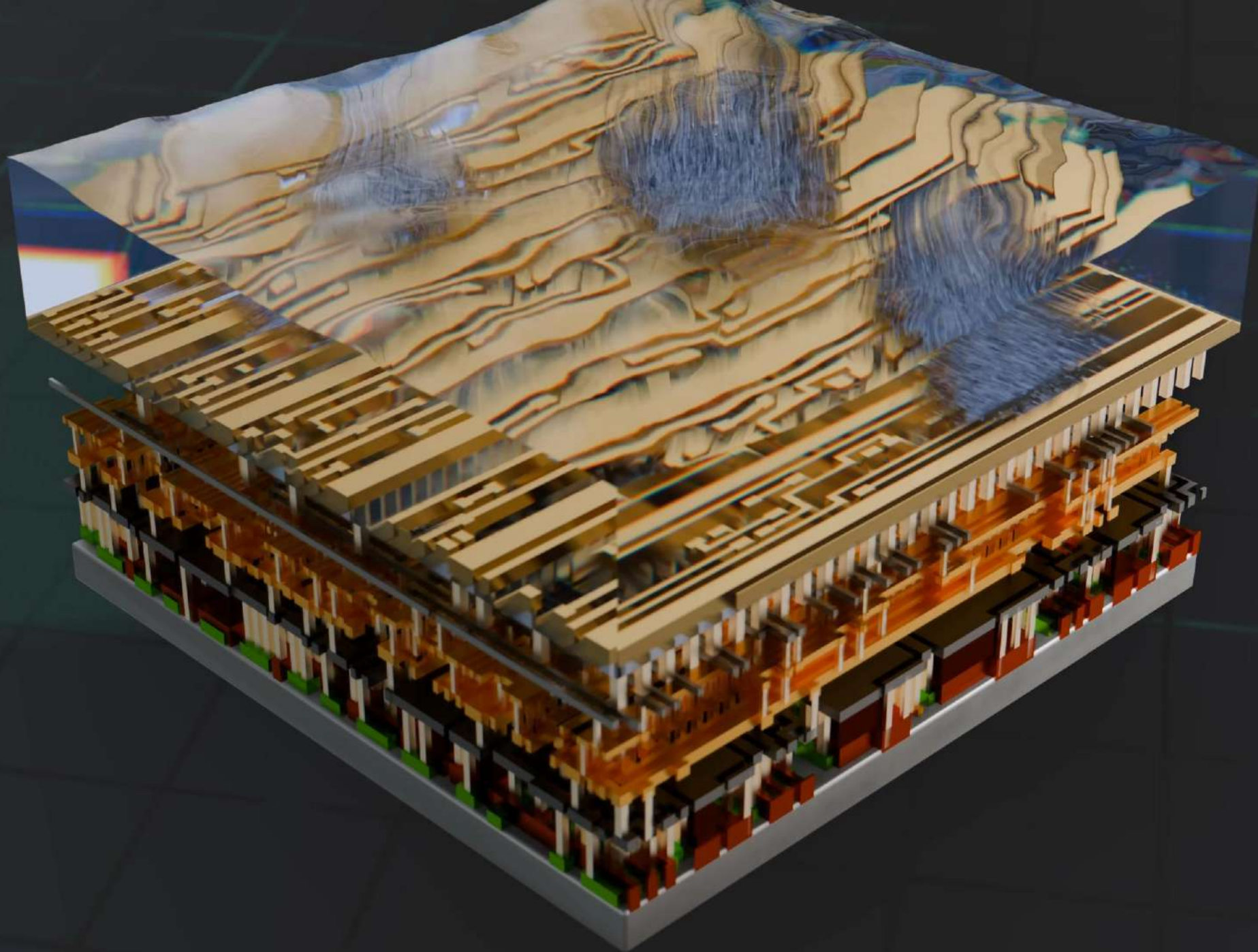
Chem



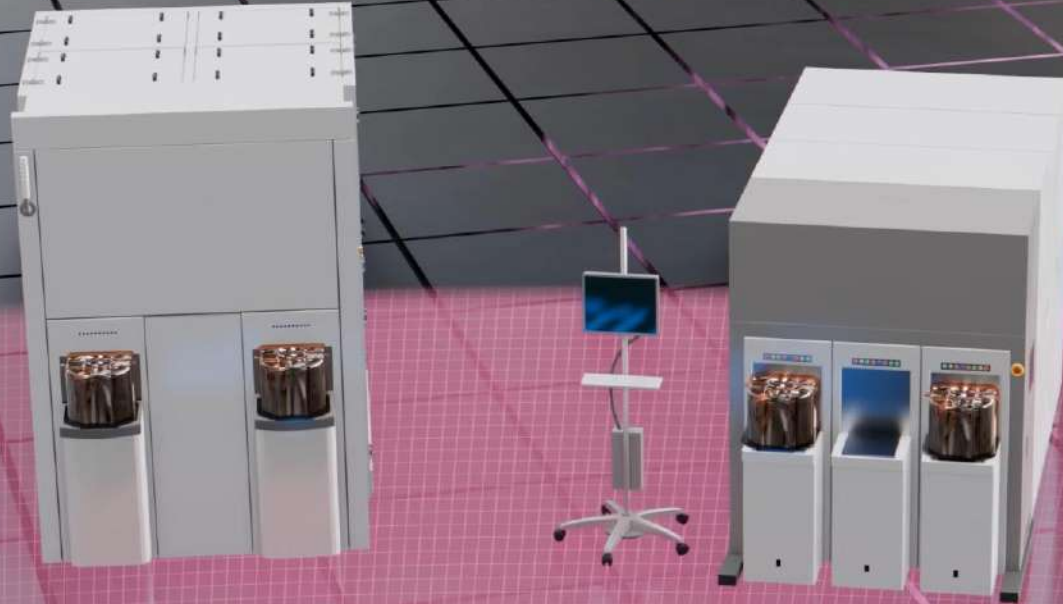






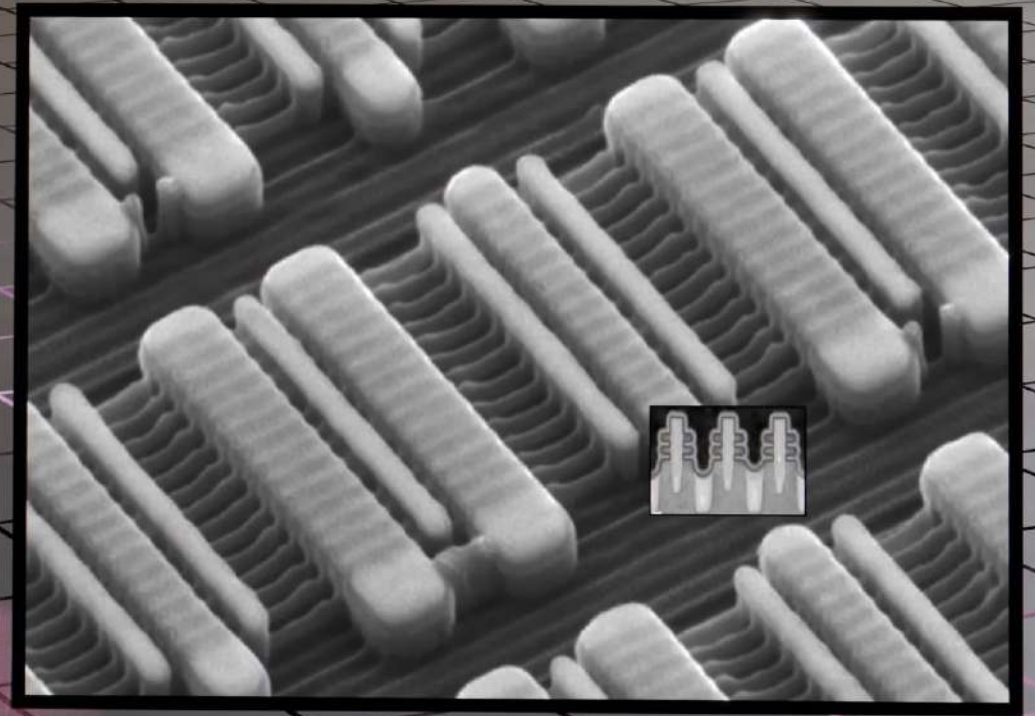
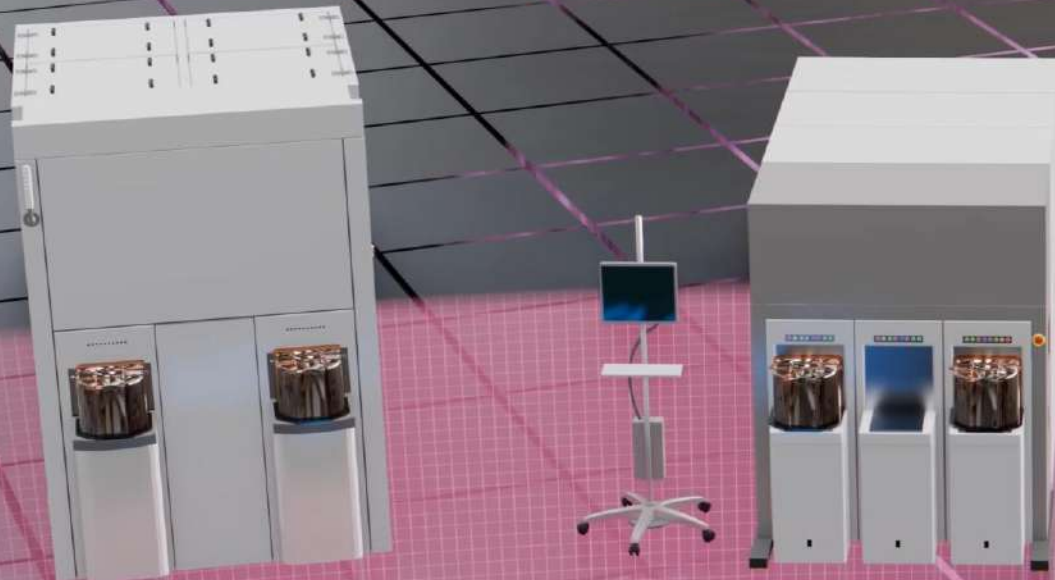


Scanning Electron Microscope



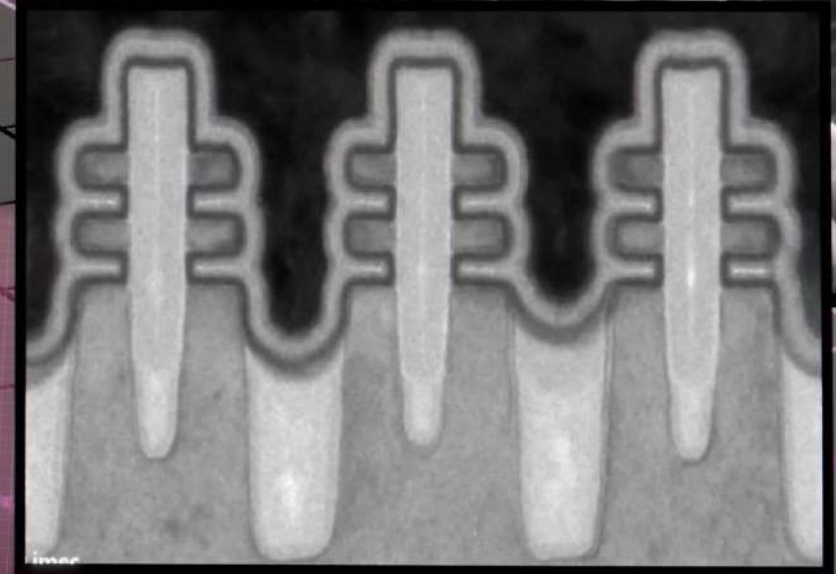
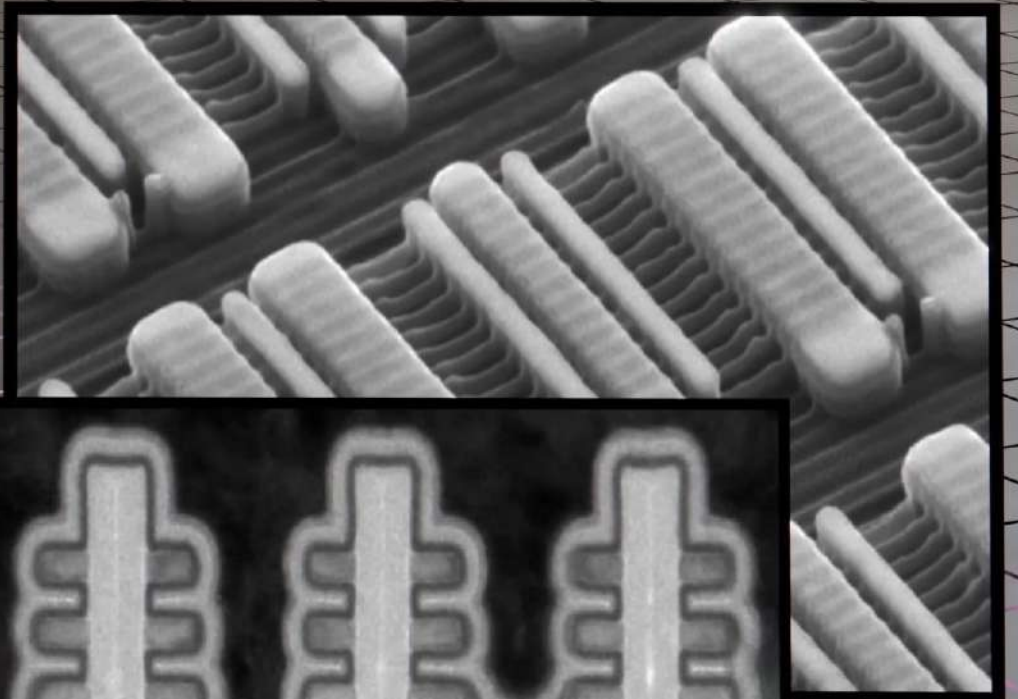
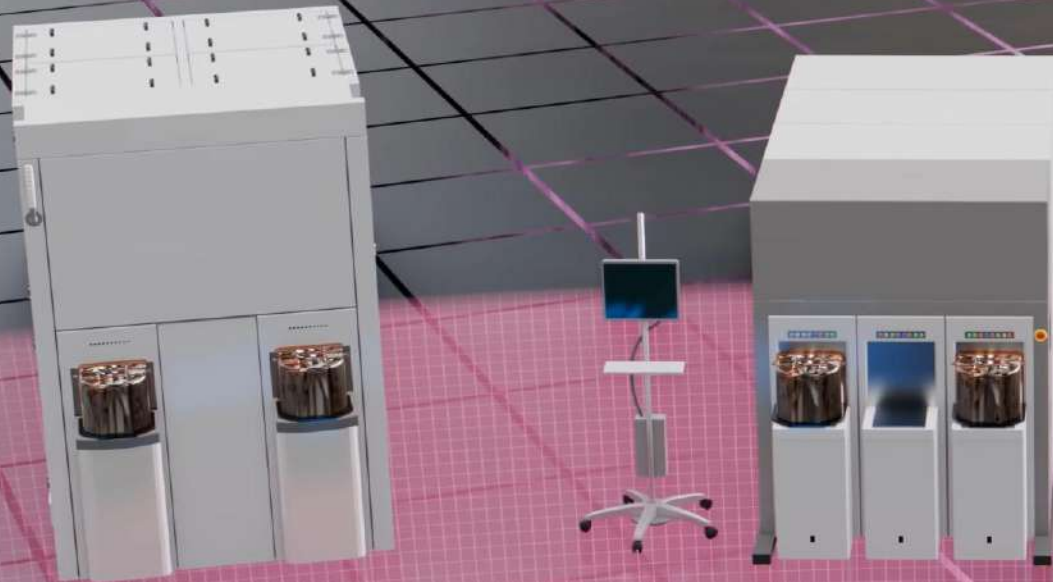
Metrology

Scanning Electron Microscope



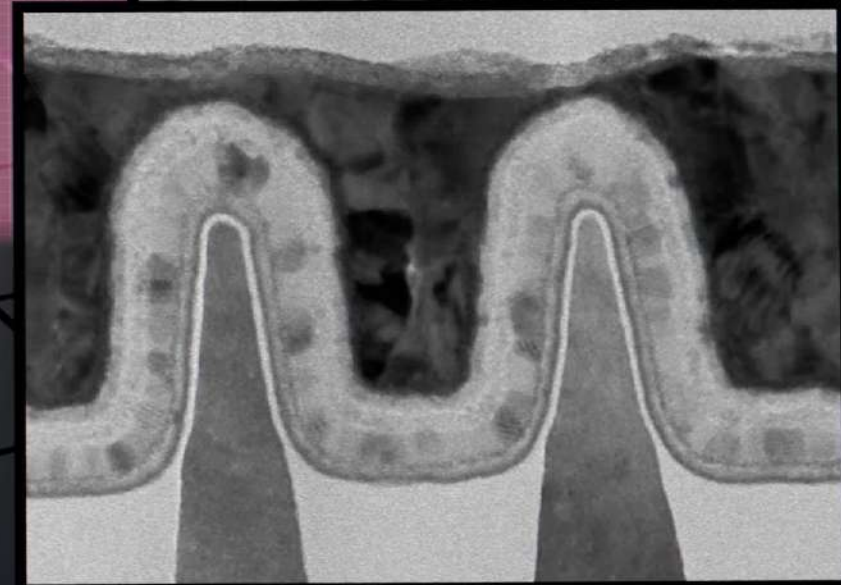
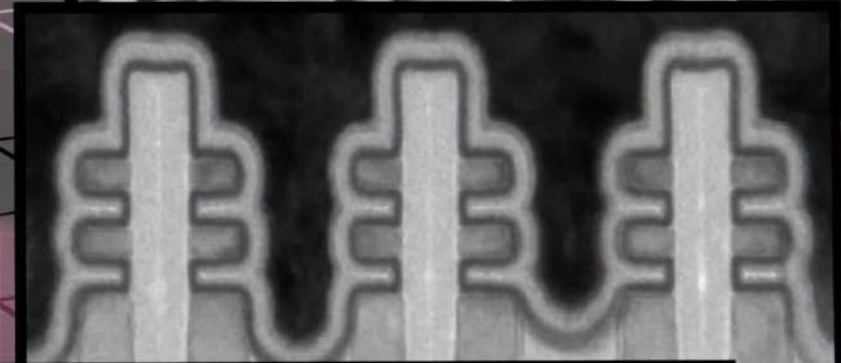
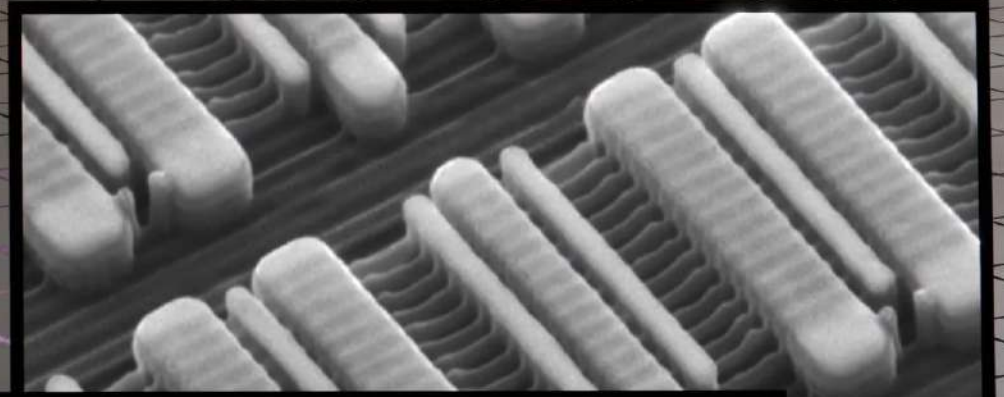
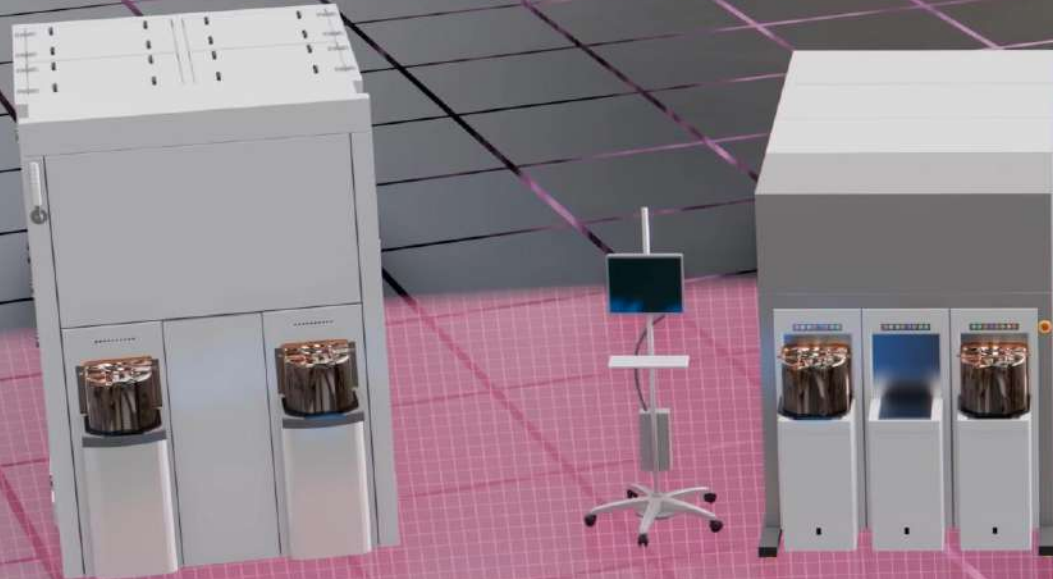
Metrology

Scanning Electron Microscope



Metrology

Scanning Electron Microscope



Metrology

Top: TechInsights

Middle: imec – nanosheets FETS

Bottom: ThermoFisher – Exploring Uncharted Realms with Electron Microscopy

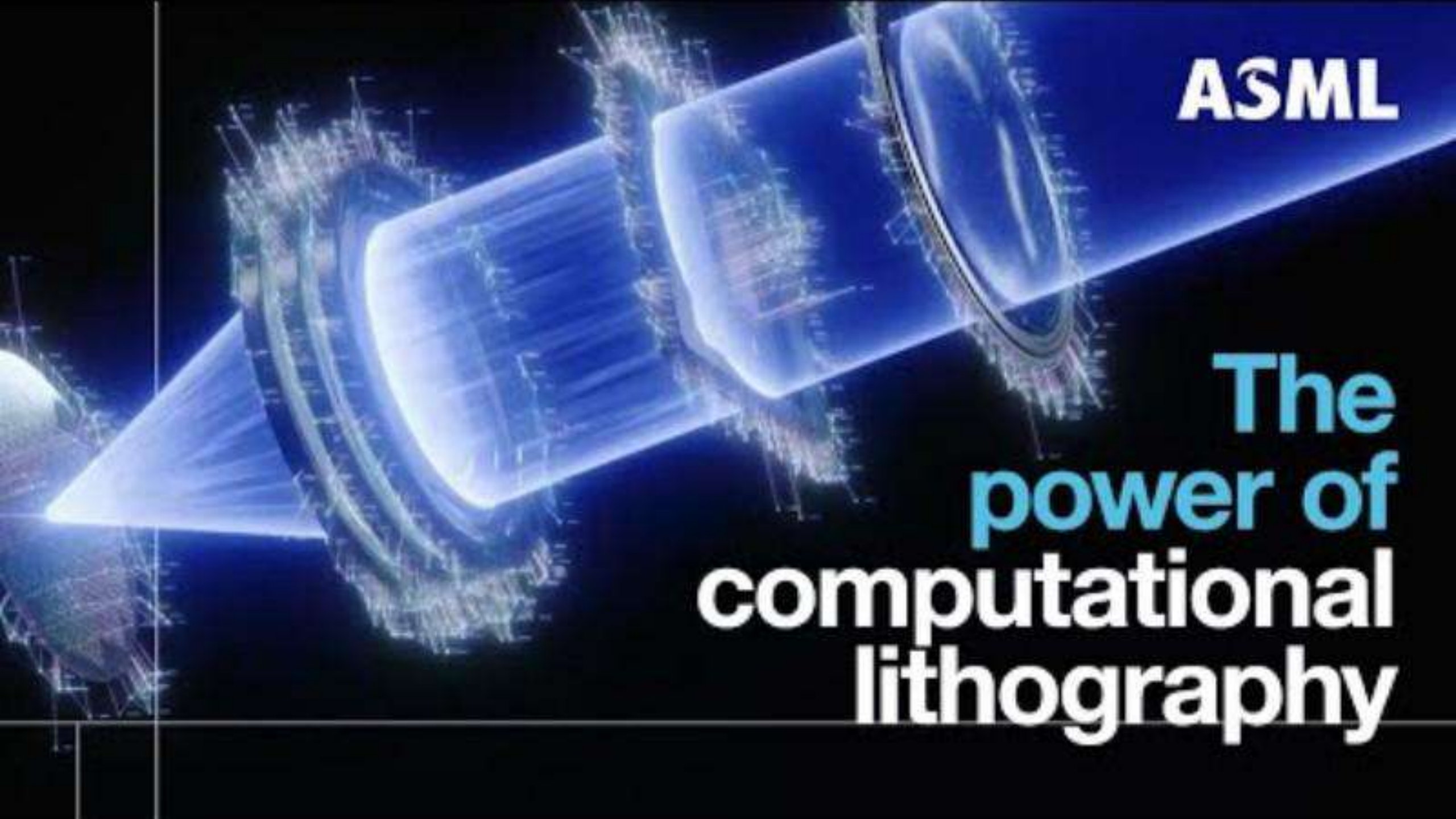
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4	5	6	7	8	9	10
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18	19	20	21	22	23	24
25	26	27	28	29	30	31

MON	TUE	WED	THU	FRI	SAT	SUN
				1	2	3
4	5	6	7	8	9	10
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18	19	20	21	22	23	24
25	26	27	28	29	30	31

MON	TUE	WED	THU	FRI	SAT	SUN
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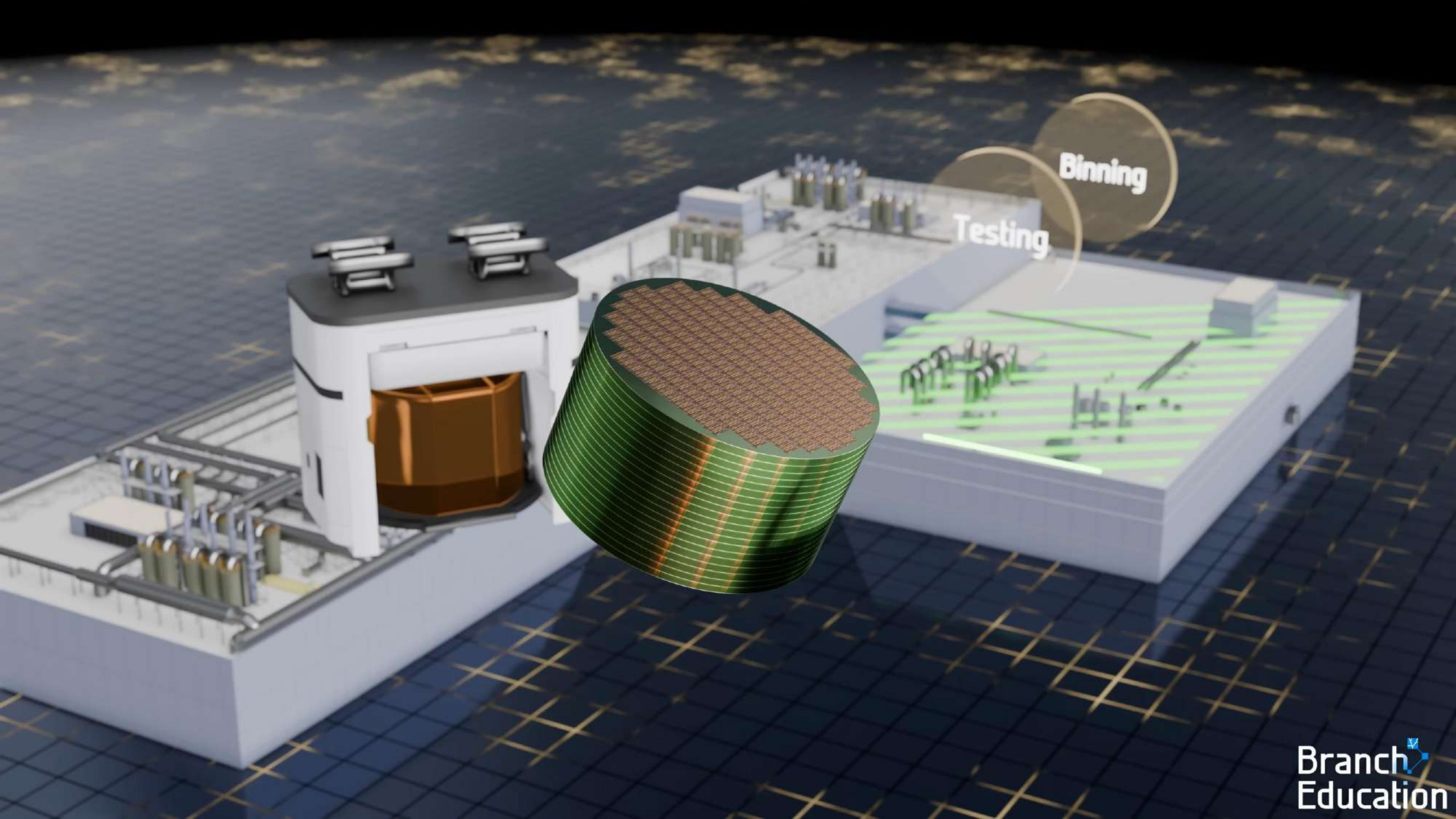
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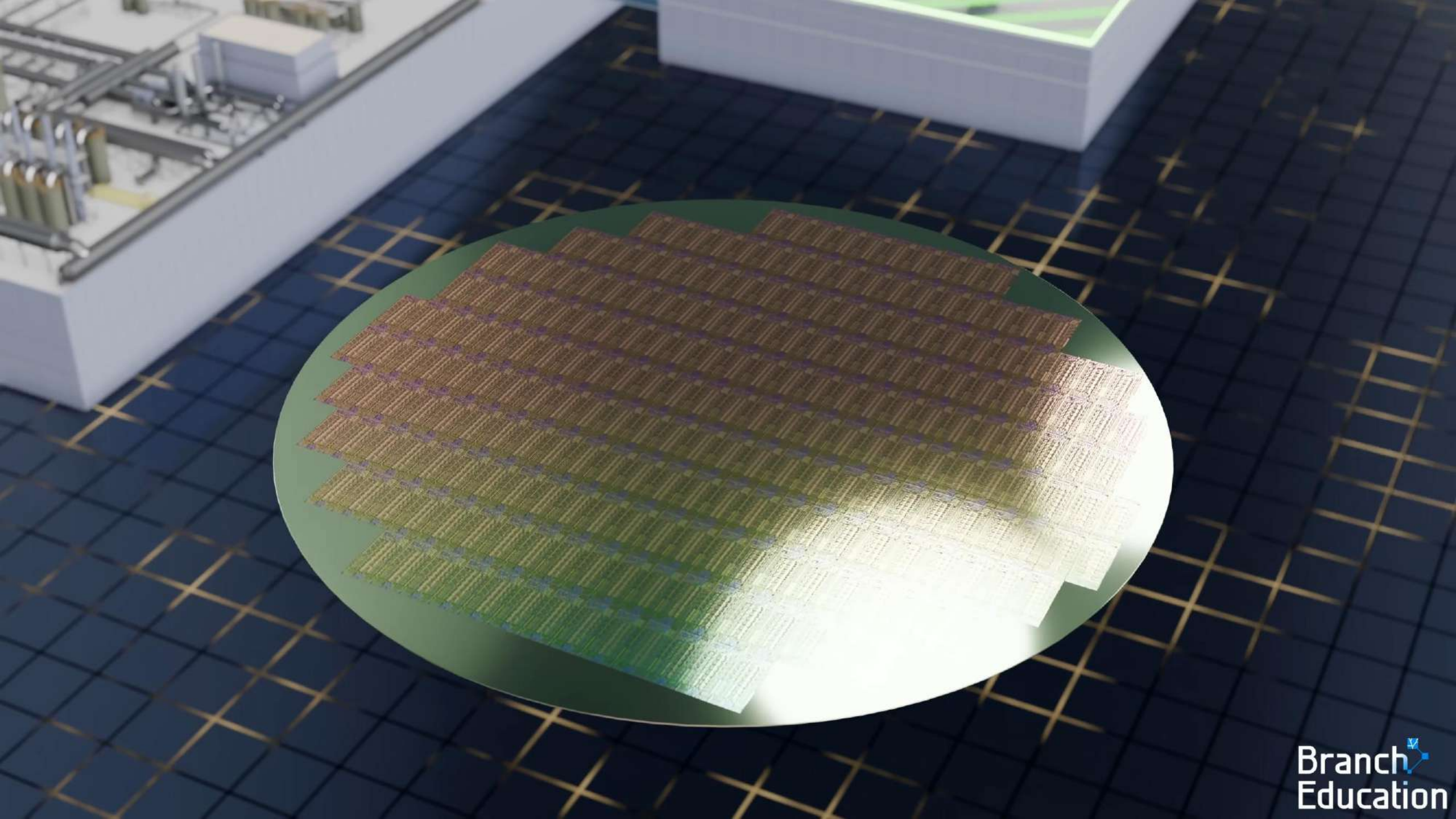


The background of the slide features a stylized, glowing blue graphic of a multi-element lens or optical system, possibly representing an EUV lithography system. The lens elements are depicted as translucent blue cylinders and rings, with a bright light source on the left creating a lens flare effect. The overall aesthetic is high-tech and futuristic.

ASML

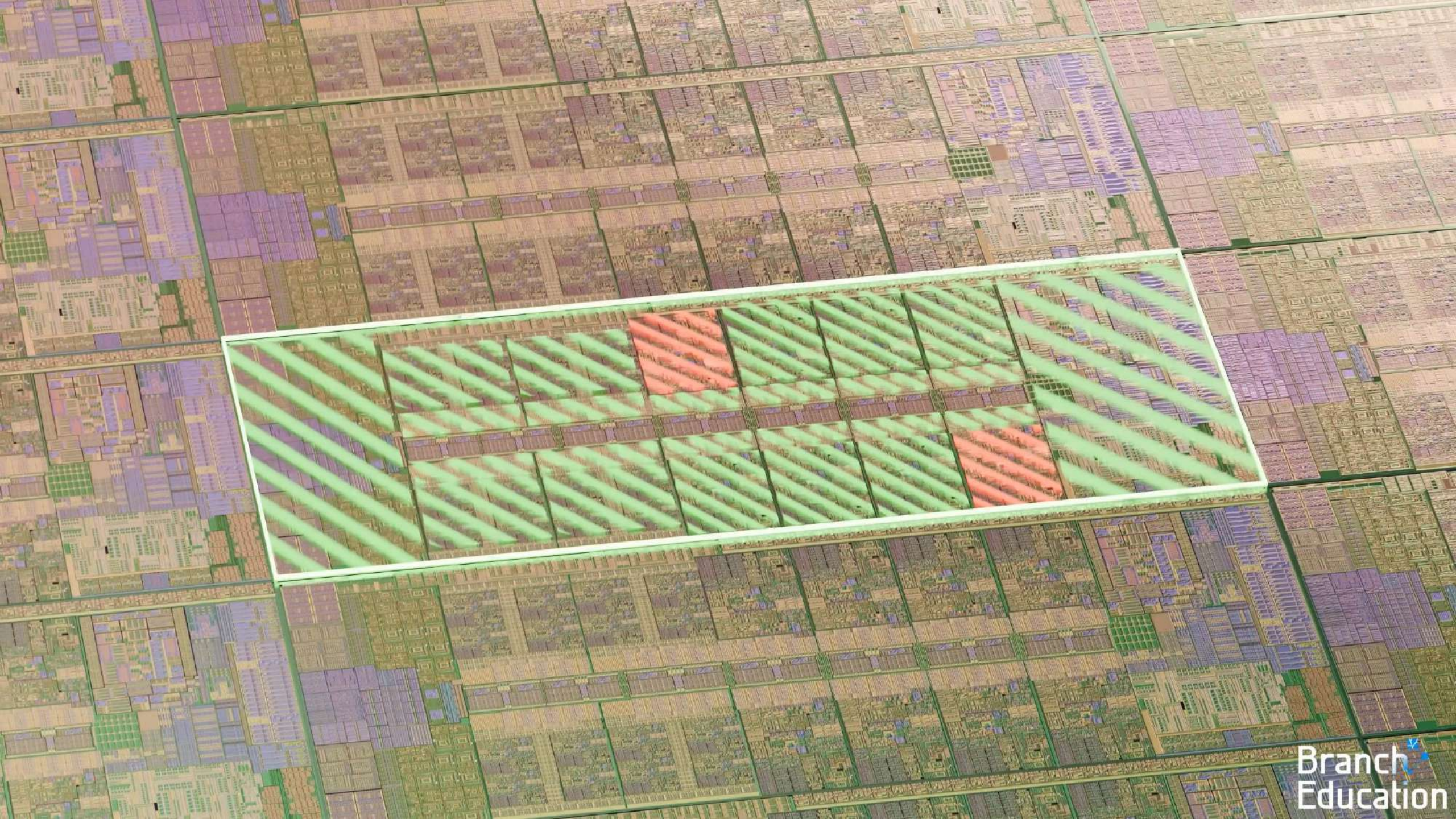
**The
power of
computational
lithography**

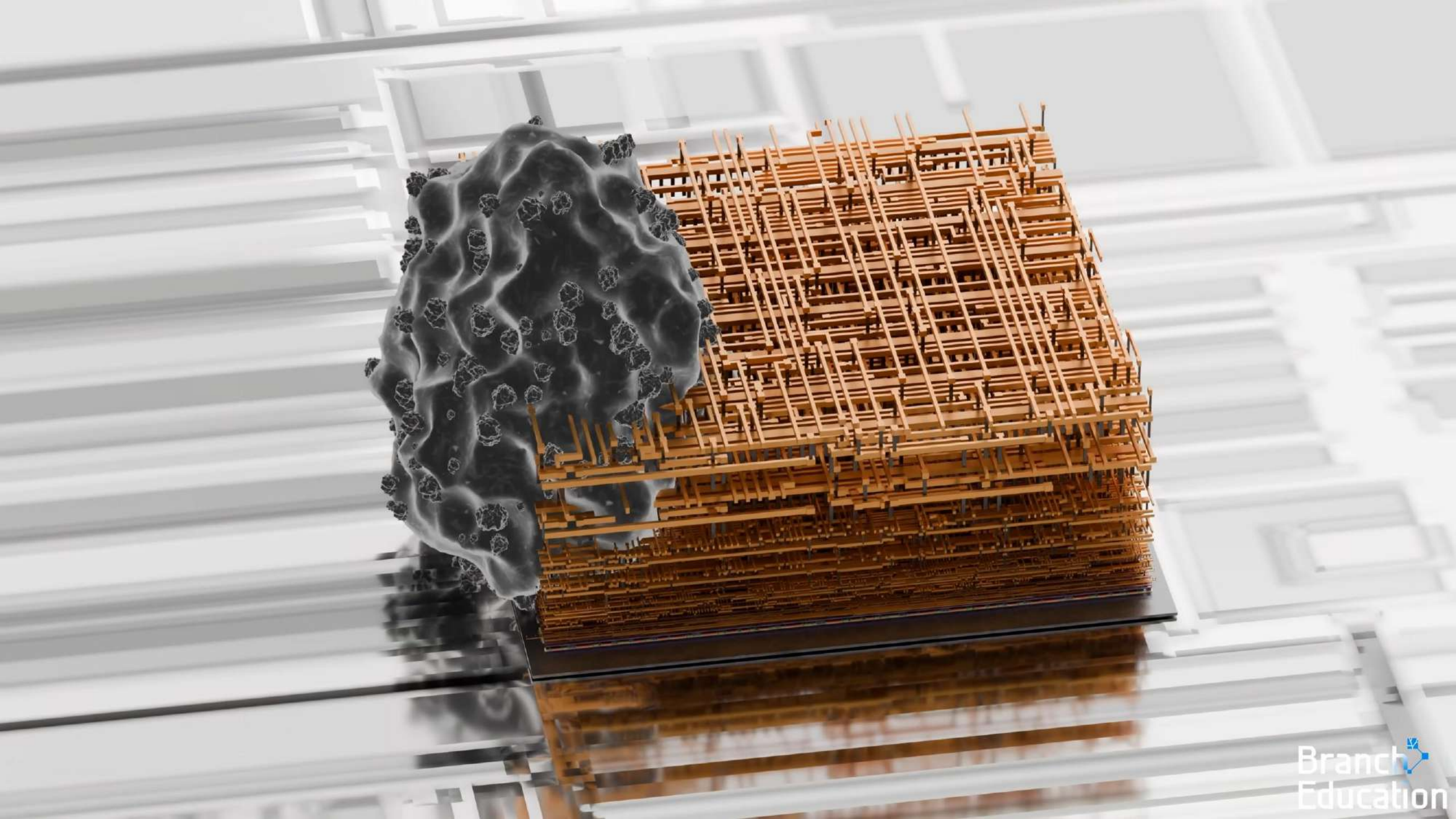


















i9



i7

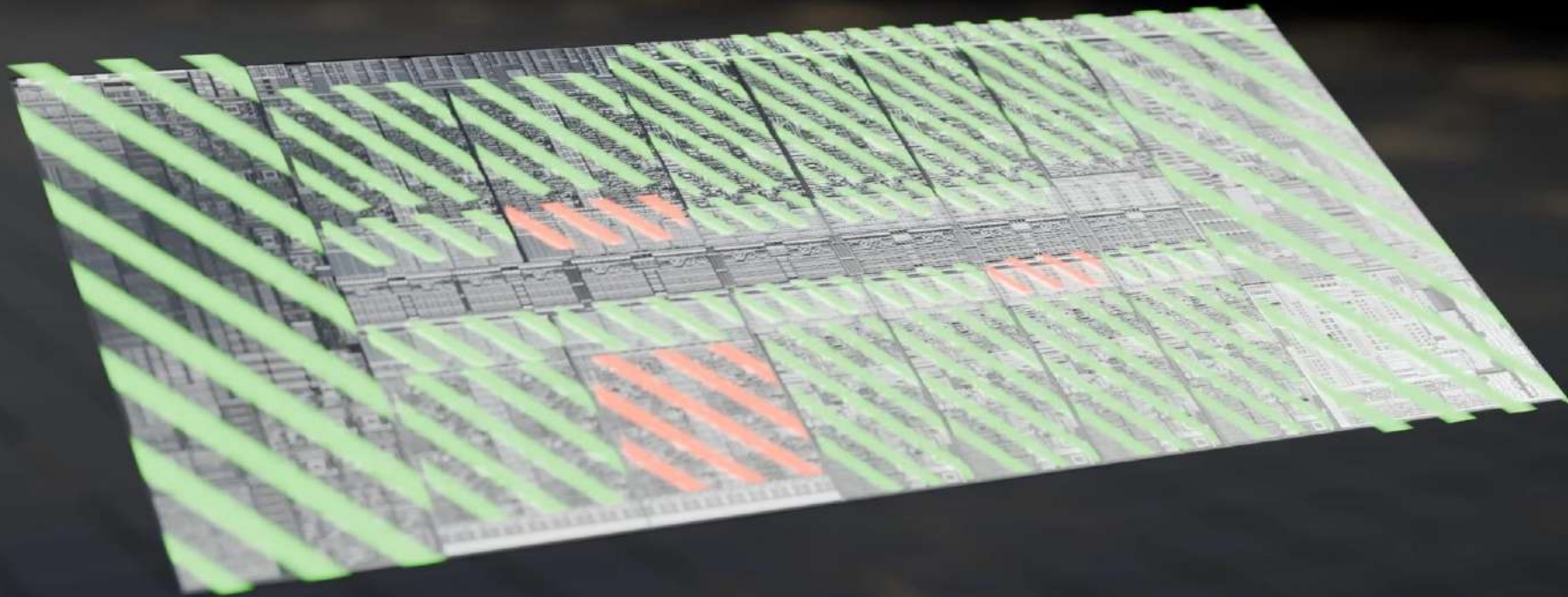


i5

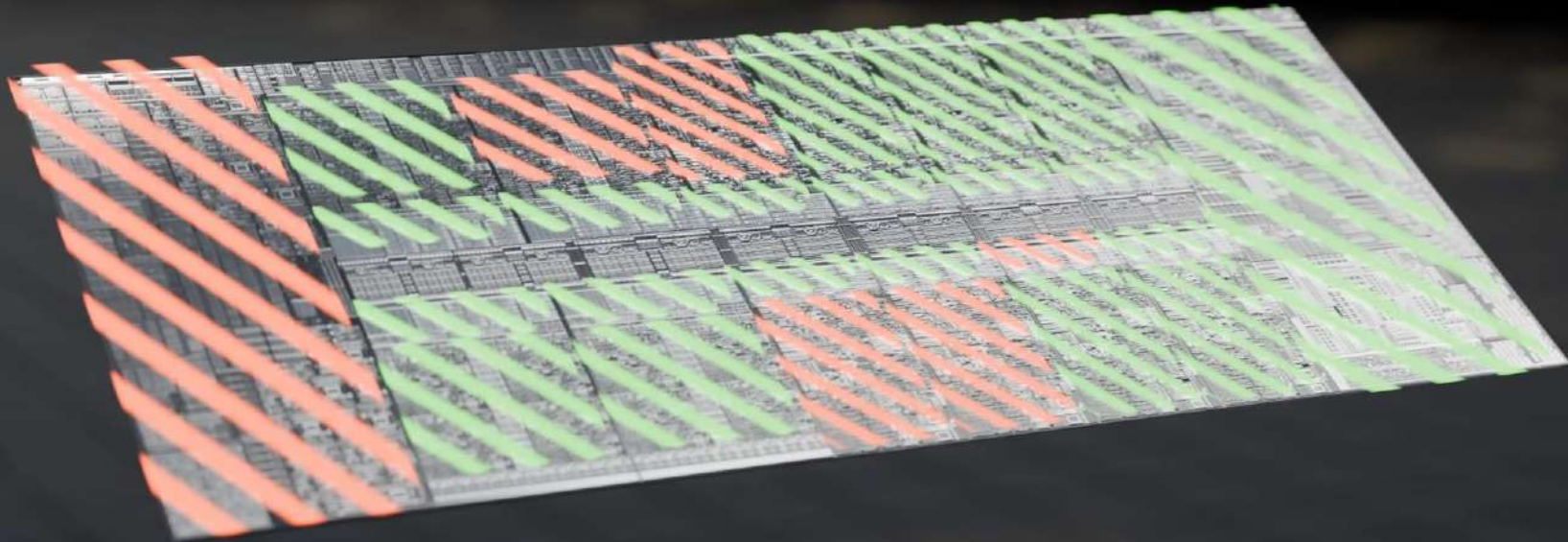


i3

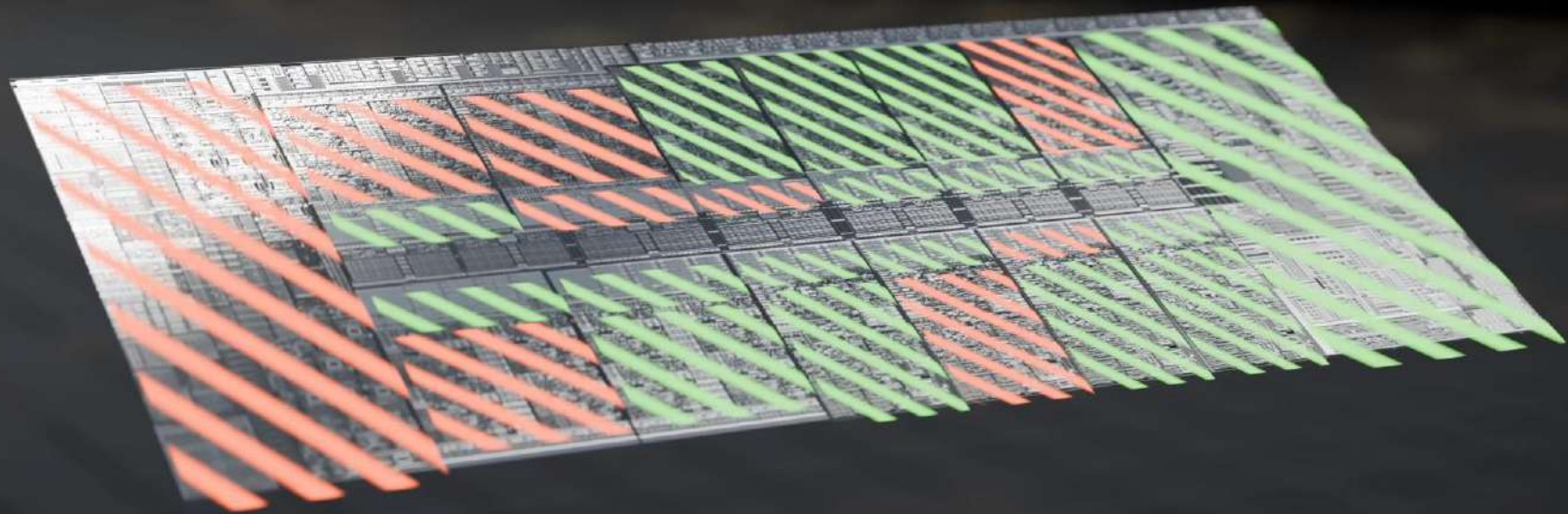
7

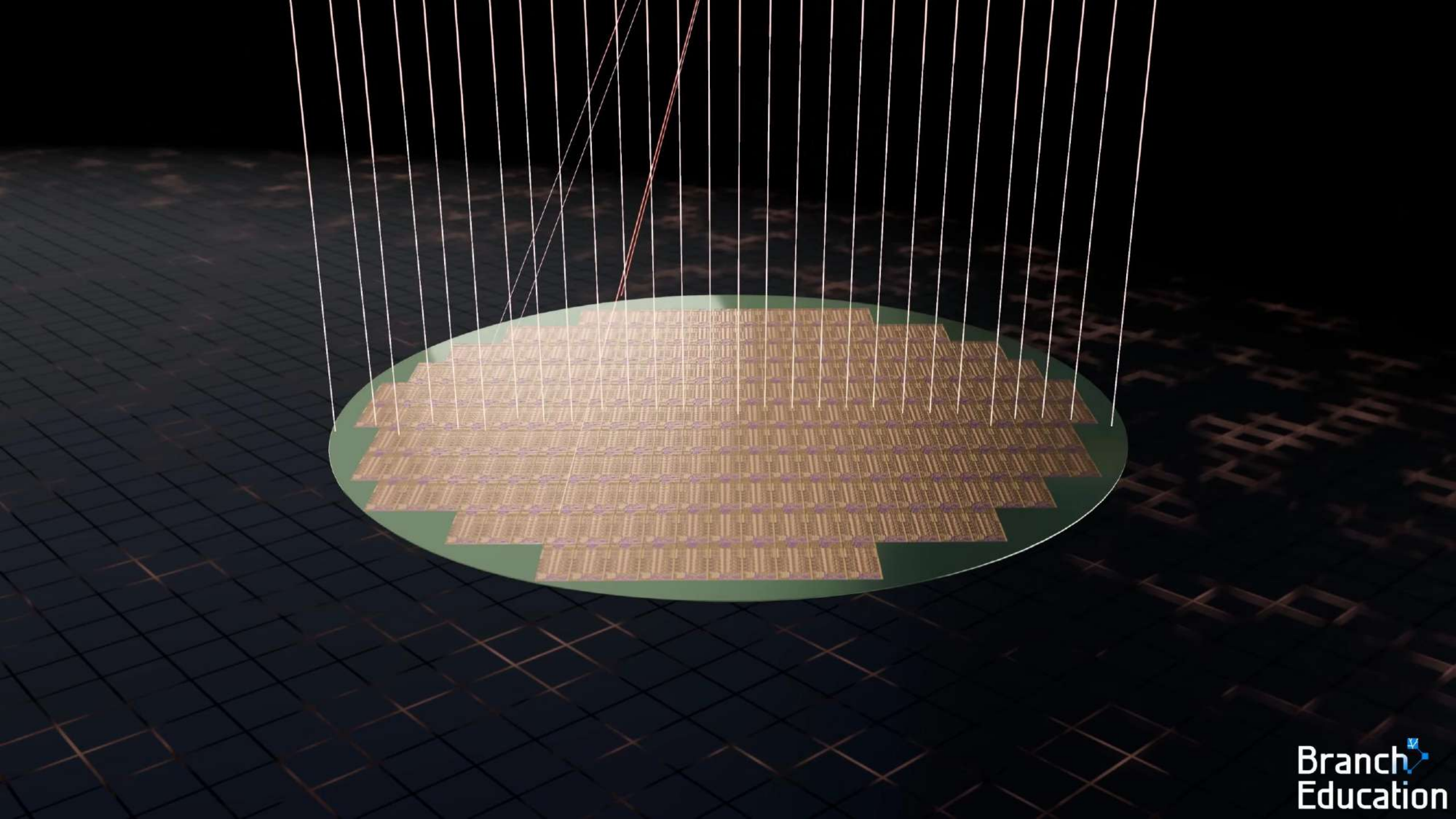


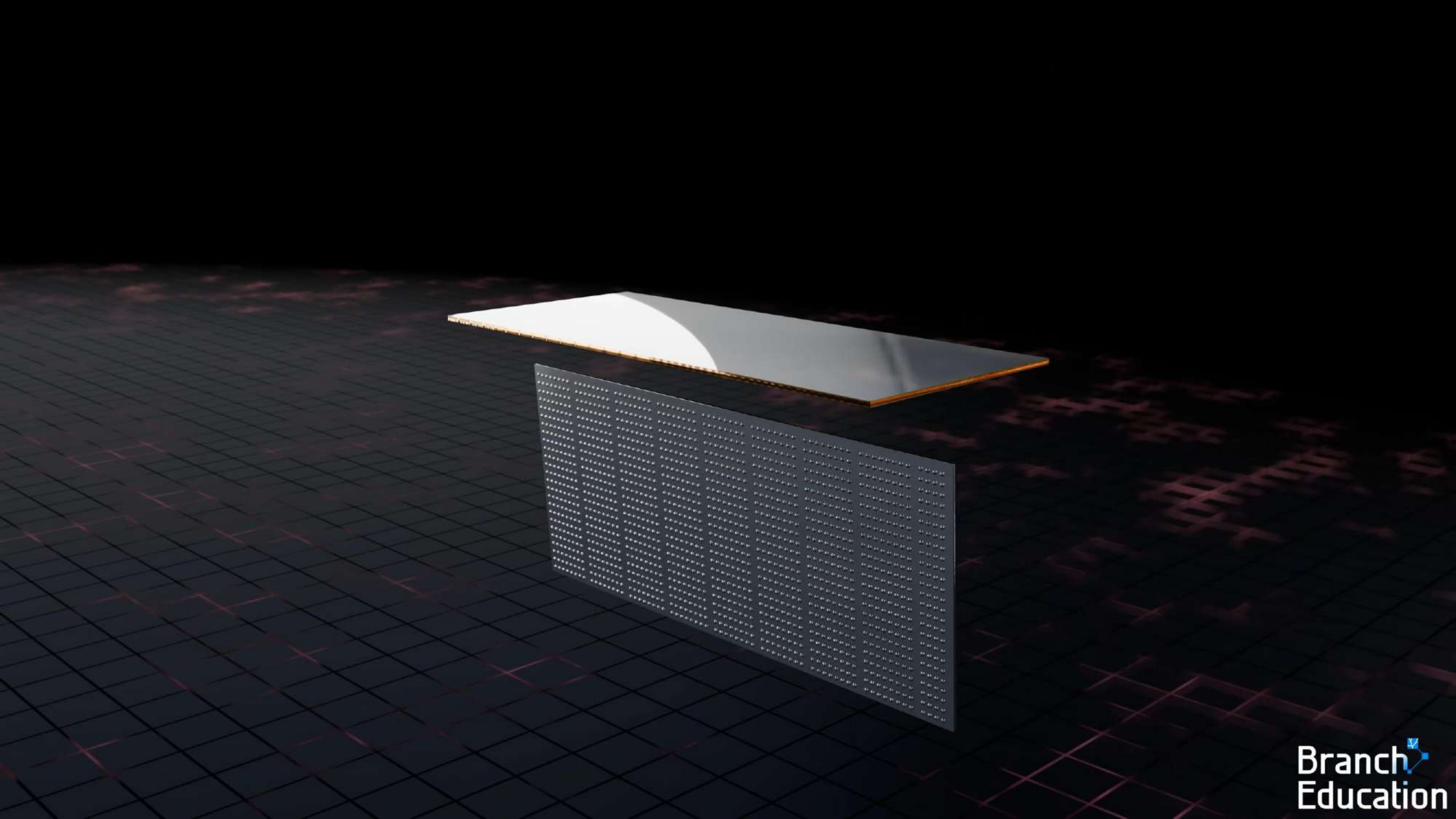
i5

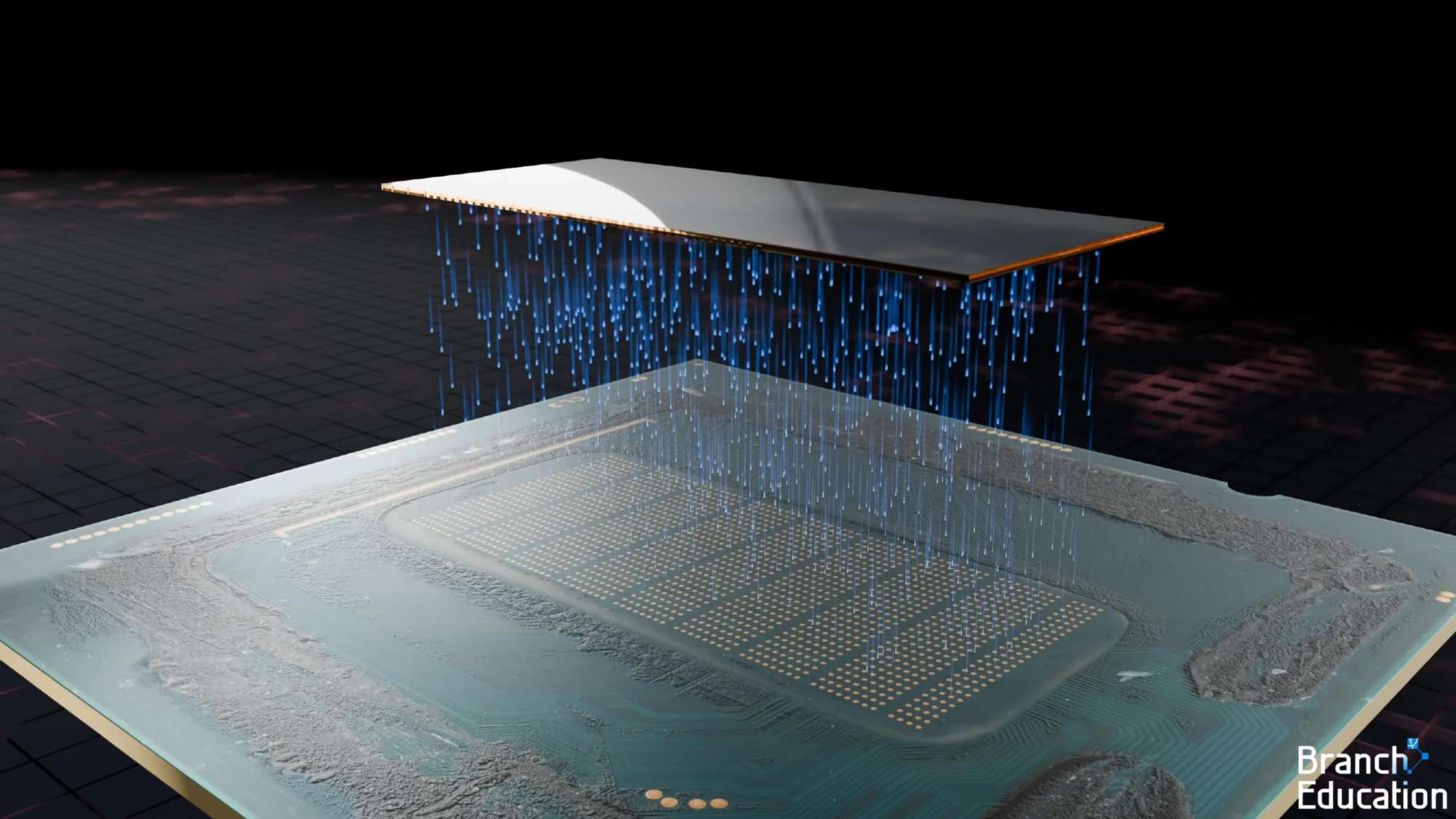


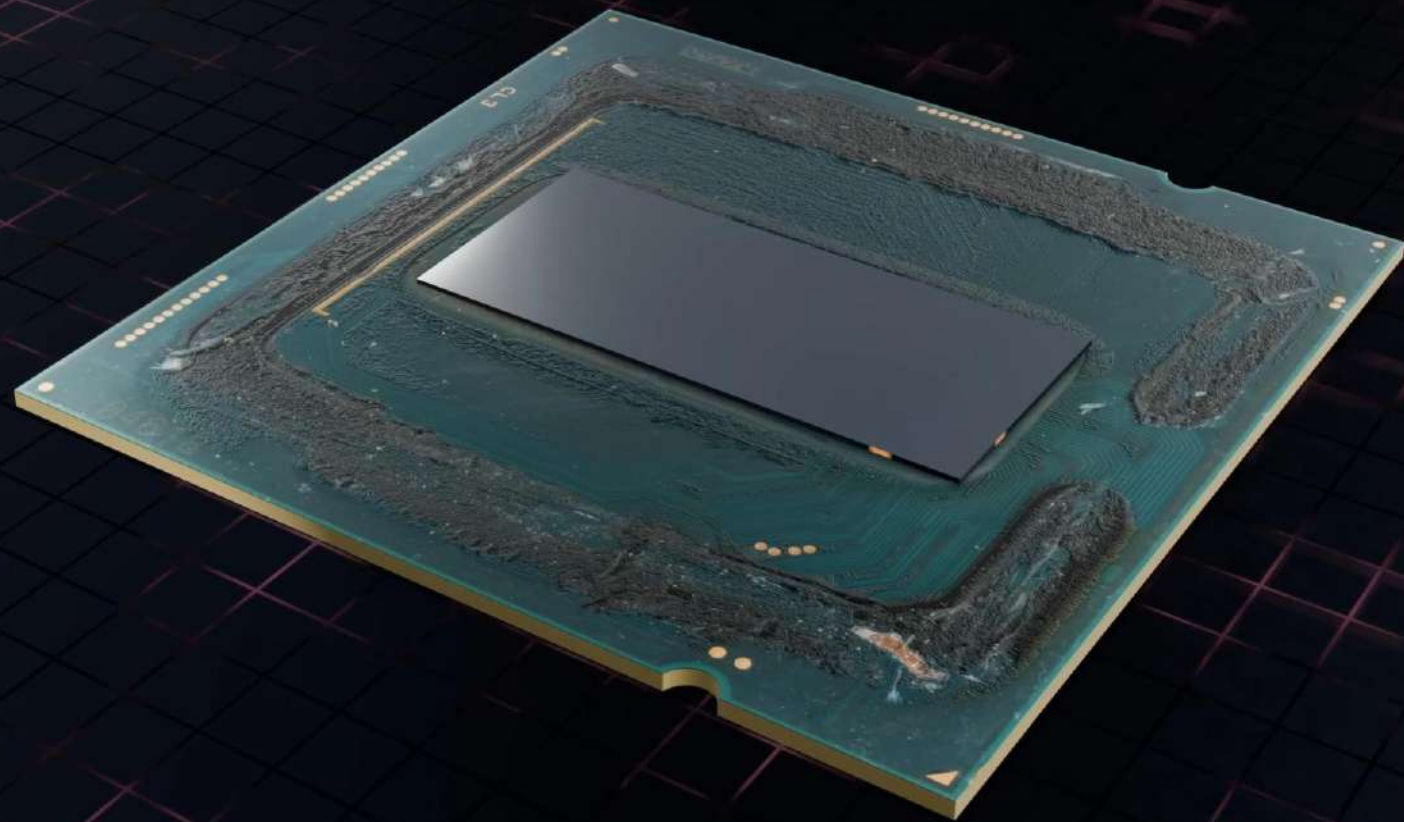
iB

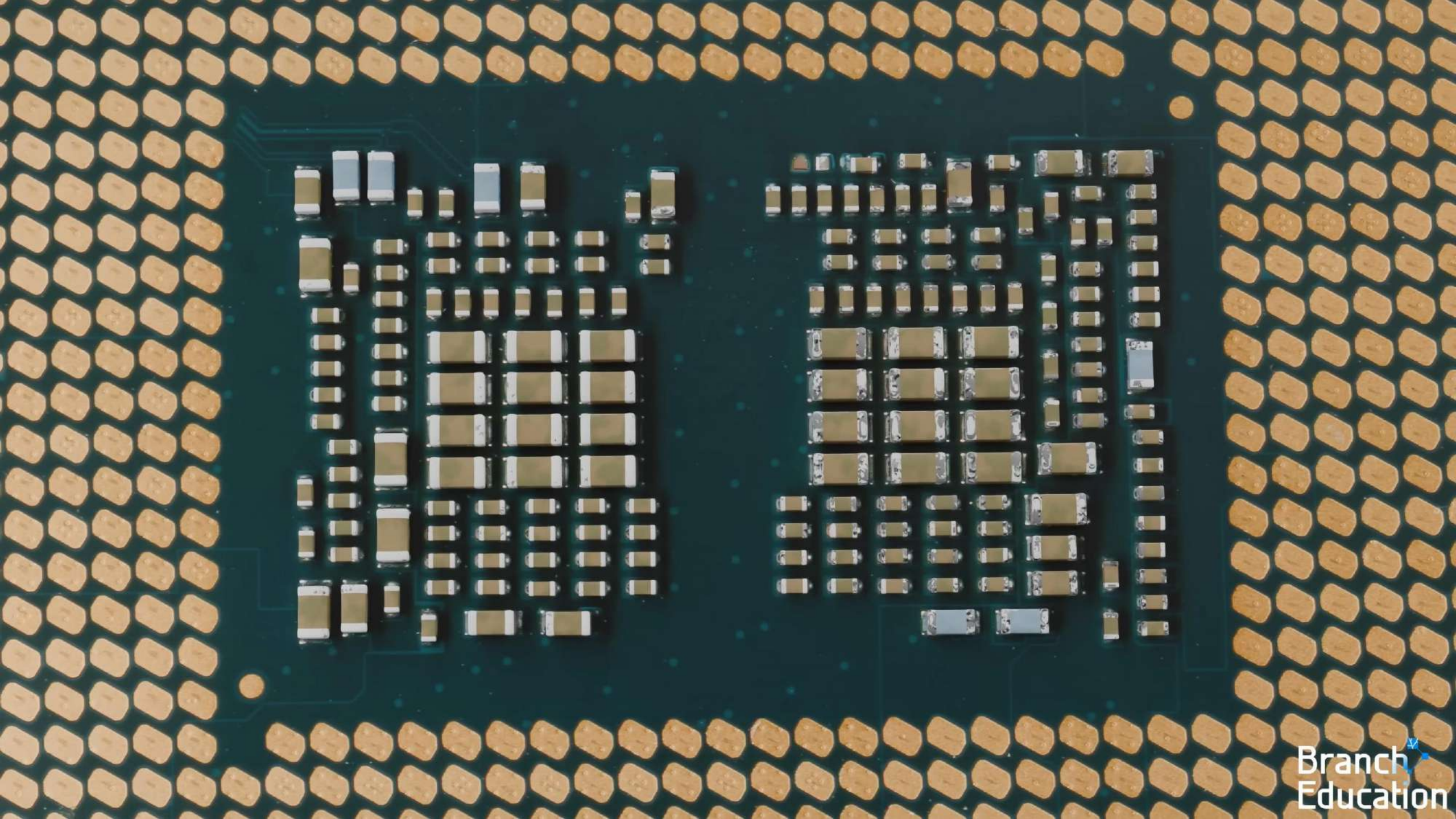
















다음 시간엔?

**우리는 기존의 폰 노이만 구조를 넘어,
더 빠르고, 효율적이며, 창의적인 미래 컴퓨팅 구조가
어떤 모습으로 발전하고 있는지 함께 살펴봅니다.**

**신경망 컴퓨팅부터 양자 컴퓨팅까지,
컴퓨팅 혁명의 새로운 패러다임을 탐구합니다.**

- 폰 노이만의 벽을 넘어서, 미래 컴퓨팅의 세계로 -